

Local Mixing of Reversible Circuits: History

This is the research history of the project: its constructions, experiments, attacks, and the reasons the method changed. The narrative and measurements below retain their original experimental context. References to the “current” or “complete” method within that narrative describe that point in the history; they are not a claim that every later implementation has been tested with those parameters. The separate current-method documentation describes the implementation available today without the experimental journey.

The historical narrative below predates the optional carrier-shuffling update of September 16, 2026. Its embedded-masking section describes fixed physical carriers. The dated implementation update records the addition while preserving the earlier discussion and results.

The following document details the ideas and experiments of Ran Canetti, Nicholas Ho, and other collaborators. The document is written by Nicholas Ho.

We begin with small local replacements and work through the attacks which motivate each construction. This leads us from linear gadgetization to product-share gadgetization, then to whole-circuit linear correlators and embedded masking. The experiments belong to the constructions and parameters described alongside them. Together, they explain what each method hides and what an attacker can still recover.

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Introduction

Circuit Obfuscation asks us to transform a circuit while preserving what it computes, so that an efficient attacker cannot distinguish the obfuscations of two equal-size circuits computing the same function. The prevalent measure of security for program obfuscation is called *Indistinguishability Obfuscation (iO)*. iO-level obfuscation is immensely powerful. For instance, it can be combined with essentially any source of computational hardness to achieve almost every other cryptographic task, such as trapdoor permutations or non-interactive zero-knowledge. Combined with lossy encryption, it is like fully homomorphic encryption can be achieved.

The history of attempts to construct iO is quite intriguing, but is beyond the scope of this document. Instead, the starting point of this document is the TCC'24 paper by Canetti, Chamon, Mucciolo, and Ruckenstein, which proposes a "first principles"

approach to constructing program obfuscation: obfuscation by way of a local mixing process applied to reversible circuits. Indeed, this approach is very different from the state of the art.

This document serves as an account of our efforts to flesh out and realize the approach outlined in CCMP'24. It describes our experiments, results, and questions that remain for local mixing of reversible circuits. We note that we solely rely on the structure of reversible circuits, but this of course is not a problem as we can reduce any arbitrary circuit to a reversible circuit.

A reversible circuit maps each input state to a sequence of unique intermediate states, and eventually a unique output state. This means that every operation can be readily undone. A state consists of a sequence of

n

bits (also called wires). We write circuits as lists of gates, where a gate applies to three wires in a state. Extra wires used to carry temporary values are called auxiliary or ancilla wires. When they start with specified values, those values are part of the function we are asking the circuit to compute.

Strategies

The below strategies will all follow the same ideas. Namely, we want to make simple actions and use minimal structure in order to obfuscate our circuits. In the pursuit of simplicity, our first circuits will thus only contain gate r57, a reversible gate which flips one target wire according to the values of two control wires. Standard gates like NOT, AND, OR, NOR, NAND, etc, can all be represented by gate r57. Thus, this simplification does not impede us from our goal of general-purpose obfuscation.

Definition: Let g be gate r57 on inputs A , B , and C , with A being the active wire, B being the positive control pin, and C being the negative control pin.

Then

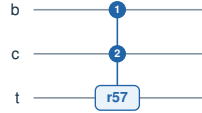
$$A = A + (B \vee \neg C)$$

In words, A is flipped unless $C = 1$ and $B = 0$. We also write this gate as $g57$; the two names refer to the same gate. Throughout, $+$ and $\oplus$ mean XOR when applied to Boolean values, while multiplication means AND. The notation $GF(2)$ refers to arithmetic on the two values 0 and 1 with these operations.

We will use the following notation for the circuit diagrams. Gates run from left to right. An $\oplus$ marks the target of an ordinary controlled XOR; filled and empty control circles read a one and a zero respectively. For $r57$, the numbered controls specify the order of its two inputs, since its control function is different from an ordinary conjunction.

Reading the circuit diagrams

r57 [t, b, c]



$$t' = t \oplus (b \vee \neg c)$$

CNOT



$$t' = t \oplus b$$

AND2 / Toffoli



$$t' = t \oplus bc$$

r57: ① is b and ② is c. Numbered circles specify input order, not AND controls.

For ordinary conjunction gates: ● reads 1, ○ reads 0, and $\oplus$ marks the wire being flipped.

Figure 1: The $r57$, CNOT, and AND2 circuit notation

For Boolean values a and b , we can use

$$a \vee \neg b = 1 \oplus (\neg a)b = 1 \oplus b \oplus ab.$$

Therefore, over $GF(2)$,

$$A = A + \neg B \wedge C + 1 = A + BC + C + 1.$$

Note the constant term. Every $r57$ gate flips its active wire unconditionally and then unflips it on one of the four control states, so a $+1$ appears in every gate's polynomial. This will matter a great deal later, when we start caring about the algebraic degree of our circuits.

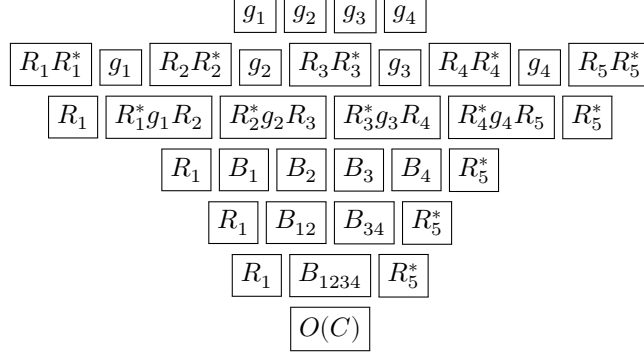
We note that the restriction to $r57$ alone does not hold all the way through the complete construction. The source circuit is built from $r57$, but the later sandwich and gadgetization use a wider gate vocabulary. Later splitting deliberately fragments the remaining complemented gates into plain conjunctions. We

describe that vocabulary when we introduce the circuit representation below. The early strategies use pure r57 unless stated otherwise.

A good question to answer is what type of actions we are allowed to do to follow the rule of “first principles”. The basic local mixing of reversible circuits strategy is thus to only allow small local and equivalent perturbations of a circuit in the hopes that with enough of them, we can get a global effect. In other words, given a circuit, we sample and replace many subcircuits of equivalent functionality. Of course, there are many ways to do this that do not actually incur a large enough effect. For example, say we have a circuit on 64 wires with 1000 gates and only make 100 replacements on 6 gate subcircuits. Then, with most replacement methods, an observer can merely find these points of replacements and potentially even undo them. We then have a number of concerns that we will need to address. Firstly, how can we sample replacements? This is a question that we did not think too deeply about at the beginning of the project, and so will be more deeply talked about later. Secondly, how can we mix gates in a way that we can not identify points of replacement?

With these questions in mind, let us introduce the obfuscation strategies that we have tried. We write R^* or R^{-1} for the inverse of a circuit R . Each gate in our vocabulary is its own inverse, so reversing the order of the gates gives the inverse circuit. Thus, RR^* is an identity: it leaves every input state unchanged.

1. **The Two Phase Strategy:** This strategy splits up the obfuscation process into two phases. The first phase is the *inflationary phase*. In this phase, we sample random subcircuits and make replacements that contain more gates than the subcircuit we are replacing. Thus, we slowly inflate our subcircuit. We then move on to the *kneading phase*. In this phase, we sample random subcircuits and make replacements with equal length random circuits in the hopes that we can spread out the randomness that we injected into our circuit in the inflationary phase.
2. **The Butterfly Method:** The butterfly method utilizes compression, something that we will talk more about later. As opposed to a two stage method, the butterfly method has many stages of (inflationary $\rightarrow$ compression). Suppose we have a circuit C . We then wrap each gate with an identity of the form $R_i R_i^{-1}$, a simple identity. We can sample R_i completely randomly and so we have a “stupid” identity. Afterwards, we group up each $R_i^* g_i R_{i+1}$ and randomize it to get B_i . This can be done by attempting to compress it, expand it, or make equal length replacements of equivalent functionality. All that remains is to then merge the B_i together, compressing as we do so. The diagram below shows one round of this in more detail. Ideally, we would use enough rounds for this to effectively “mix” our gates.



3. **Pair Replacement Methods:** The pair replacement methods are different strategies that utilize one type of replacement: the pair replacement. Given two gates g_i and g_{i+1} , we can replace them with some identity I_i by relabeling the first two gates to be equal to the pair. In other words, we have $I_i = g_i g_{i+1} B_i$, which gives us $B_i^* = g_i g_{i+1}$. These strategies will make repeated replacements of this type before moving onto a compression phase. These two phases will then repeat. This is meant to be seen as a smaller version of the butterfly method.
4. **Fragmentation:** Fragmentation deliberately stops requiring every gate to remain r57. Keeping one gate type makes the rainbow tables much easier to use, but it also leaves the transformed circuit with one repeated gate shape. We begin with an exact split. The r57 gate $a \oplus = (b \vee \neg c)$ can be split in either of the following two ways:

$$\{a \oplus = b, \quad a \oplus = \neg b \wedge \neg c\}$$

or

$$\{a \oplus = \neg c, \quad a \oplus = b \wedge c\}.$$

In either split, the two firing conditions are disjoint and their XOR is the firing condition of the original gate. This makes the replacement exact. The two pieces are plain conjunctions which share a target and commute with one another. More importantly, some pieces of two parent gates which originally collided may now commute because their firing conditions are disjoint. We can therefore move those pieces into, and sometimes past, one another. The initial split still has at most two controls. Later crossing splits can create gates with three, four, or more controls and produce more varied polynomial forms. The later fragmentation section explains the two mixing methods which build on this identity.

Sampling Subcircuits

In the above strategies, it is important for us to sample random subcircuits and then replace them. We will discuss how to make these replacements in a later section. Before we talk about our methods, it is useful to define collisions between gates.

Definition: Two gates $g_1 = (A_1, B_1, C_1)$ and $g_2 = (A_2, B_2, C_2)$ collide if one of the following holds:

1. $A_1 == A_2$
2. $A_1 == B_2$
3. $A_1 == C_2$
4. $B_1 == A_2$
5. $C_1 == A_2$

The first obvious way to sample subcircuits is by sampling a random contiguous subcircuit within our circuit. If we view our circuit as an array of gates, then this is merely taking a window of some size. This is, of course, very easy to do.

The other method is to take “convex subcircuits”. The meaning of convex here directly lies in finding convex subgraphs of a directed graph. In order to turn a circuit into its graph representation, let each gate be a node i and for every other gate j that collides with i , is such that $i < j$, and there is no k with $i < k < j$ that already collides with i , then there is a directed edge from i to j . We do this for all nodes in order to get our skeleton graph. From this, we can sample a convex subgraph and then take the circuit version of it in order to get what we call a **convex subcircuit**. As we have convexity, we can then rearrange gates in the original circuit to make this convex subcircuit a contiguous subcircuit. As a reminder, we note that gates can be swapped without changing the functionality of the original circuit if they do not collide. Unfortunately, finding convex subcircuits is expensive when repeated hundreds of millions of times. In our measurements, it took about 40% of the total computation time.

Random Circuits

Before anything, it is important to establish a shared language to discuss our circuits. Below is a table for any circuit-wiring language that we use for circuits with only r57.

Table: Wire Index Encoding Used by Circuits of r57

Wire range	Encoded as	Notes
0–9	0–9	Decimal digits
10–35	a–z	Lowercase letters
36–61	A–Z	Uppercase letters

Wire range	Encoded as	Notes
62–71	! @ # \$ % ^ & * ()	Special symbols (set 1)
72–82	- _ = + [] { } < > ?	Special symbols (set 2)
≥ 83	$\sim^k X$	k tildes followed by base-83 digit, giving wire $83k + X$

A gate consists of three pins, each on a distinct wire. We represent a gate by listing the wires in the order (active pin, positive control pin, negative control pin). For example, 123 represents a gate with the active pin on wire 1 and control pins on wires 2 and 3. A circuit is represented as a sequence of gates separated by ;.

Notation:

A gate $g = [a, b, c]$, where a is the active pin and b, c are the control pins, is written as $\mathbf{abc}$.

A circuit is written as a list of such gates separated by ;.

While this is not the notation we will be using to represent our mixed circuits, this is the notation that is used when storing circuits as we will strictly be using r57 gates in our rainbow tables.

Beyond r57

The encoding above can only ever describe r57 circuits. A gate is three wire tokens and nothing else, so there is nowhere in it to say *which* control function the gate uses: the control function is fixed by the format, and any circuit written this way is r57 by construction. For most of this document that is exactly what we want, and we will lean on it heavily.

As we said at the outset, it does not hold all the way through the complete construction. Later stages deliberately emit gates that are *not* r57, and once they have run the circuit is not expressible in the three-token format at all. Why we are willing to give up r57, and what it costs us, is an argument that belongs with the attack material and is taken up later in *Fragmentation: leaving r57 behind*. What we need here is only a notation general enough to write the results down.

That notation is a single gate type: a single-target controlled XOR whose control is a conjunction of mixed-polarity literals, optionally complemented as a whole.

$$\mathbf{fires}(x) = \mathbf{comp} \oplus \bigwedge_i \mathbf{lit}_i(x), \quad x[\mathbf{target}] \oplus = \mathbf{fires}(x)$$

Each $\mathbf{lit}_i$ is a wire or its negation. The members of this family that actually occur in our circuits are worth naming, since we refer to them by name throughout.

Table: The wider gate vocabulary

Name	comp	Controls	Effect
X	0	none	$t \oplus = 1$
CNOT	0	one, positive	$t \oplus = x_c$
NCNOT	0	one, negative	$t \oplus = \neg x_c$
AND2	0	two	$t \oplus = \text{lit}_a \wedge \text{lit}_b$
conj wK	0	$K \geq 3$	$t \oplus =$ the K -literal conjunction
r57	1	two, opposite polarity	$t \oplus = (x_b \vee \neg x_c)$
comp wK	1	K , any polarity	$t \oplus =$ $\neg(\text{the conjunction})$

The important row is the r57 one. An r57 gate $[a, b, c]$ is **comp** = 1 with the two literals $\neg b, c$, since $1 \oplus (\neg b \wedge c) = b \vee \neg c$. So r57 is a *member* of this family rather than something sitting outside it, which is what makes the notation worth adopting at all: one vocabulary holds plain r57 and non-r57 material at the same time, and a circuit that is only partly converted is still a single object in a single format. At the other end of the table, an empty conjunction is true, so the **comp** = 0 gate with no controls fires on every input and is an unconditional NOT.

Circuits in this vocabulary are written in a second, plainer text format we call **mpmct1**, for mixed-polarity multi-controlled Toffoli. The three-token encoding gets its density from knowing the gate type in advance, which is precisely what **mpmct1** cannot assume, so it spells everything out instead: a header line naming the wire and gate counts, then one gate per line, all in decimal.

```
mpmct1 <num_wires> <num_gates>
<target> <comp> <k> <wire> <pol> ... (k wire/polarity pairs)
```

The wire numbers in this format are ordinary zero-indexed decimal numbers. A polarity bit of 1 denotes the positive literal x_w , while a polarity bit of 0 denotes the negative literal $\neg x_w$. If the conjunction of the k literals is L , then **comp** = 0 flips the target when $L = 1$, while **comp** = 1 flips the target when $L = 0$. In other words, the stored gate performs

$$x_{\text{target}} \oplus = \text{comp} \oplus L.$$

The following gives one stored-line example for every gate form in the table above. We use wire 1 as the target throughout.

- **X:** 1 0 0 gives $x_1 \oplus = 1$. Here $k = 0$, and the empty conjunction is true.
- **CNOT:** 1 0 1 2 1 gives $x_1 \oplus = x_2$.
- **NCNOT:** 1 0 1 2 0 gives $x_1 \oplus = \neg x_2$.
- **AND2:** 1 0 2 2 1 3 0 gives $x_1 \oplus = x_2 \wedge \neg x_3$.
- **Conjunction of width 3:** 1 0 3 2 1 3 0 4 1 gives $x_1 \oplus = x_2 \wedge \neg x_3 \wedge x_4$.
- **r57:** 1 1 2 2 0 3 1 gives $x_1 \oplus = 1 \oplus (\neg x_2 \wedge x_3) = x_2 \vee \neg x_3$.

- **Complemented conjunction of width 3:** 1 1 3 2 1 3 0 4 1 gives $x_1 \oplus = 1 \oplus (x_2 \wedge \neg x_3 \wedge x_4)$.

Below is an example of a 3-gate circuit written in this format.

```
mpmct1 32 3
15 0 2 7 1 20 1
4 0 2 2 1 31 1
4 0 1 20 1
```

The header declares physical wires 0 through 31 and three gate instructions. The value 32 cannot be reduced without relabeling the gates, since the second gate refers to wire 31. Reading the instructions from top to bottom, they give

1. $x_{15} \oplus = x_7 \wedge x_{20}$,
2. $x_4 \oplus = x_2 \wedge x_{31}$,
3. $x_4 \oplus = x_{20}$.

We will use the simpler three-pin notation when a circuit only consists of r57 gates.

Sampling Circuit Replacements

The simplest way to sample a random circuit with a particular functionality, would be to search it up in a precomputed table of circuits. If we do this carelessly, then our table will not take into account circuits that are essentially the exact same circuit, but differ in trivial ways. For instance, the below circuits are all equivalent circuits 123;456;789 123;789;456 789;123;456

Recalling that our definition of collisions require two gates to share a wire between an active pin and any other pin, we notice that all of the 3 above circuits depict 3 gates that share no wires, which means there can exist no collision. This shows that we can actually order the gates in any way that we want. Many of our replacements in the kneading stage were therefore just trivial replacements as seen here. This gives us a new problem of defining circuit equivalences beyond trivial gate swaps.

Canonicalization

There are two types of trivialities we can identify. First, it is the one described in the last section where we can swap commuting gates. For the second, let us first consider the following two circuits. 123;345 and 145;523;.

Before we discuss the second type of canonicalization, we first establish the idea of circuit permutation. Besides the gates of a circuit, a circuit can be identified by its functionality. Using this form of identification can be useful for us as circuits can share the same functionality while sharing a different sequence of gates. For instance, 140;214;250;014;250;205;014;205;140; and

136;063;031;013;136;016;061; both compute identities. A circuit representation is thus a bijective mapping of all input states to all output states. For instance, a circuit on 3 wires can have the following input states: 0, 1, 2, 3, 4, 5, 6, 7. We call these states because each of these can be thought of as the states of each wire on the 3 wire circuit, with 0 = 000 and 7 = 111 when thought of as bit strings.

Going back to our two circuits, we can see that the gates now collide and so we can not just swap things around to make them equivalent. In fact, they aren't even equivalent at all. However, the structure that they have is identical. Namely, there is one collision between the second control pin of the first gate and the active pin of the second gate. We can see that mapping the wires

$$\begin{aligned} 1 &\rightarrow 1 \\ 2 &\rightarrow 4 \\ 3 &\rightarrow 5 \\ 4 &\rightarrow 2 \\ 5 &\rightarrow 3 \end{aligned}$$

allows us to map the first circuit to the second one. Let us see what is actually happening when we consider the states as bits.

$$\begin{aligned} 001 &\rightarrow 001 \\ 010 &\rightarrow 100 \\ 011 &\rightarrow 101 \\ 100 &\rightarrow 010 \\ 101 &\rightarrow 011 \end{aligned}$$

The two circuits are equal up to a shuffle of the right two bits. Thus, while the circuits are effectively different circuits, we can see that they are extremely similar. By considering only one of these circuits in our rainbow table of possible circuits, our random sampling is only amongst truly different and random looking circuits. So while the first type of circuit canonicalization is based on gate ordering and collisions, the second type of canonicalization is based on the bit shuffles of the circuit's permutation.

It is important for our canonicalization algorithms to be efficient as we will be constantly computing them. One extremely simple algorithm for circuit canonicalization is by just taking the lexicographic ordering, with respect to colliding gates. This is very efficient already as it is essentially just a sorting algorithm. On the other hand, computing permutations is much harder. A direct method is to try every possible bit shuffle in order to find the lowest lexicographical ordering. This becomes expensive very quickly, and motivates the polynomial method below. Once the bit shuffle that corresponds to the

smallest permutation lexicographically, we can just apply the bit shuffle to the circuit and of course undo the shuffle whenever we want so that we don't lose any circuits in this canonicalization, just like how we can merely shuffle gates around again to “undo” circuit canonicalization.

Canonicalizing via Polynomials

The two canonicalizations above got us a long way, but relying solely on permutations can get quite expensive if we wish to canonicalize circuits on 7+ wires. To identify a circuit's functionality by its permutation, we must save an entry for every possible input. It is rare for a random circuit to only span 7 wires, and when we make replacements we would like to be able to replace both deep (many gates) and wide (many wires) circuits.

The fix is to stop identifying a circuit by its permutation and start identifying it by its list of polynomials. Recall from the definition of `r57` that a gate $[a, b, c]$ gives $a' = a + \neg b \wedge c + 1$. If we seed each wire i with the degree-1 monomial x_i and apply that substitution gate by gate, then after the last gate every wire holds a polynomial in the inputs over $GF(2)$. A circuit on n wires is therefore a list of n polynomials, and two circuits are functionally equal exactly when their polynomial lists agree. This is often a far cheaper object than a permutation for the short windows we sample, and it is the one we now use. We use Boolean polynomials, so $x_i^2 = x_i$ and repeated monomials cancel over $GF(2)$. These reduced polynomials are the algebraic normal form of the function. They can still grow exponentially, which is why we bound the work of a local lookup below.

The remaining problem is the same one we had before: we want a canonical *wire labeling*, so that two circuits which differ only by a relabeling collapse onto one entry. Except now we can rank the wires by their polynomials rather than by brute forcing bit shuffles. The wire with the highest ranking gets mapped to wire 0, the next highest to wire 1, and so on. The algorithm is as follows.

Input: a list of n polynomials over $GF(2)$, one per wire (represented as sets of monomials, each monomial a bitmask of variables).

Output: the canonical reordering of those polynomials, plus the wire permutation that achieves it.

1. **Degree profile partitioning** — For each wire i , compute its degree profile: a vector where entry k is the number of monomials of degree k in P_i , sorted highest degree first. Sort wires by degree profile (descending) and group wires with identical profiles into equivalence classes $C_1, C_2, \dots$
2. **Build class polynomials** — For each class C_i , build $P_{\{C_i\}}$: the sum (with integer coefficients) of all polynomials of wires in that class, counting how many times each monomial appears across the class. This gives us a new polynomial with coefficients in $\mathbb{N}$.

Degree profile partitioning simply means we group all of the polynomials by the number of monomials with some highest degree. Any tied polynomials, we then look at the number of monomials with the next highest degree, etc. For instance, consider the polynomials:

$$P_1 = x_0 + x_1 + x_2 + x_0x_1 + x_2x_3 + x_0x_2x_3$$

$$P_2 = x_2 + x_3 + x_2x_3x_4x_5$$

$$P_3 = x_3 + x_5 + x_0x_1x_3x_5$$

Their degree profiles are:

Polynomial	Degree 4	Degree 3	Degree 2	Degree 1
P_1	0	1	2	3
P_2	1	0	0	2
P_3	1	0	0	2

Since P_2 and P_3 both have 1 monomial of degree 4 while P_1 has none, P_2 and P_3 are placed in a higher class than P_1 . Since P_2 and P_3 have identical degree profiles, they remain in the same equivalence class. Thus, we have C_1 , the first polynomial class, holding P_2 and P_3 , and C_2 is then just simply P_1 . To build the class polynomials, we now just sum the polynomials within each class together, noting that we have coefficients over $\mathbb{N}$, rather than our polynomial being in $GF(2)$. P_{C_1} is thus $P_2 + P_3 = x_2 + 2x_3 + x_5 + x_2x_3x_4x_5 + x_0x_1x_3x_5$ and $P_{C_2} = P_1 = x_0 + x_1 + x_2 + x_0x_1 + x_2x_3 + x_0x_2x_3$.

We note that the degree profile is used only to build these class polynomials. It does *not* seed the ranking of the wires. The refinement below starts with every wire tied and produces the entire ranking itself.

Before we go through how to use these P_{C_i} to rank our wires, we first must know how to rank two monomials.

Given M and M' and some partial ranking σ of the variables, we have $M > M'$

- **(a)** If the degree of M is greater than the degree of M'
- **(b)** If the degrees are equal, then if the highest ranked variable, based on σ , of M is ranked higher than the highest ranked variable of M' .
- **(c)** If the degrees are equal and the variables are equally ranked, then if the coefficient of M is greater than the coefficient of M' .

Let us take an example. Suppose we have $x_0x_1x_3$ vs x_2x_4 with current partial ranking $0, 1 < 4 < 3, 5, 6$. Then from rule (a), we have $x_0x_1x_3 > x_2x_4$ as the degree of $x_0x_1x_3$ is greater than that of x_2x_4 . Let us keep this partial ranking and move onto examples of when we would use (b) and (c). So let us consider the two monomials x_0x_4 and x_1x_5 . These monomials have equal degree, so rule (a) tells us these two monomials are tied. We now consider rule (b). The highest ranked variable in x_0x_4 is x_0 . The highest ranked variable in x_1x_5 is x_1 . As our partial ranking currently can not differentiate between 0 and 1, then we have that x_0 and x_1 are ranked equally. Thus, we move on to the next variables. The next highest ranked variable in x_0x_4 is x_4 and for x_1x_5 is x_5 . As we have $4 < 5$, then x_4 is the higher ranked variable. Thus, $x_0x_4 > x_1x_5$. Let us finally take an example where rule (c) would be called. Let us have the monomials $5x_0x_1$ vs x_0x_1 . As the degrees are equal, then rule (a) fails to differentiate them. As the variables are exactly the same, then rule (b) fails to differentiate them. We note that the variables do not have to be equal for this case to fail. For instance, x_0 vs x_1 would also fail since our partial ranking does not differentiate between 0 and 1. Thus, we use rule (c) and find that $5x_0x_1 > x_0x_1$ based on the coefficients.

Given this definition, we continue with our algorithm.

3. Start with an empty partial ordering (all wires tied). Repeat until no ties remain:

- **Phase 1** — For each $P_{\{C_i\}}$, starting from $P_{\{C_1\}}$, look at the highest ranked monomial. In the case that there are multiple, then we consider all of them. The frequency of variables that appear in the highest monomial define which wires should be ranked higher than other wires. The initial partial ordering is created looking at the highest degree monomials of $P_{\{C_1\}}$. Any ties, which are variables that we have not yet been able to determine a ranking for, are then attempted to be broken by looking at the subsequent monomials. If every monomial of $P_{\{C_1\}}$ has been observed and ties remain in σ , then continue to $P_{\{C_2\}}$, etc. Whenever a tie is broken, restart from $P_{\{C_1\}}$. We note that the highest monomial may change as our partial ordering becomes more complete. In other words, we have less ties between variables. If we exhaust all $P_{\{C_i\}}$ and ties remain, then continue to the next step.
- **Tiebreak 1 (per-polynomial key)** — For each tied group, consider a ranked version of each P_i (the original polynomials associated with the tied wires) within that group: replace each variable with its current rank, sort ranks within each monomial, sort monomials by degree, then lexicographically. The lexicographically smaller ranked polynomial is then deemed to have a higher rank. Whenever a tie is broken, restart from $P_{\{C_1\}}$ in phase 1.
- **Tiebreak 2 (dynamic class polys)** — Build dynamic class polynomials $P_{\{D_r\}}$ from the current tied groups as opposed to from degree profiling. In other words, for all polynomials P_j for $j \in J$

where J is the list of tied wires in a group, then construct P_{D_s} as the sum of all P_j as a polynomial with coefficients in $\mathbb{N}$. Apply the same Phase 1 frequency splitting. Whenever a tie is broken, restart from $P_{\{C_1\}}$ in phase 1.

- **Rule L (exhaustive)** — If ties still remain, find the highest-ranked tied group. For each candidate wire w , hypothetically denote w as higher than all others in the group, continue the loop, and compute the resulting canonical form. Repeat for all candidates. Pick the w that yields the lexicographically smallest result.
4. **Remap polynomials and monomials** — Using the final wire ordering, move the output polynomial of the wire at position p to position $\mathbf{p}$, rename that wire's input variable to variable $\mathbf{p}$, and apply this renaming to all monomials. The same ordering is used for the inputs and outputs.
 5. **Trim trailing identity wires** — Remove wires from the end where $P_i = \{x_i\}$ and x_i appears in no other polynomial. These are fully decoupled identity outputs.

Below are some examples for (3).

Phase 1 — Suppose wires $\{x_0, x_1, x_2\}$ all share the same degree profile and form class C_1 , with $P_{x_0} = P_{x_1} = x_0x_1x_2 + x_0x_1 + x_2$ and $P_{x_2} = x_0x_1x_2 + x_0x_2 + x_1$. Their class polynomial is $P_{C_1} = 3x_0x_1x_2 + 2x_0x_1 + x_0x_2 + x_1 + 2x_2$. We start with the empty partial ranking, which just means all wires are considered equal. The highest monomial $3x_0x_1x_2$ contains all three variables equally, so no tie is broken. Moving to the next highest ranked monomial, $2x_0x_1$, which tells us $0, 1 < 2$. Since we made progress, we restart from the highest ranked monomial, but learn nothing. We then go to the next highest monomial, etc. We note that after the initial partial ranking, we are now just trying to tie-break. In this case, we are trying to determine the true ranking of 0 and 1.

Tiebreak 1 — Suppose Phase 1 has established $x_2 < x_3$, but x_0 and x_1 remain tied, with $P_{x_0} = x_0x_2 + x_3$ and $P_{x_1} = x_1x_3 + x_2$. Replacing variables with their current ranks and sorting within each monomial gives $x_0x_2 + x_3$ for P_{x_0} and $x_0x_3 + x_2$ for P_{x_1} . Since we have that $x_2 > x_3$ from our partial ordering, we find that $P_{x_0} > P_{x_1}$ and so $0 < 1$. The intuition is that x_0 co-occurs with x_2 , while x_1 co-occurs with x_3 , and $x_2 < x_3$.

Tiebreak 2 — Suppose x_0 and x_1 remain tied after Phase 1, with $P_{x_0} = x_0x_1 + x_0x_2 + x_0x_3$ and $P_{x_1} = x_0x_1 + x_1x_2 + x_0x_3$. Tiebreak 1 fails here as they are equal when we set $x_0 = x_1$. The tied group we are considering is $0, 1$, and so we build the dynamic class polynomial with P_0 and P_1 . Building the dynamic class polynomial $P_D = P_{x_0} + P_{x_1} = 2x_0x_1 + x_0x_2 + x_1x_2 + 2x_0x_3$. We now do what we did in Phase 1. The highest ranked monomial, if we assume partial ranking $0, 1 < 2 < 3$ is $2x_0x_1$. This does not help us break $0, 1$. The next highest monomial is $2x_0x_3$. As there is x_0 but not x_1 , this tells us that $0 < 1$.

Rule L — Suppose x_0 and x_1 are still tied after all prior steps. We take hypotheses $x_0 < x_1$ vs $x_1 < x_0$. Whichever hypothesis produces the lexicographically smaller canonical form is committed to. Unlike the earlier tiebreaks, Rule L is global: it considers the full resulting representation rather than any local polynomial property. When two branches give the same form, their wire labelings reveal a symmetry. We remember this symmetry and skip later equivalent branches when the symmetry preserves the current tied groups. Thus, we need not visit every symmetric branch separately.

Highest monomials depending on σ — Suppose we have polynomial $P = x_0x_1 + x_1x_3$ with partial ranking $1 < 2 < 0, 3$. Here the monomials would be tied. However, if we break $0, 3$ and have $0 < 3$ from some rule above, then the highest monomial would be x_0x_1 . Thus, it is important to remember that the rules above are all relative to the current partial ordering of the variables.

We note that Rule L can be quite expensive if we are not careful. After all, it is the same method of wire relabeling that we had before. However, it will only run if all above methods have failed and ties remain between wires. This means we may only need to consider a couple branches, as opposed to a full $n!$ different wire relabelings. Thus, in practice, this is still efficient. There is an optional cap on the total number of Rule L candidates across the entire recursive call. For instance, a budget of 512 candidates limits how much work we spend on a difficult window. Exceeding the budget makes the canonicalization fail rather than return a partial answer. A failure skips the database lookup; it does not establish that the function has no friends.

Polynomial expansion needs a budget for the same reason. A monomial can be represented by a bit mask, with one bit for each variable it contains. A 64-bit representation therefore supports at most 64 distinct wires in the sampled window, even when the surrounding circuit is much wider. We can also bound the number of reduced monomials per wire and the work spent multiplying polynomials. For example, a limit of 200,000 reduced monomials per wire keeps one difficult r57 window from taking over the entire mixing process. If a window exceeds a limit, we leave it unchanged. These limits control the cost of searching for a replacement; they do not change what it means for two circuits to be equivalent.

Step 5 is what buys us the most. It allows us to relate the functionalities of circuits on a different number of wires together, which is something that we were not able to easily do before. Thus, our tables no longer need to be separated by wires. Instead, we can have a table for 1 gate circuits, for 2 gate circuits, etc. In fact, we can then combine all of these together into one large database, which then allows us to relate circuits of both differing gates and differing wires together.

Besides only including canonical circuits within our tables, we have two more restrictions. Firstly, we do not include any circuits that have two adjacent and equivalent gates. Suppose we were generating our 5 gate circuits and one of

them had 2 identical and adjacent gates. By the definition of our gate, this means those two gates are just an identity and so this circuit is equivalent to a 3 gate circuit. As we would already have generated the 3 gate circuits, then this is a redundant circuit. In the regular table, the second restriction is to not store a circuit and its reversal separately. Any canonical circuit can be mapped to its inverse, which makes storing both redundant. Rather than probing for the reversal, we get this for free from the key itself: we canonicalize the circuit in both directions and key it by whichever of the two comes out smaller. A circuit and its reversal therefore collapse onto a single entry automatically. We look up the smaller canonical form. If it came from the reverse direction, we reverse the replacement again before inserting it. The curated lookup, described below, uses its forward form.

The trimming at the end means that circuits that are equivalent in functionality, but differ in the total number of wires, will still be met with the same wire canonicalizations. This means that we can actually store our rainbow solely by their functionality and generate them based on the number of gates we have already generated. We no longer need to separate our tables by wires.

Rainbow Tables

With our new notions of canonicalizations (we will be using both gate-level and polynomial-level canonicalizations), we can greatly decrease the number of possibilities stored in our rainbow tables, as we saw that many circuits are already equivalent. It turns out that this means that many permutations only have a single circuit that corresponds to it. It is actually hard to find many circuits that have many “friends”, where two circuits are friends if they differ in both canonicalizations above and still have the same permutation (which corresponds to the same functionality).

There are two main things to now consider: storage and speed. Storage is greatly helped by canonicalizations. Let n be the number of wires and m be the number of gates. Let us consider the rainbow table for $n = 5$ and $m = 5$. This has $(n * (n - 1) * (n - 2))^m = 777,600,000$ total circuits before canonicalizations. After permutation canonicalization, we can bring this number down to about 2.5 million circuits.

A general-purpose database can store these circuits by their functionality and let us retrieve or randomly select equivalent circuits. However, the number of circuits grows exponentially in both n and m , and the tables can grow past a terabyte. Polynomial canonicalization removes the need to separate them by wire count. We also have a second simplification: the tables are constructed ahead of time, so mixing only needs to read them. This lets us use the simpler storage described below.

The Frozen Table

We first note what the table is meant to store. We take the canonical polynomial representation of a circuit and hash it to obtain a 128-bit key, a compact identifier used to find the entry. The value associated with that key is the list of canonical circuits we have found with the same polynomial representation. Thus, a lookup begins with the functionality of a sampled subcircuit and returns other circuits that may be used in its place.

The table is constructed ahead of time and is only read during mixing. We never need to insert, update, or delete an entry at mixing time. A general-purpose database maintains machinery for changing entries while other operations are in progress. Since mixing does not need that machinery, we can store the same mapping in a simpler form.

The distinction in names is important here. The *rainbow table* is the logical mapping from a canonical functionality to the circuits that compute it. The *frozen table* is the physical file format we use to store that mapping. It is a static, read-only, compressed point-lookup store, meaning that it is designed to answer a request for one exact key at a time. The regular and curated tables described below contain different circuits, but both use this same frozen file format.

We do not retain all 128 bits of each hash. The hash is first serialized as 16 little-endian bytes, and we divide the bits of that byte sequence as follows. A shard is one of several files which divide up the table; a bucket is a smaller group of entries within a shard.

Part of the hash	Number of bits	Purpose
Shard	8	Selects one of 256 shard files
Bucket	20	Selects one of 2^{20} buckets within that shard
Tail	48	Identifies the entry within the bucket
Omitted	52	Not stored

In other words, the first 28 bits tell us where to look, and the next 48 bits tell us which entry we want once we get there. The remaining 52 bits are discarded. Here, “first” refers to the serialized bytes, not the most significant bits of the numeric 128-bit hash. Within each bucket, the 48-bit tails are sorted and compressed using Elias–Fano coding, which compresses a sorted list of integers while still allowing us to search it. Each tail then has one value: a compressed list of circuits, with each circuit preceded by its length so that we can locate it.

A lookup then proceeds as follows:

1. Canonicalize the polynomial representation of the sampled circuit and hash it.
2. Use the first 8 bits of the serialized key to choose a shard.

3. Use the next 20 bits to choose a bucket within that shard.
4. Read that bucket’s two offsets from an in-memory index and load only the corresponding byte range.
5. Search for the 48-bit tail and, if it is present, decode the associated list of circuits.

Thus, we do not need to descend through a tree or search unrelated entries. We read only the bucket selected by the key, and an empty bucket requires no disk read.

Keeping only 76 bits means that two different 128-bit hashes can produce the same stored key. For the table of approximately 22.6 billion entries described below, in a 2^{76} -element key space, the probability that a random absent key agrees with a stored key on all 76 retained bits is approximately 3×10^{-13} . A campaign making approximately 10^{10} absent lookups would therefore expect only a few thousandths of a false hit. A collision may return a circuit from the wrong entry or hide the entry we wanted to find. We therefore verify every proposed replacement before inserting it. We compare every input when the combined set of wires has at most 24 wires, and compare the exact output polynomials above that width. If we cannot complete the exact check within its budget, we decline the replacement. Thus, the lookup can suggest a wrong candidate, but that candidate cannot change the circuit’s functionality because it must pass the independent equivalence check first.

The circuit lists themselves are compressed using Huffman coding, which gives shorter bit strings to frequently occurring symbols. We treat one complete three-pin gate as a symbol because recurring gate patterns are more useful for compression than their wire indices considered separately. The symbol frequencies depend on the circuit width and the gate’s position in the circuit. On the table measured here, the encoding used approximately 9 bits per gate.

The resulting frozen table occupied 320.3 GB, or approximately 14.15 bytes per entry. The same content occupied 1.1 TB in a general-purpose B-tree database and approximately 700 GB in a compressed general-purpose database. This reduction comes from three changes: removing the unused write machinery, compressing the circuit values, and truncating the stored hashes. We have not measured the contribution of each change independently.

Finally, we note that the two bucket offsets immediately tell us when a bucket is empty, so that case requires no disk read. If the selected bucket is not empty, an absent key ordinarily requires reading the bucket before we can determine that its tail is not there. We can avoid most of these reads by keeping an optional approximate membership filter over the retained 76-bit keys in memory. If the filter says that a key is absent, we skip the disk lookup. If it reports that the key may be present, we still perform the exact bucket lookup. For a filter built from the same store, a false positive only causes an unnecessary exact lookup and cannot select an incorrect value. We build the filter from the frozen table and check that it recognizes every retained key. In our measurement,

the filter reduced the average miss time from approximately 120 microseconds to approximately half a microsecond. An exact hit in the frozen table took approximately 33 microseconds. These timings depend on the store, hardware, and cache state.

What the Database Covers

We generate the source tables by gate count rather than by wire count, and then merge them into the frozen store. A circuit with m gates can use at most $3m$ distinct wires because each r57 gate touches three distinct wires. Polynomial canonicalization relabels only the wires that the circuit actually uses, so additional unused wires in the surrounding circuit do not affect the key. Thus, a source table for m gates can include circuits taken from any total circuit width.

The following coverage figures describe a table built from circuits of 1 through 11 gates. Its generation bounds matter: a table containing some 11 gate circuits need not contain every shorter circuit.

For gate counts 1 through 6 in this construction, generation was run without a minimum used-wire cutoff. These tables were intended to cover every canonical functionality reached by an r57 circuit with at most 6 gates, regardless of how many distinct wires those gates use. This is already a large improvement over our old limit of 7 wires and 4 gates.

Starting at 7 gates, the tables become too large to generate exhaustively. However, this does not give us a simple cutoff where we have every circuit up to one gate length and no circuits after it. We generate a longer table by taking shorter circuits and adding another gate. If the new gate uses several wires that have not appeared before, then many choices become equivalent after canonicalization. Permuting those new wire labels produces the same canonical circuit, so we can skip those repeated choices. We refer to this as *fresh-wire symmetry*.

If the new gate instead reuses wires that already appear in the circuit, then the different choices can produce different polynomials and do not collapse in the same way. Thus, after 6 gates, our tables mainly cover circuits that use many distinct wires. Circuits with the same number of gates on fewer distinct wires are much harder to generate. Here, *used wires* refers to the number of distinct wires touched by the gates, not the total number of wires in the surrounding circuit.

The following table summarizes what was generated at each gate count in that construction. The first four rows are small enough to state exactly. We round the larger rows because their exact-looking build totals are not useful here, and we only rely on their approximate scale.

Gate count	Stored circuits	Unique polynomials*
1	1	1
2	23	23

Gate count	Stored circuits	Unique polynomials*
3	895	894
4	87,002	85,667
5	$\approx 13.2\text{M}$	$\approx 13.2\text{M}$
6	$\approx 2.88\text{B}$	$\approx 2.87\text{B}$
7	$\approx 9.82\text{B}$	$\approx 9.82\text{B}$
8	$\approx 3.60\text{B}$	$\approx 3.60\text{B}$
9	$\approx 5.76\text{B}$	$\approx 5.76\text{B}$
10	$\approx 287\text{M}$	$\approx 287\text{M}$
11	$\approx 290\text{M}$	$\approx 290\text{M}$

Unique polynomials counts the distinct canonical polynomial keys among the stored circuits. Two stored circuits can compute the same polynomial, in which case they give us another replacement candidate under the same key rather than another key. Here, M means million and B means billion. After merging and deduplicating all eleven source tables, the frozen store had approximately 22.6 billion unique keys.

The used-wire coverage in that build differed after 6 gates. The 7 through 11 gate tables began at 12, 16, 19, 23, and 26 used wires, respectively. Thus, the larger totals do not mean that these tables contain every circuit of those gate counts.

The 10 and 11 gate tables can still provide replacements, but their usefulness is limited for the narrow windows we most often sample. A 10 gate circuit can use at most 30 wires, while that 10 gate table only contained circuits using at least 23 wires. Similarly, an 11 gate circuit can use at most 33 wires, while that 11 gate table began at 26 used wires. These circuits therefore reuse very few wires between gates.

This missing region directly affects the lookup rate for longer, narrow windows. A window outside the generated used-wire range can only match the regular table if its functionality also has a circuit in one of the ranges that we did generate, including a shorter circuit. The polynomial-work and used-wire limits described above can also cause a lookup to be skipped. When sampling pure-r57 circuits, windows matched 100% of the time through 5 gates and 94% of the time at 6 gates. The measured rates for 7 through 12 gates were then 56%, 31%, 20%, 8%, 3%, and 0%. These rates are specific to that sampling and lookup configuration. Thus, we normally use short windows when we need a consistently high lookup rate.

This does not mean that every longer window will miss. The lookup key represents the functionality of the window rather than its number of gates. For example, a 20 gate window can still match if the same functionality has an equivalent circuit of 11 gates or fewer in the table. In an experiment making replacements in convex windows with the curated-first lookup described below,

the match rate decreased by roughly a factor of 0.75 per additional gate after length 7, but remained at 1.72% at length 20. This later result uses a different sampling and lookup configuration from the pure-r57 measurements above. Thus, longer windows still have a use, even though their matches are much less common. Finally, these longer gates can be useful in generating *minimal* identities and supply the *curated* table.

The Curated Table

Alongside the general table we keep a second, much smaller one, which we call the *curated* table. The motivation is that the general table answers “what else computes this function”, and the overwhelming majority of equivalent circuits aren’t too random, even after canonicalization. What we actually want, when we are trying to inflate a circuit rather than compress it, is a replacement that is structurally *unlike* what it replaces.

An identity I is *minimal* if every subcircuit, other than the entire circuit, is incompressible. For instance, consider an identity $[[0,1,2], [0,1,2], [0,1,3], [0,1,3]]$. This is not minimal because the subcircuit $[[0,1,2], [0,1,2]]$ is compressible to 0 gates. We can thus use the general table to create a large list of identities, and then filter out the non-minimal ones. Given the minimal identities, we can then extract every single circuit in order to create a list of *curated* circuits. Consider the following identity $[[2,1,0], [1,5,2], [3,0,4], [5,3,2], [2,0,3], [2,4,3], [5,3,2], [2,4,3], [2,0,3], [3,0,4], [1,5,2], [2,1,0]]$. We can extract a prefix $[[2,1,0], [1,5,2]]$ which will have equivalent circuit $[[2,1,0], [1,5,2], [3,0,4], [2,0,3], [2,4,3], [5,3,2], [2,4,3], [2,0,3], [5,3,2], [3,0,4]]$. Extracting prefixes and suffixes in this manner, for all prefixes and for all cyclic rotations and both directions of the identity, is what fills the curated table.

The half-window minimality test used for this identity collection is narrower than the definition above. Rather than checking every proper subcircuit, it checks every contiguous window of exactly half the identity’s length, and “incompressible” operationally means that the window’s canonical key is *absent* from the general table, i.e. that we know of no alternative spelling for it. This is a table-relative test of selected windows rather than the full definition, and identities that pass it can still contain a compressible subcircuit of some other length.

One curated table held approximately 11.9 million keys in about 1.72 GB, which was roughly one key per 1,900 general keys and about half a percent of the general table’s bytes. Each key retained at most 20 candidates and at most 512 bytes of circuit representations. The bound limits how much time and memory one lookup can take, since a single functionality can have an enormous number of friends. In these measurements, curated lookups were about an order of magnitude cheaper than general ones, around 3.8 microseconds against 33.

We can instead retain every distinct function/circuit pair that we generate. This

gives us more choice, but we must avoid constructing every friend before choosing one. We first scan the stored circuit lengths, select a friend, and only then construct that circuit and map its wires. A very large list can still cost substantial time and memory. Limiting the list changes the available diversity and mixing choices, even though every retained candidate remains correct.

Curated lookup uses the window’s forward canonical form. When no curated candidates are found, the regular fallback follows the minimum-direction rule described above; some mixing policies try several curated window sizes before falling back.

Using the Rainbow Tables

The main use we have for our tables is to make replacements, and in particular to shorten or lengthen subcircuits.

This allows us to both compress and expand circuits and gives us additional randomness. This randomness comes from the fact that a single circuit functionality can be shared amongst x different circuits up to canonicalization, and an obfuscator can easily select any one of these to make a replacement. For instance, a circuit with 100 gates and 64 wires could have a subcircuit with 10 gates and only 6 wires. We could then compress this down to 5 gates, to which the original 10 gates have all been changed. This allows us to fundamentally change a circuit, as the states between the 10 original gates has been lost and instead put into only 5 gates. Of course, the state before and after our new 5 gates will remain equal to the state before and after the old 10 gates that we compressed. The same can be said about expansion.

We now state some things about compressibility. Stupid identities such as 214; 251; 123; 123; 251; 214, created by just reversing an identity and then concatenating them, is extremely easy to compress. In addition, it is easy to compress identities created by combining friends from the rainbow table. However, it is extremely hard to compress completely random circuits. These facts are important to consider when looking at a new strategy for local mixing of reversible circuits.

In these earlier mixing experiments, accepting both compressions and expansions, the best gate-length to start with appeared to be 9 gates. Making cascading replacements, i.e. making replacements that include gates of the previous replacement, rather than random replacements, led to better mixing, but were slower. Contiguous circuits with few collisions very rarely had shorter replacements, which motivated our use of convex subcircuits. The generation-mixing section describes how we combine this overlap with a schedule for circuit size.

How We Measure Success

As we discuss our strategies, it will of course be important for us to measure how well we are doing. At first, we mainly used heatmaps and incompress-

ibility. However, these only test two particular kinds of structure, and later attacks showed that passing them is not enough. We now use several different measurements, each of which asks a different question about the circuit.

Before applying any attack, we first check that the transformed circuit still computes the required function. Some constructions use a *fixed slice*: we fix specified auxiliary inputs and require the chosen output wires to compute the original function on that restricted set of inputs. We describe this explicitly when we reach slicing. Full-state and endpoint comparisons give sampled evidence at large sizes, while exhaustive comparison is only practical at small sizes. Correctness is separate from hiding: an incorrect circuit can look excellent under every attack below.

The main measurements introduced here are:

1. state heatmaps, which ask whether the states reached by the original and transformed circuits remain correlated;
2. compression attacks, which ask whether an attacker-computable compressor can undo our expansion and recover a much smaller equivalent circuit;
3. algebraic-degree and differential tests, which ask whether the intermediate states remain low-degree functions of the original input;
4. affine and bounded-degree reconstruction heatmaps, which ask whether an original intermediate state can be recovered as a low-degree function of the transformed circuit’s current wires;
5. whole-circuit linear correlators, which combine observations from different times to recover or predict source values. We collect these and the local tests in the gadget gauntlet; and
6. Boolean satisfiability (SAT) tests, which turn the published circuit into a Boolean formula and ask for an input which produces a chosen output. We use this later to search for preimages in the trapdoor permutation challenge.

We note that none of these measurements replaces the others. A circuit can look random in an ordinary heatmap while still having low algebraic degree. Its physical wires can have high degree in the input while its logical values remain affine functions of those wires. A circuit can also look well mixed under these measurements while its SAT formula remains easy to solve. Thus, whenever we give a result below, it is important to state which of these attacks the result actually tests.

State heatmaps. Let C_1 and C_2 be two circuits on n wires. For each pair of prefixes i, j , we run both circuits on the same set S of sampled inputs and compare the two intermediate states. The displayed heatmap value is

$$D(i, j) = \frac{1}{n|S|} \sum_{x \in S} \text{HD}(C_{1,i}(x), C_{2,j}(x)),$$

where HD is Hamming distance. In other words, each cell averages the number of

differing state bits across the shared inputs, and we normalize by n for display. This ordinary heatmap assumes that the physical wire positions of the two circuits are comparable. A wire relabeling can therefore change the map even when it has only moved the bits around.

We use *green* to denote “random” and *red* to denote “equal”, although some of our older heatmaps use *purple* and *green* instead. Two identical circuits should have a red diagonal along $y = x$. Two completely random circuits should give a green heatmap. If two circuits look random internally but compute the same function, then their input and output corners should agree while the interior is green. Figure 1 below is a sketch of this.



Figure 2: Sketch of an ideal heatmap

Figure 1: Sketch of an ideal heatmap.

We can calibrate this measure by comparing actual random circuits. This allows us to see how many gates are needed before random circuits look random in the heatmap. Below, we vary both the number of gates and the number of wires.

Figure 2: Heatmap two completely random 100 gate circuits on 32 wires.

Figure 3: Heatmap two completely random 500 gate circuits on 32 wires.

Figure 4: Heatmap two completely random 100 gate circuits on 64 wires.

Figure 5: Heatmap two completely random 500 gate circuits on 64 wires.

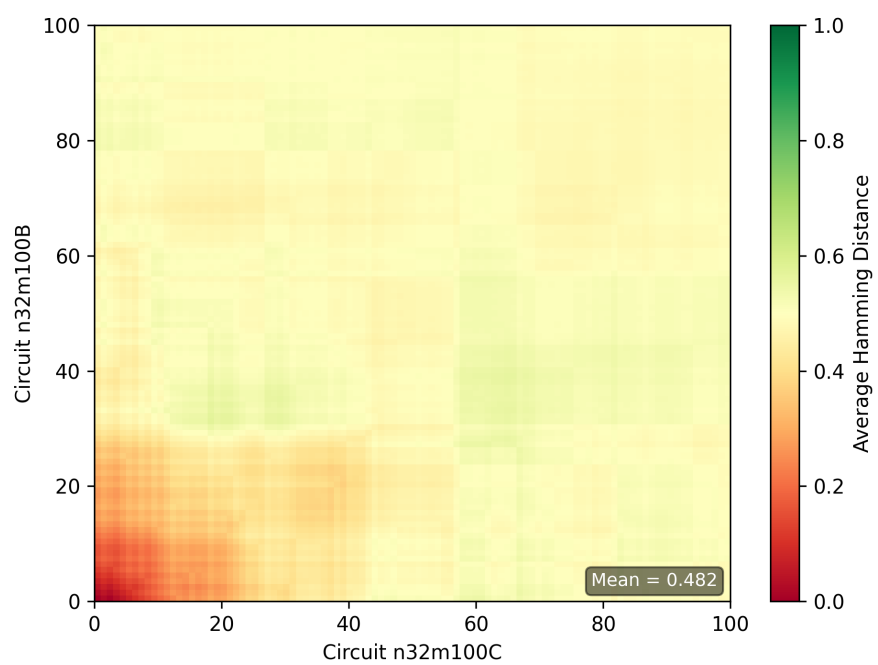


Figure 3: Heatmap two completely random 100 gate circuits on 32 wires

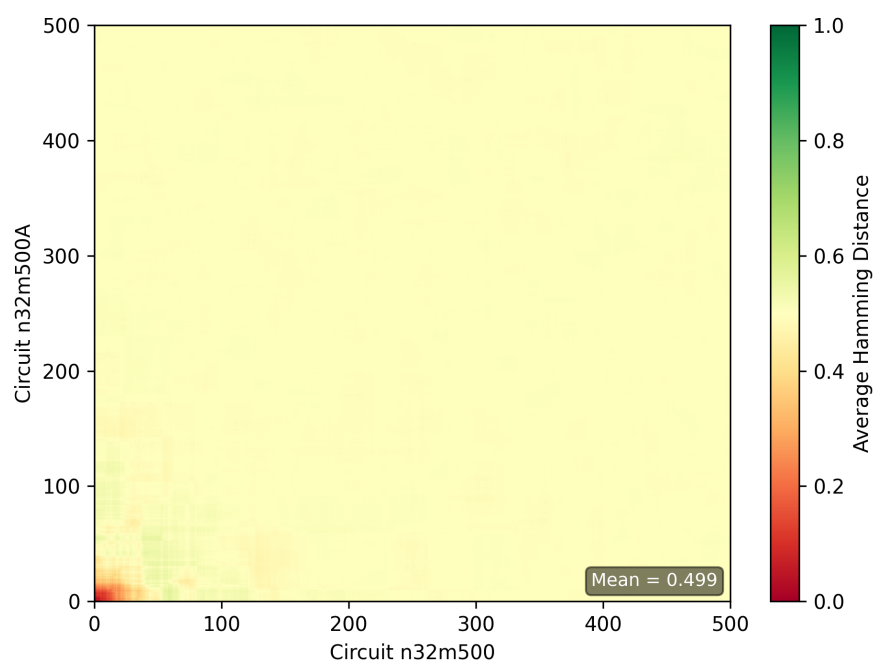


Figure 4: Heatmap two completely random 500 gate circuits on 32 wires

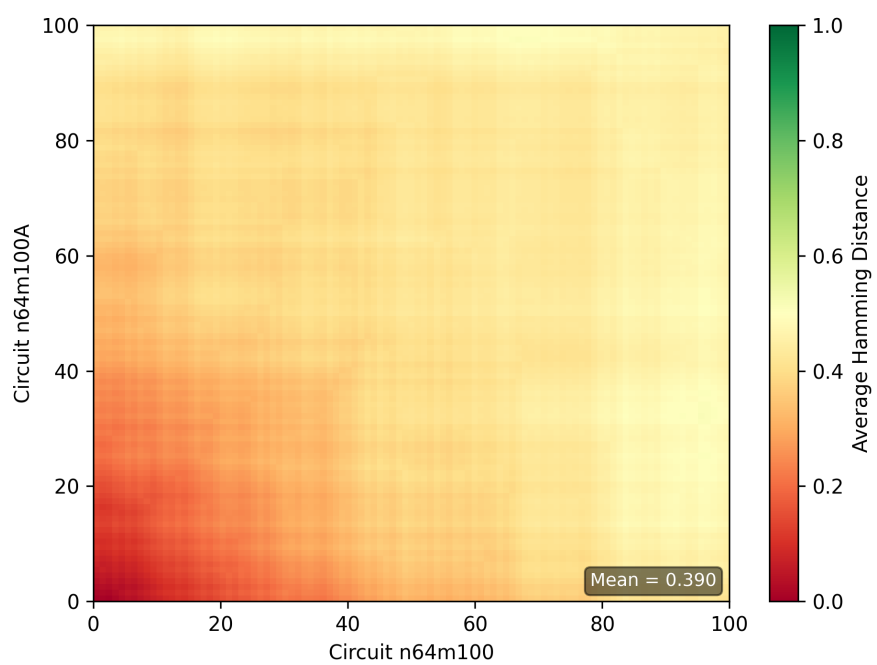


Figure 5: Heatmap two completely random 100 gate circuits on 64 wires

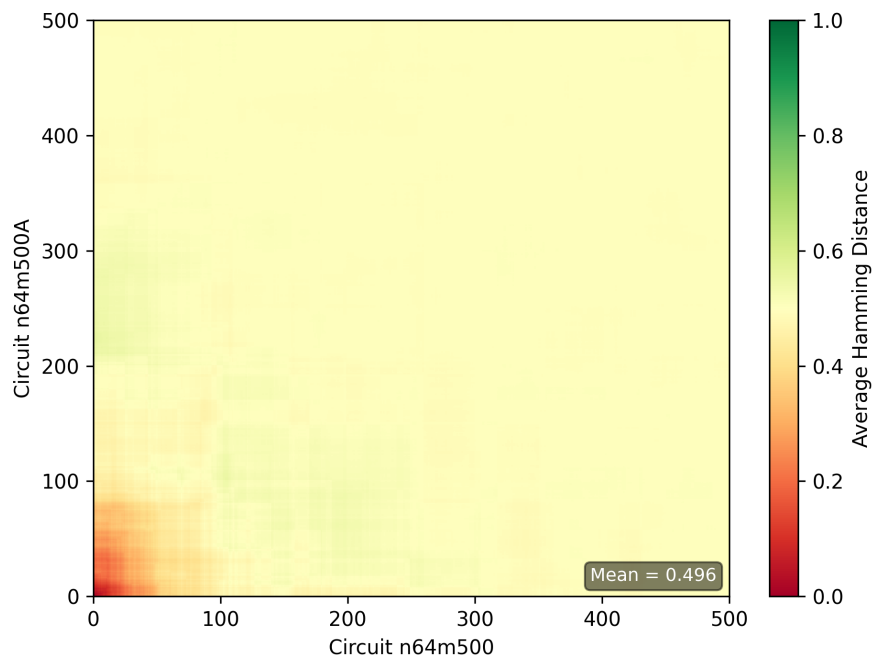


Figure 6: Heatmap two completely random 500 gate circuits on 64 wires

From Figures 2 and 3, we can see that on 32 wires, two completely random circuits already look uncorrelated at 100 gates. On the other hand, Figures 4 and 5 show a good bit of correlation on 64 wires at 100 gates. This correlation is gone when we use 500 gates. Thus, a random-circuit comparison needs enough gates before it becomes a meaningful baseline. Without restricting the circuit to $r57$ gates, this number is $O(n \log n)$.

Figure 6 shows a circuit compared with itself. This is the diagonal that we are attempting to escape from.

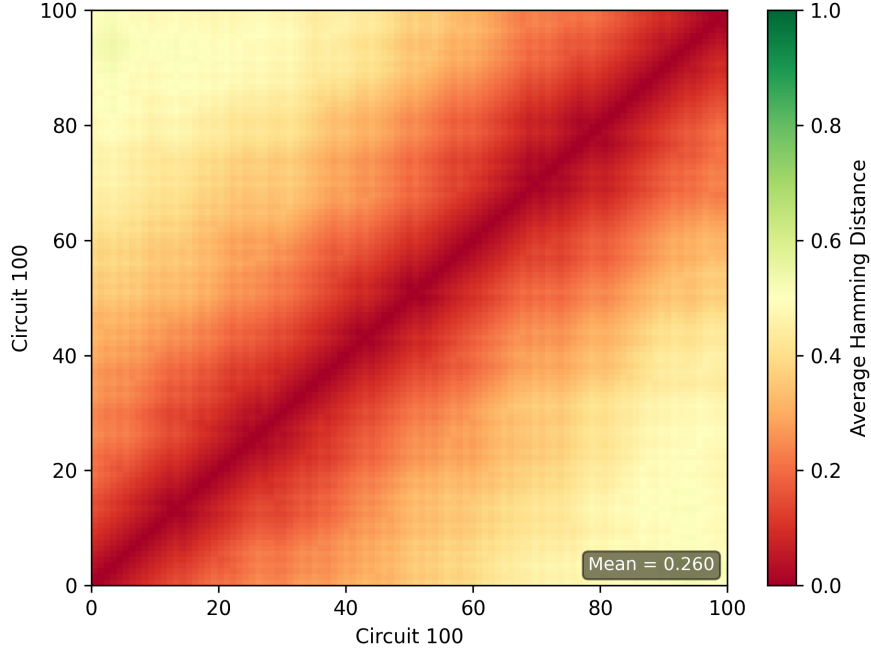


Figure 7: Heatmap showing a 100 gate circuit with itself

Figure 6: Heatmap showing a 100 gate circuit with itself.

The ordinary heatmap is not the only state heatmap that we use. A pure wire permutation can make the ordinary map look random without actually hiding the computation. Since a permutation preserves Hamming weight, we also compare

$$|\text{wt}(C_{1,i}(x)) - \text{wt}(C_{2,j}(x))|.$$

This Hamming-weight heatmap catches the simple wire-shuffle case, although bit flips and more general affine encodings can defeat it. We can also make either heatmap on isolated pieces of a circuit, build up longer prefixes one piece at a

time, or choose inputs which are more likely to expose structure. Our chosen inputs include low-Hamming-weight inputs, inputs close to one fixed random input, and pairs which differ in exactly one bit. A flat global map on random inputs only rules out the statistic and input distribution that we tested.

Compression attacks. Incompressibility asks how well an attacker can sample subcircuits and replace them with fewer gates. If we expand a 100 gate circuit on 64 wires to 1000 gates, then a compressor which brings it back to 100 gates or fewer has undone the expansion for this purpose. This is an empirical attack, not a proof of incompressibility: a stronger compressor may find replacements that ours misses. Our deterministic compressor tests whether the additional structure survives repeated equivalent simplifications. Showing that no practical compressor can undo the construction remains an open problem.

Differential attacks and algebraic degree. Heatmaps compare sampled states, but a circuit can pass them while every intermediate wire is still a simple function of the original input. To test this, we begin with input variables $x_0, \dots, x_{n-1}$ and apply each gate as a substitution over $GF(2)$. Every wire at every prefix is then an algebraic-normal-form polynomial in the original inputs. Its degree is the largest number of distinct input variables in any of its monomials.

If a coordinate has degree at most d , then its $(d + 1)$ st derivative vanishes. Equivalently, XORing its value over the 2^{d+1} points of a suitable affine cube gives zero. This gives a differential distinguisher when d is small. We therefore inspect the degree of every wire throughout the circuit, rather than only the maximum: one remaining low-degree coordinate can leak even if another wire has high degree. The end-to-end degree is fixed by the function that we must preserve. Only the intermediate degree profile is ours to change.

Affine reconstruction heatmaps. The differential test above asks how complicated each transformed wire is as a function of the original input. The affine reconstruction heatmap asks a different question: given all of the wires in the transformed circuit at one moment, can an attacker combine them to recover the original circuit's state at some moment?

For instance, two carrier wires c_0 and c_1 may each be high-degree functions of the original input while still encoding a logical value as

$$v = c_0 \oplus c_1.$$

The differential test sees two complicated physical wires. The affine reconstruction test sees that the logical value is still recovered by one XOR.

Let C be the original circuit on n wires and let G be the wider transformed circuit on N wires. Row i of the heatmap represents the state of C after i gates, while column j represents the state of G after j gates. To fill the cell (i, j) , we run both prefixes on the same collection of logical inputs. We also follow the auxiliary-input policy of the construction: the public slice is fixed, while

the remaining helper wires are fixed or sampled as required by that experiment. This policy is part of the result. A flat map under one choice of helper inputs does not tell us what happens under every other choice.

We then take one target wire t from the state of C_i and ask whether there are fixed coefficients $a_0, \dots, a_N$ such that

$$C_i(x)_t = a_0 \oplus a_1 G_j(x)_1 \oplus \dots \oplus a_N G_j(x)_N$$

for every sampled input x . Here all the features come from the single state G_j . The heatmap compares many states, but each cell fits its own equation. The whole-circuit attack later removes this restriction and combines observations from different times.

The coefficients do not change with the input. Each coefficient is either 0 or 1: choosing 1 includes that wire in the XOR, while choosing 0 leaves it out. For example, if

$$C_i(x)_t = 1 \oplus G_j(x)_3 \oplus G_j(x)_{11},$$

then the attacker has recovered that logical bit, even if neither physical wire 3 nor physical wire 11 contains it by itself. We repeat this for every one of the n target wires of C_i . In other words, the test does not ask whether the two physical states look alike. It asks whether the original state can be decoded by XORing wires of the transformed state, with an optional constant.

We find these coefficients with Gaussian elimination over $GF(2)$, which is ordinary linear algebra with XOR used as addition. We use one set of inputs to find a relation and a separate holdout set to test it. If a target wire has no affine relation on the training set, it receives error 0.5, which is the error of a random guess. This 0.5 is assigned by convention; it does not mean that we measured the best affine predictor and found that it guesses randomly. If we find a relation, it receives its actual error on the holdout set. The heatmap value $H(i, j)$ is the average of these errors over the n target wires. Thus, $H(i, j) = 0$ means that the complete state of C_i was recovered exactly on the holdout inputs. A value near 0.5 means that this test did not find exact affine reconstructions for most of the target wires. It does not mean that no biased or more complicated predictor exists. An intermediate value usually means that the test recovered only part of the original state.

A low- H cell therefore says that the state of C at row i can be read from the state of G at column j . If the location of these low- H cells moves forward as i moves forward, they form a ridge across the map. This ridge tells the attacker where the successive stages of the original computation occur inside the transformed circuit. We call it a ridge because we can plot recoverability as $0.5 - H$. In the raw error map, the same feature is a low- H valley.

We do not read this map using its overall mean. Most cells compare unrelated stages and therefore sit near 0.5. Also, when G gets longer, the leaking region can occupy a smaller fraction of the map even when the leak itself has not changed. We first remove the forced input and output ports and then measure the low- H cells in the interior. *Coverage* tells us how many original-circuit rows contain a meaningful dip. *Depth* tells us how far that dip falls below the background of its own row. Once enough rows have meaningful dips, ρ , the Spearman correlation between the row and the location of its dip, tells us whether those locations move forward with the original circuit. We can also compare ρ with shuffled ridge locations to determine whether the observed ordering is stronger than chance.

Bounded-degree reconstruction heatmaps. The affine test only allows a constant and XORs of individual wires. We can give the attacker more power by also allowing products of the current wires. For example, the degree-2 test can use terms such as

$$G_j(x)_3 G_j(x)_{11}.$$

Here, multiplication is an AND of the two wire values.

It then asks whether each target wire of C_i can be written as an XOR of a constant, individual wires of G_j , and products of two wires of G_j . The degree- d test does the same thing with products of up to d distinct wires. Thus, the degree here describes the power of the reconstruction attacker, not the degree of G 's wires as functions of the original input.

The difficulty is that the number of terms grows very quickly. On N wires, a complete degree- d test has

$$R(N, d) = 1 + \sum_{k=1}^d \binom{N}{k}$$

possible terms. For reference, the counts are:

wires N	$d = 1$	$d = 2$	$d = 3$
48	49	1,177	$\approx 18,000$
128	129	8,257	$\approx 350,000$
512	513	131,329	$\approx 22,000,000$

The number of training samples must exceed the number of possible terms. After that, the Gaussian elimination itself also becomes expensive. For this reason, a practical degree-2 or degree-3 test may use only some of the possible products. A flat map from one of these restricted tests only means that this particular

set of terms did not reconstruct the original state. It does not rule out every attacker of that degree. We should also run a control with a degree- d relation that the restricted test is supposed to find. If it cannot recover that control, then its flat result tells us nothing useful.

SAT tests. A reversible circuit is already a straight-line Boolean program, so we can translate it directly into a SAT formula. We begin with one Boolean variable for each input wire and keep a table recording the variable which currently represents each physical wire. For every gate, we allocate a fresh variable for the gate’s updated target wire and add CNF clauses which force that variable to equal the result of applying the gate to the current target and control variables. A conjunctive normal form (CNF) formula is an AND of clauses, where each clause is an OR of literals. For a gate, we choose clauses which rule out exactly the assignments where the fresh variable disagrees with the gate’s truth table. We then update the target’s entry in the table. The other wire entries do not change. Repeating this for the complete gate list produces a CNF formula whose satisfying assignments are exactly valid executions of the circuit under the chosen input restrictions.

We can then add unit clauses which fix the public auxiliary inputs and the chosen output bits. A SAT solver returns values for every remaining variable, including the free input wires. If the constrained output is a target image, the returned input is a preimage of that image. We verify the answer by running the circuit on the returned input. A successful model is therefore an unambiguous break of that instance, while a timeout is only evidence about the particular solver, encoding, and resource limit that we used. We use this test for the trapdoor permutation challenge, which we discuss later.

The Two Phase Strategy

The two phase strategy was the initial method that we used in order to try and achieve local mixing of reversible circuits. We first inflate the circuit by replacing short subcircuits with longer equivalent ones, then knead it with equal-length replacements. The hope is that kneading spreads the added randomness until the original replacement boundaries disappear. This is the method proposed in the paper by Canetti, Chamon, Mucciolo, and Ruckenstein; here, we will provide some results.

While we were experimenting with this method, we did not use compression nor heatmaps in order to determine how well we were doing. However, the loose proof given in that paper was enough motivation to believe that this method could work well. We tested numerous things here and got many different results. We remind that the heatmaps here use a different coloring scheme, with *purple* meaning the two circuits appear random to each other and *green* to mean the two circuits are equal.

One essential thing for us to figure out, as long as we assume that the two phase method works, is the correct parameters to choose. We need to answer how

much we need to inflate, as well as how many rounds of kneading is necessary. The principal idea behind the inflationary phase is to first add padding that allows room for our circuit to be randomized. Afterwards, it is the kneading that stops attackers from undoing our replacements.

When it comes to making our replacements in each phase, the easiest method would be to simply precompute a table of all possible circuits of a given n wires and m gates. For now, this is what we will be moving forward with.

The \$10,000 Bounty

Trusting that the inflationary and kneading phases would be enough to stop any attacks on our circuit, we started up a bounty. We started with a completely random circuit with 64 wires and 1014 gates. Afterwards, we repeated the inflationary phase until we reached 12,111 gates and then did hundreds of thousands of rounds of the kneading phase. If the kneading phase was done correctly, then all replacements in the replacement phase should be irreversible. The goal of the bounty is to find an equivalent circuit on 1014 gates or less, essentially finding the original circuit and showing that the obfuscation was ineffective. The bounty can be found [here](#).

The bounty was quickly broken by Killari. There were a number of techniques that were used. One of them, was to use a precomputed rainbow table, much like we did to make our own replacements. However, Killari would try to make sampled subcircuits smaller. In other words, by sampling random subcircuits, Killari was able to compress the circuit back down. In fact, the circuit was compressed to less than 1000 gates. This is when it became obvious that we needed to strongly consider compression attacks. For more details, see Killari's blog post on breaking the bounty. A detailed discussion can be found [here](#).

After the Bounty

Now that we knew that our two phase method was incomplete, we moved on to looking at heatmaps. We wanted to better understand the circuits that we could “obfuscate” with this method. As a result, we generated numerous heatmaps to understand what was happening with different changes in parameters. Namely, how the circuits look after more rounds of kneading.

Keeping to 64 wires, it was clear that a couple thousand number of rounds for the kneading stage was insufficient. The heatmap in Figure 7 below shows that there is still an extremely clear correlation between the original circuit and the “obfuscated” circuit as there is a clear green line across $y = x$.

Figure 7: Heatmap showing the two phase method with 1,000 rounds of kneading.

The two heatmaps in Figures 8 and 9 below show what the same circuit heatmaps look like with 100,000 rounds of kneading and 1,000,000 rounds of kneading.

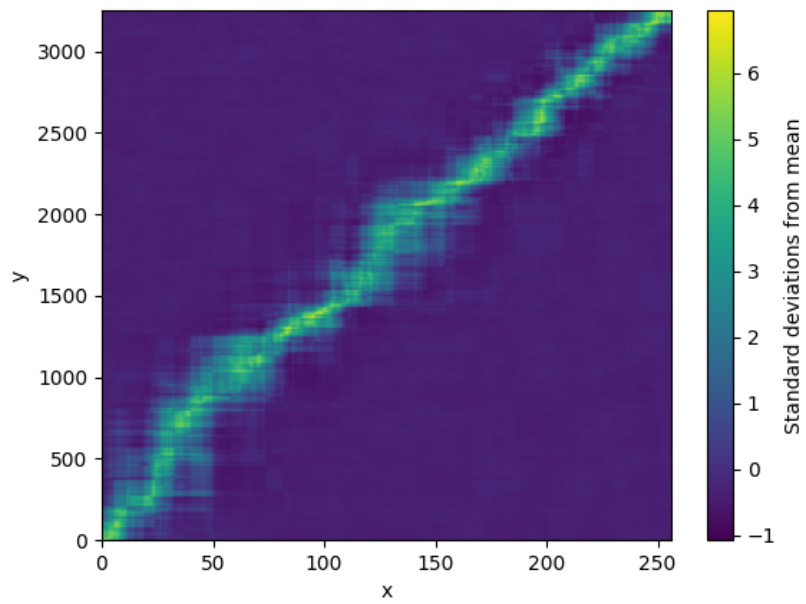


Figure 8: Heatmap showing the two phase method with 1,000 rounds of kneading

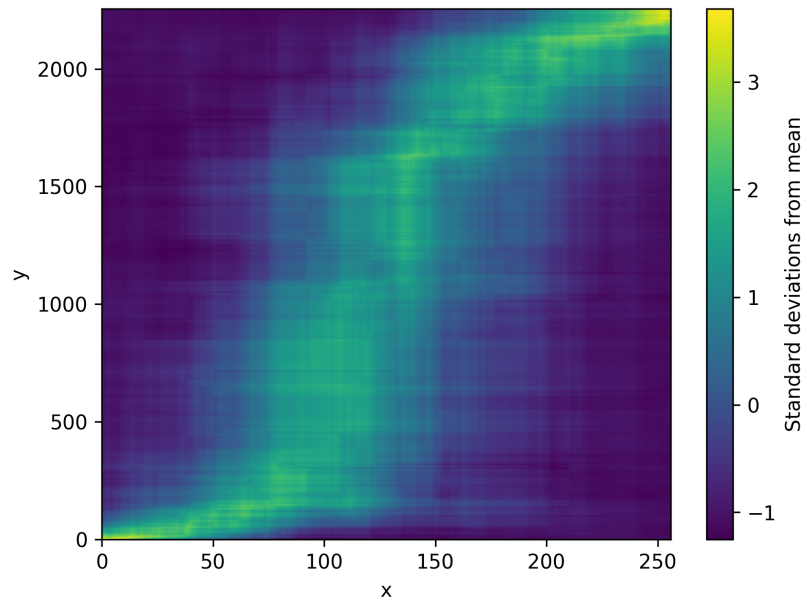


Figure 9: Heatmap showing the two phase method with 100,000 rounds of kneading

Figure 8: Heatmap showing the two phase method with 100,000 rounds of kneading.

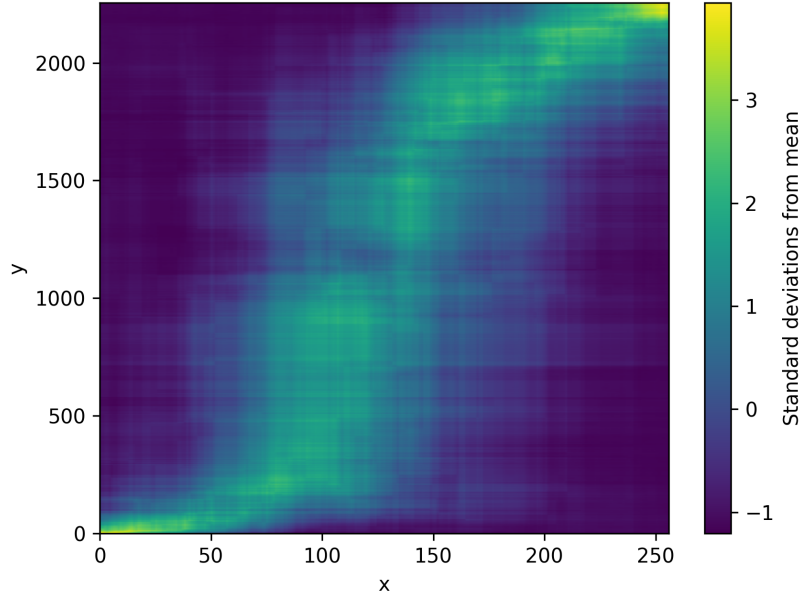


Figure 10: Heatmap showing the two phase method with 1,000,000 rounds of kneading

Figure 9: Heatmap showing the two phase method with 1,000,000 rounds of kneading.

While it is clear that our heatmap is indeed “spreading out”, which means that we are indeed seeing more randomness, we also see that it would require tens or even hundreds of millions of kneading rounds. Thus, it was obvious that something needed to be changed.

The Butterfly Methods

The main motivation for these methods, is to get incompressibility. There are two versions of the butterfly method that we will talk about. Namely, they are the asymmetric butterfly method and the symmetric butterfly method. The asymmetric butterfly method is just the butterfly method but with each R_i being completely random. If we look at the blocks in asymmetric butterfly, they are of the form $R_i^* g R_{i+1}$. As the left and right circuits are random to each other, then this block is essentially completely random. As mentioned before, this makes the blocks extremely incompressible. On the other hand, the symmetric

butterfly method is with each R_i being equal. While the symmetric butterfly method gives us much less randomness, the additional structure of each block R^*gR makes them more compressible.

We note that $5OA$ will represent 5 rounds of symmetric butterfly on circuit A , and $5QA$ would be the same but with asymmetric butterfly.

We first test our experiments on a small number of wires, as we know that we can simply take contiguous subcircuits without sampling subcircuits with more wires than we are able to compress with our rainbow tables. Thus, we test on 5 to 7 wire circuits. We did many of our tests on the two identities, circuit A , a 55 gate identity on 5 wires, and circuit B , a 22 gate identity. We are interested in numerous heatmaps here, such as A vs OA , B vs OB , A vs B , and OA vs OB . The first two gives us an idea of how well we obfuscate in relation to the original circuit, while the last two give us an idea of how well we obfuscate equivalent, yet different circuits.

There was a problem we quickly ran into though. The butterfly methods were ineffective on a small number of wires. Even with many rounds of the butterfly methods, we would always be able to compress back down to the original circuit. If we started with an identity, we would always be able to compress back down to zero gates. This means that we are largely unable to compress blocks and the gates g in R^*gR are not effectively being hidden. This means that we may need more “room” for the gates, which means we may need to use more wires. We of course can still start with the 5 wire circuits, but the R that we sample should be over 64 wires. This would allow us to sample circuits from 3 wires to 7 wires, giving us much more freedom.

Moving on to more wires

One change becomes immediate as we move onto more wires. We can no longer sample contiguous circuits of size greater than 2 gates. This is because if two gates share no wires, then 2 gates will already span 6 wires and 3 gates will span 9 wires, exceeding the limitations of our rainbow table.

Before showing off some heatmaps, it is important to note that these results use a compressor restricted to replacements on at most 7 wires. They show what the butterfly method achieves against that compressor. We then strengthen the compressor and see what survives.

Below gives our first heatmaps of A vs B , A vs OA , A vs $2OA$. The results for B are very much the same and so will not be shown.

Figure 10: Heatmap showing the circuit A vs circuit B .

Figure 11: Heatmap showing the circuit A vs circuit OA .

Figure 12: Heatmap showing the circuit A vs circuit $2OA$.

From the above heatmaps in Figures 10, 11, and 12, two things are clear. Firstly,

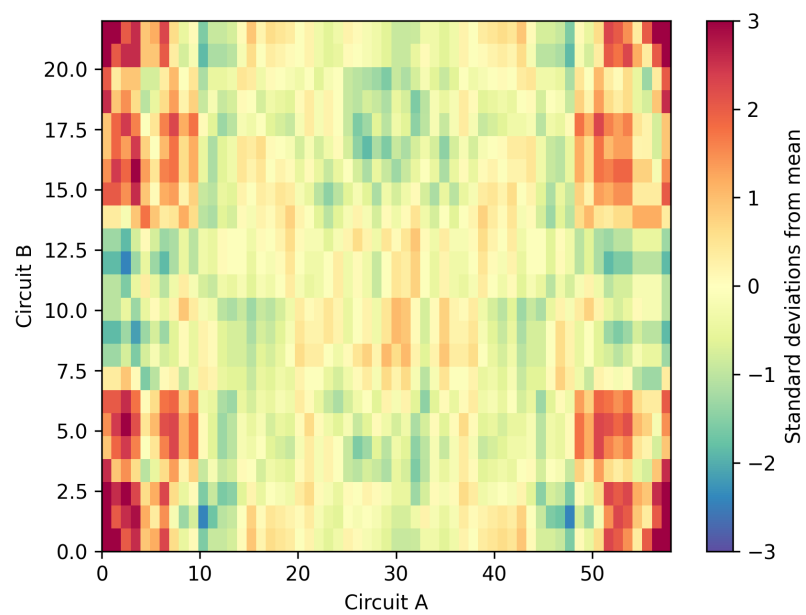


Figure 11: Heatmap showing the circuit A vs circuit B

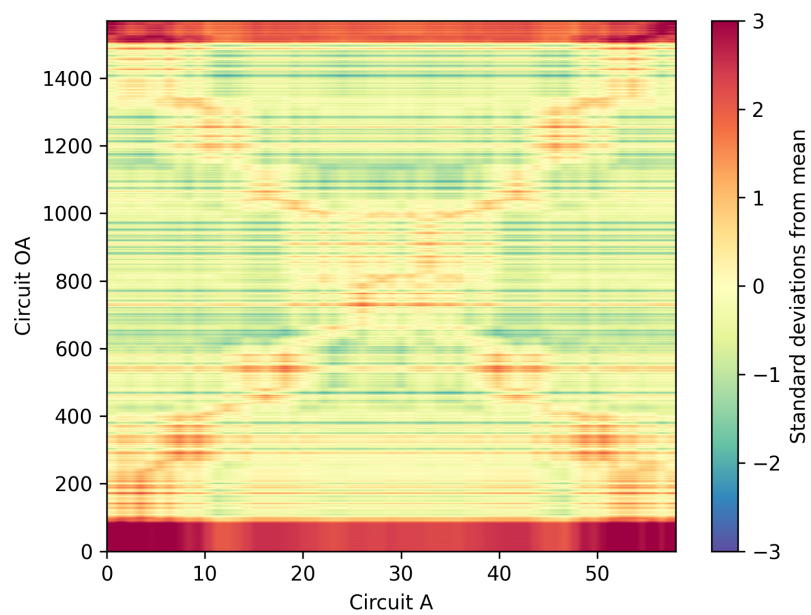


Figure 12: Heatmap showing the circuit A vs circuit OA

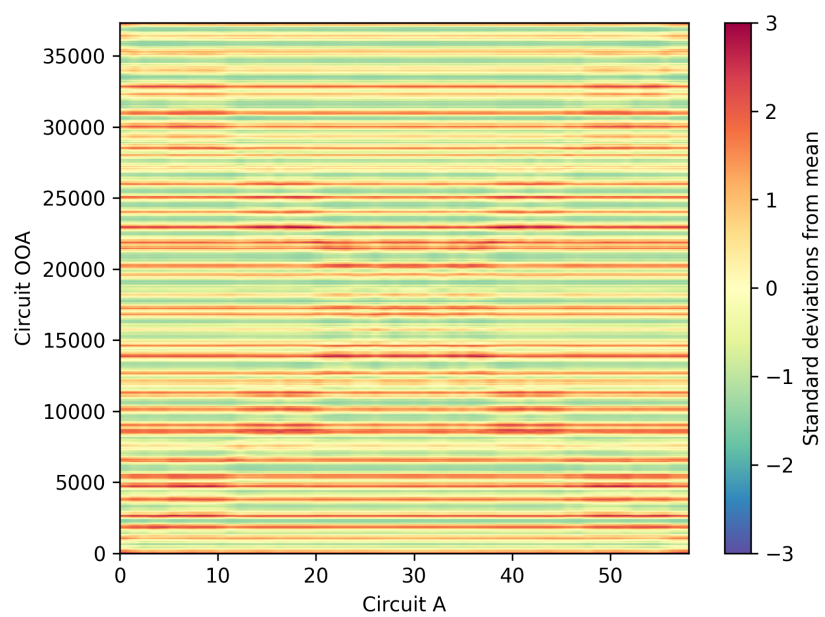


Figure 13: Heatmap showing the circuit A vs circuit $2OA$

there remains an obvious diagonal. Secondly, the number of gates has blown up greatly. The number of gates in each R is randomized between 6-25 gates, and with the weak compressor, it struggled to compress down at all, resulting in a large blow up in gates. However, there is still an obvious diagonal. In addition, due to runtime constraints, we would be unable to run another round as the number of gates would blow up even more. The final nail in the coffin, is if we were to look at heatmaps on raw hamming distance data, rather than standard deviations. Standard deviations are useful to look at as they enlarge small differences for us to see, but when we look at the raw data, it tells a much more somber story.

Figures 13, 14, 15, and 16 are heatmaps where R , which we will now call *wings* is small (5-15 gates), medium (30-70 gates), large (100-150 gates) and large x2 (200-300 gates). We see that there is a gradual decrease in redness, which shows that indeed, longer wings will create more randomized obfuscated circuits. This is obvious as the gates we are trying to hide in each block are more “buried” within the arbitrary gates in the wings.

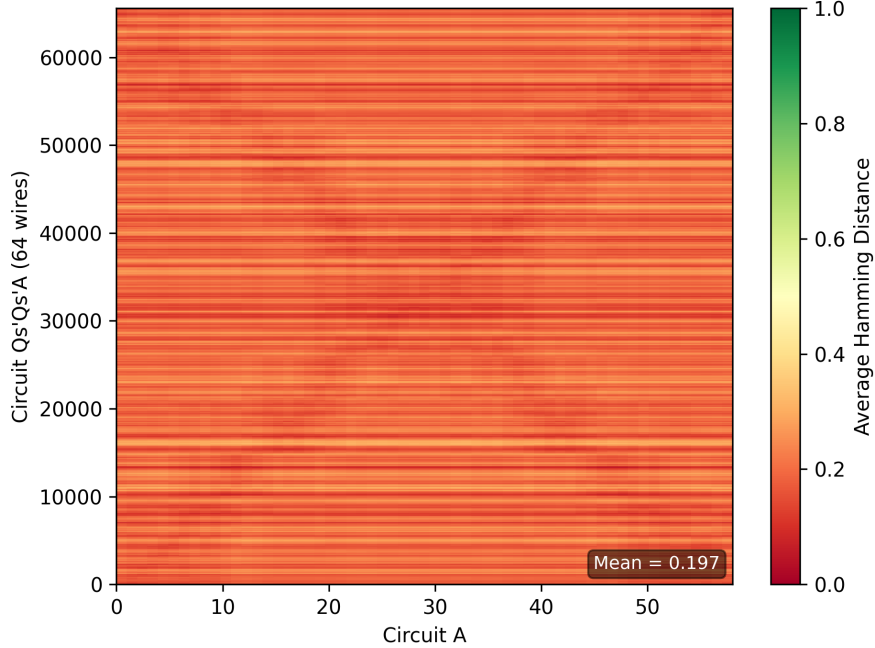


Figure 14: Heatmap showing the circuit A obfuscated with small wings

Figure 13: Heatmap showing the circuit A obfuscated with small wings.

Figure 14: Heatmap showing the circuit A obfuscated with medium wings.

Figure 15: Heatmap showing the circuit A obfuscated with large wings.

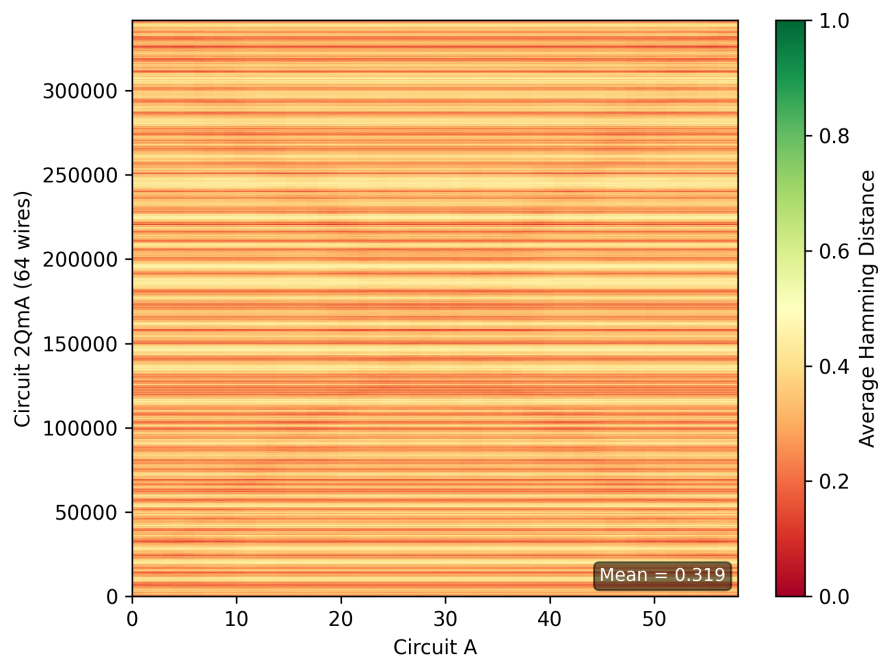


Figure 15: Heatmap showing the circuit A obfuscated with medium wings

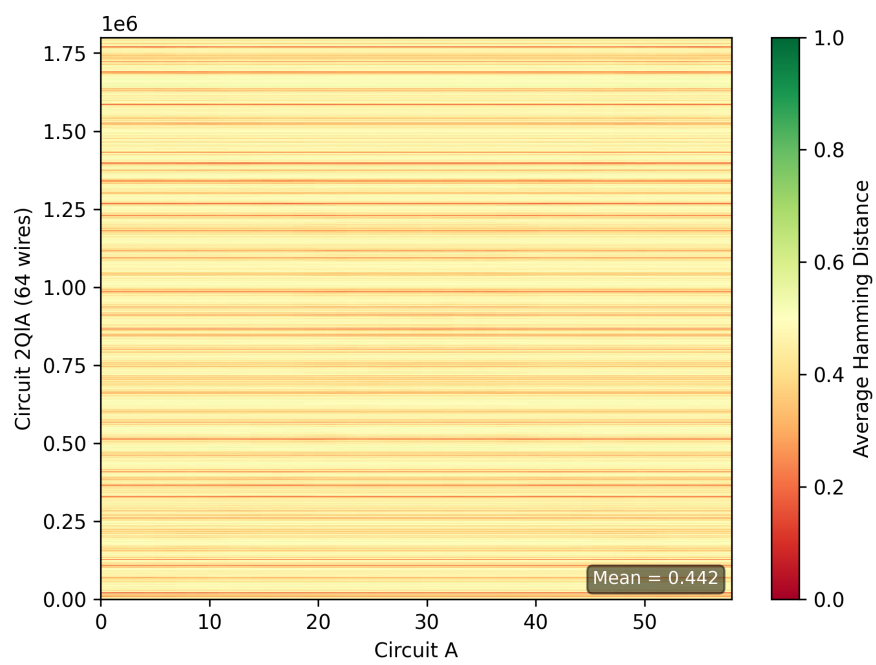


Figure 16: Heatmap showing the circuit A obfuscated with large wings

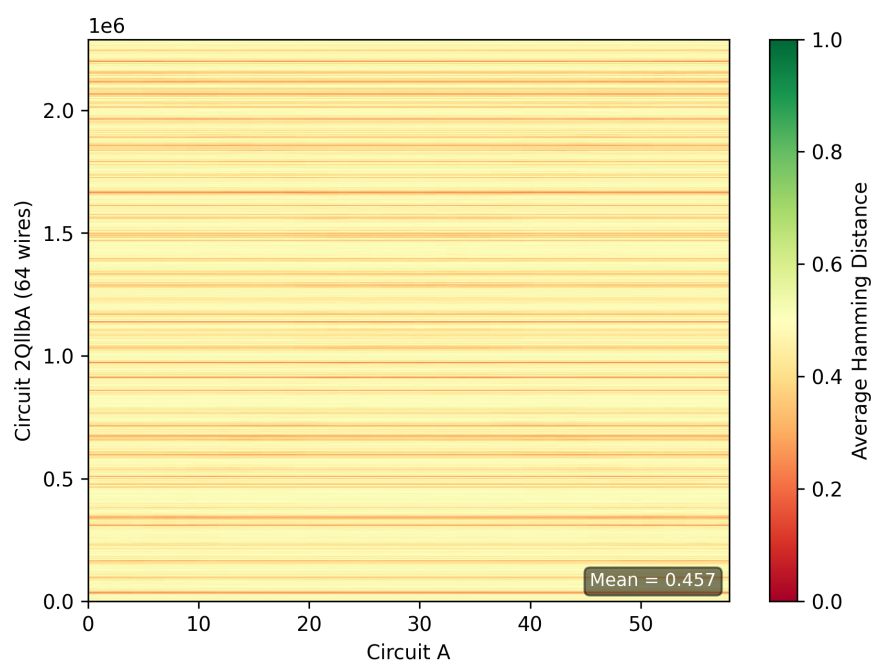


Figure 17: Heatmap showing the circuit A obfuscated with very large wings

Figure 16: Heatmap showing the circuit A obfuscated with very large wings.

As there seemed to be little to gain from the heatmaps in Figure 15 and Figure 16, we kept with 100-200 gate wings and continued testing. We then made compression faster and allowed it to use ancilla wires during a replacement, giving it access to equivalent circuits on more wires than the sampled window. For example, when searching for a 5 wire circuit, we would find the corresponding circuit permutation on 6 or 7 wires and then find a replacement in our 6 or 7 wire rainbow tables. With this stronger compressor, the expansion in those experiments could be undone. This made the butterfly method, with those parameters, ineffective. Suppose we have A and attempted to generate OA . Well this would compress completely and we would just have $A = OA$. If we then tried to get $2OA$, then this would be equivalent to as if we were trying to generate OA A . In other words, $A = OA = 2OA = \dots$. One thing that would be useful, is understanding at what point during compression our circuits stop looking random. The heatmaps below in Figures 17, 18, 19, and 20 answer this question, to an extent. We start with a completely random circuit on 64 wires and 500 gates, and then take heatmaps as it is being compressed.

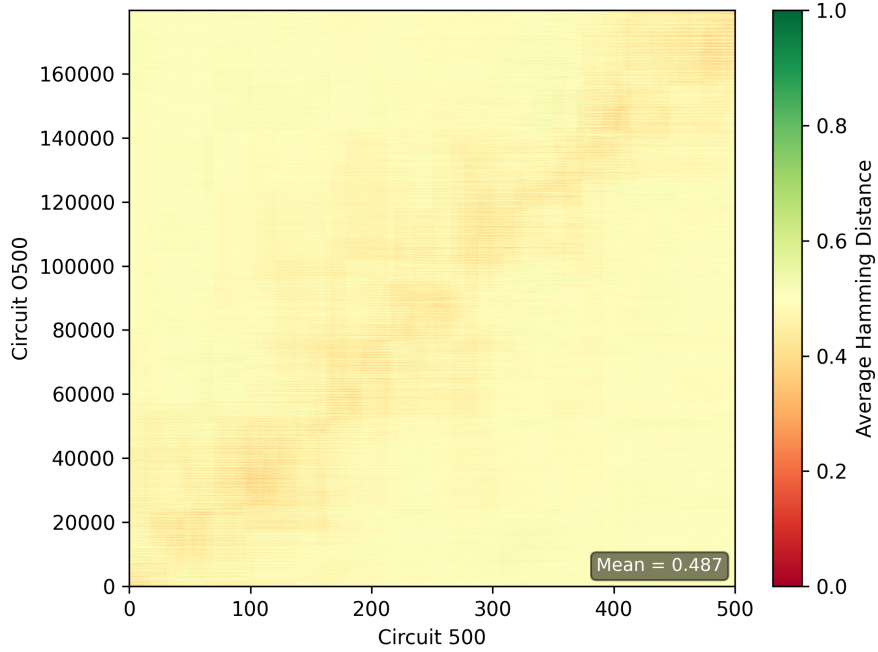


Figure 18: Heatmap showing circuit 500 before compression: 170,000 gates

Figure 17: Heatmap showing circuit 500 before compression: 170,000 gates.

Figure 18: Heatmap showing circuit 500 during compression: 50,000 gates.

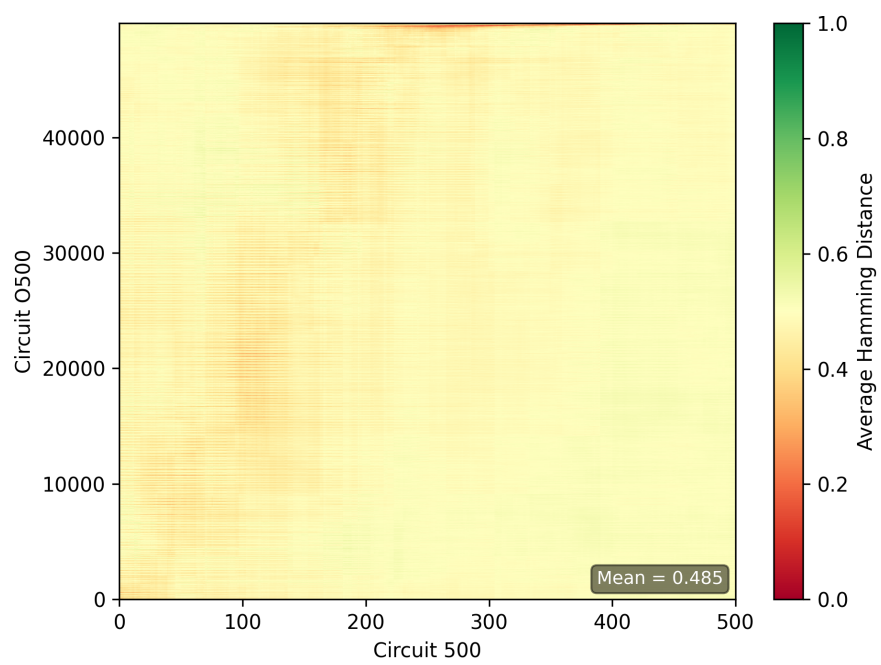


Figure 19: Heatmap showing circuit 500 during compression: 50,000 gates

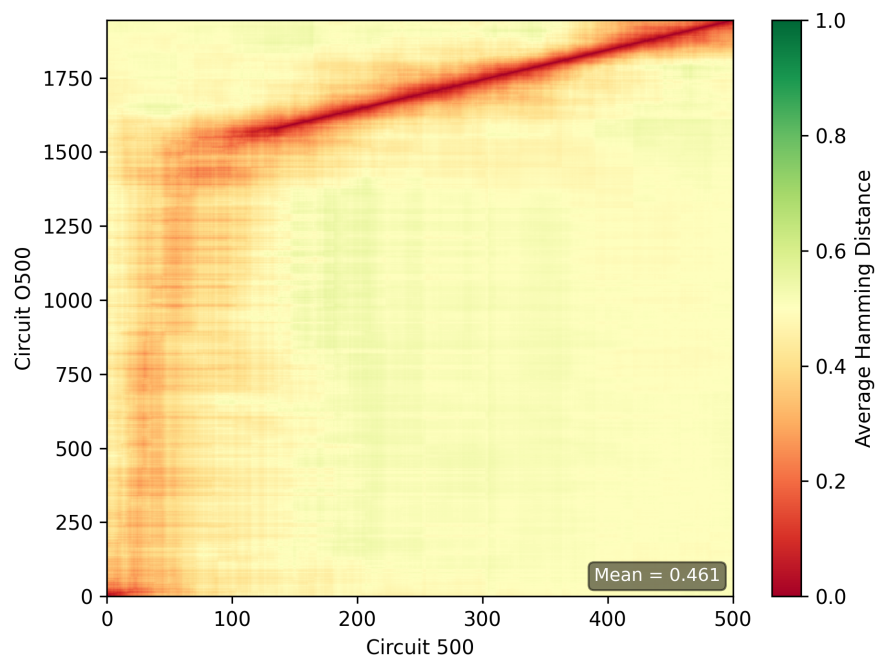


Figure 20: Heatmap showing circuit 500 during compression: 1900 gates

Figure 19: Heatmap showing circuit 500 during compression: 1900 gates.

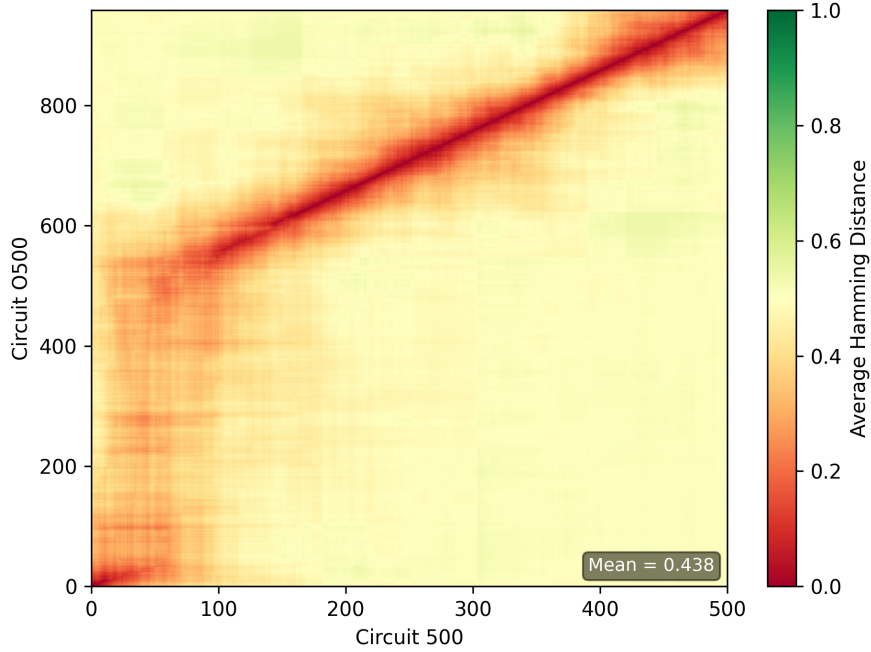


Figure 21: Heatmap showing circuit 500 after compression: 900 gates

Figure 20: Heatmap showing circuit 500 after compression: 900 gates.

These results led us to instead stopping compression early, at around 1,000 gates. After each round, we would increase this by another thousand. So for the 10th round of compression, we would stop compression at 10,000 gates. It was only in the final round would we attempt to completely compress a circuit. This finally gave us incompressibility, without the extreme blowup in the number of gates. However, these heatmaps did not look the most random. The bigger problem was that if we changed our compressor to allow ancilla wires, which means when we sample a 5 wire subcircuit, we allow the compressor to look into the 6 or 7 wire rainbow table, we could actually compress these down completely. Thus, we needed to include ancilla wires to our randomization in some way.

Ancilla Wire Replacements

When we are making replacements, we would add randomization in two ways. The first way is by randomly sampling a subcircuit. The second way is by randomly choosing how many wires our subcircuit could be on. For example, we may try to only sample a 5 wire subcircuit as opposed to the much easier 7 wire one. One change that we could make to this, is to sample a 5 wire

subcircuit, but then find how the circuit's permutation on 7 wires, and then use the 7 wire rainbow tables to make the replacement. In other words, we are adding ancilla wires to our replacements. To illustrate how we do this, consider a circuit on 3 wires. To compute its permutation, we would consider bits such as 000, 001, 010, However, nothing stops us from treating a 4th wire as a “do nothing” wire, which then allows us to consider input bits 0000, 0001, 0010, ... in order to compute a permutation on 4 wires as opposed to the original 3. We then need to search for this permutation in the 4 wire rainbow table, which allows us to find a functionally equivalent replacement of a 3 wire circuit with a 4 wire circuit.

Ancilla wire replacements alone gave us incompressibility. One useful thing about this process, is that it gave us even smaller incompressible circuits. The heatmap in Figure 21 below shows that we still have the red beam in the top right corner, which tells us that we still need more randomization.

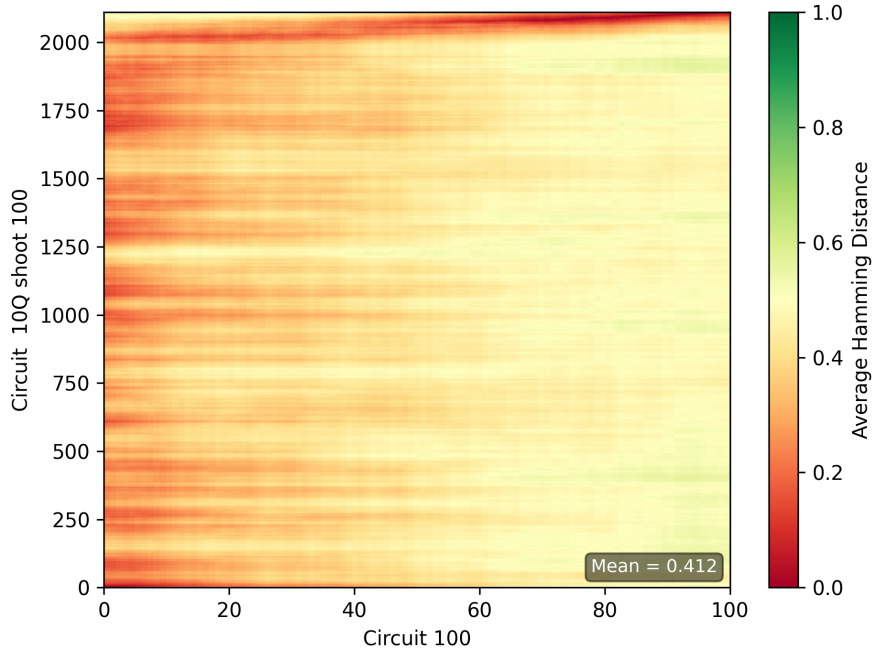


Figure 22: Heatmap showing a 100 gate circuit on 64 wires after the butterfly method with ancilla replacements

Figure 21: Heatmap showing a 100 gate circuit on 64 wires after the butterfly method with ancilla replacements.

Blurring

With our new goal of randomization, we will consider various methods. As we are trying to make the heatmaps look more random, this means that we want to make our heatmap look more “blurry”. Hence, we call this section “blurring”. One easy way to blur is to randomize the gate ordering of non-colliding gates and then “locking them in” via the incompressibility of the butterfly method. Another way to do so is to replace gates entirely with rewired identities. For instance, if we have g and an identity I , and then shuffle the bits of I such that the first gate of I matches with g , then we have $I = gB$, for some B . This means that $B = g^{-1} = g$, since every gate is its own inverse (and in fact every Toffoli gate) is its own inverse, and so we can replace g with B . These two ideas will be the pillars of our discussions moving forward.

Random Shooting vs Random Walking

This is our first, and more basic, method to blurring. The processes are described below.

Random Shooting:

1. Select a random gate g .
2. Select a random direction d , left or right.
3. Send g all the way d until it meets a gate that it collides with.
4. Repeat.

Random Walking:

1. Create the skeleton tree of the circuit
2. If any nodes have no parents in candidates, then add them to candidates (for the first iteration this is all nodes at level 0)
3. Select a random node from candidates, add it to c , and remove from candidates (nodes can not be re-added).
4. Repeat from step 2 until candidates is empty and all gates have been added to c .

The basic idea of these two methods is the same: to randomize the ordering of gates. The first one is extremely simple. There is no problem with sending a gate all the way left or right, because any gates in the same level can still be sent past it. For instance, say we have a circuit abc such that no gates collide. If we send c all the way to the left, then we have cab . We do not have to worry about c getting “stuck” there, as if we send b to the left, then we have bca . However, there remains a concern that the very last gate that we shoot will be at the end, or the farthest it can be. It is unclear whether this causes any damage or not. Random walking is meant to circumvent this problem as we are truly randomizing the nodes in each level of the skeleton tree. However, this is much

more algorithmically complex as we need to convert our circuit to the skeleton graph. With a different data structure, it is possible that this is more efficient, but for now, we choose to use random shooting.

To further convince that this choice is not wrong, we have three tests to compare the two. For the first test, let circuit N be a circuit with no colliding gates on 64 wires. This circuit has 62 gates. There are three ways we can randomize this and retain functionality. First, we can just randomize all the gates by simply selecting the gates in a completely random order. This can be done because no gates collide and so any ordering of the gates will yield the same functionality. This essentially serves as our goal for the next two. Second and third are our proposed random shooting and random walking. As we are only on 62 gates, there is high variance, which means that heatmaps won't be the best way to view the results. Instead, we can record the mean hamming distance across hundreds of random states between N and the randomized circuits, and then randomize them 100 times to get a new mean each time. Taking the average gives us a better estimate on how we are doing.

For the second test, let us now consider a completely random 100 gate circuit and then do the same process. Note that we no longer are able to complete randomize the circuit as gates will now collide.

For the third and final test, we can consider a 500 gate random circuit. The results of all the tests are below.

Scenario	Method	Average
Non-colliding	Random ordering	0.4532510031
	Walking	0.4522365294
	Shooting	0.4535883884
100 gate	Walking	0.4286999006
	Shooting	0.4226936795
500 gate	Walking	0.2003583436
	Shooting	0.1988034497

Average mean Hamming distance by scenario and method.

As we can see from the table, we see only very small differences between random shooting and random walking. Thus, we conclude that there is little gain in using random walking over random shooting given the additional algorithmic complexity in random walking.

For an idea of how well we can actually mix, below are two heatmaps. Figure 22 is a random 100 gate circuit on itself as we saw earlier, which we expect to have a “red beam” along $y = x$ as the circuits are equal. Figure 23 is the same 100 gate circuit but with random shooting. We note that the heatmap for random walking looks extremely similar.

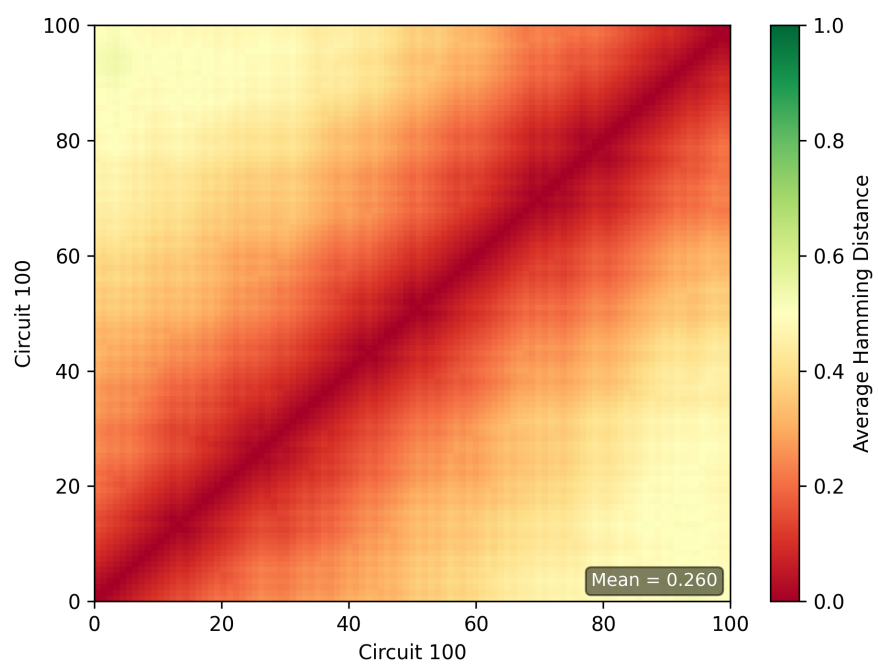


Figure 23: Heatmap showing a 100 gate circuit with itself

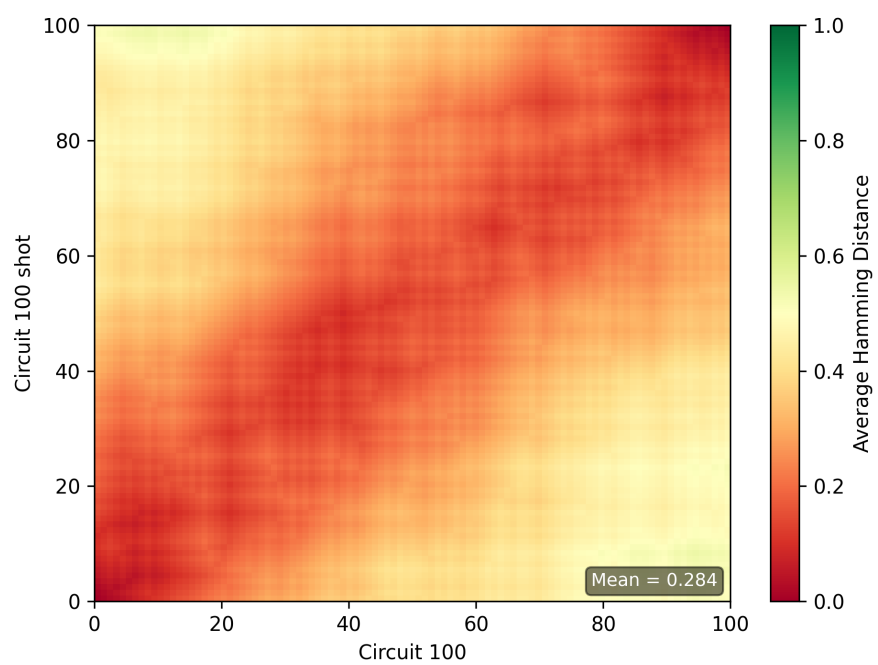


Figure 24: Heatmap showing a 100 gate circuit before and after random shooting

Figure 22: Heatmap showing a 100 gate circuit with itself.

Figure 23: Heatmap showing a 100 gate circuit before and after random shooting.

Replacing Gates with Rewired Identities

A more complex method of achieving blurring is to make single and pair gate replacements. The process for both is given below.

Single Gate Replacement

1. Select a random gate g .
2. Create a random identity I by finding two friends from our rainbow table.
3. Rewire the very first gate of I to match g to get I' . We note that this is still an identity as bit shuffles of identities are still identities.
4. Remove the first gate of I' , which should be equal to g , to get B .
5. Replace g with B .
6. Repeat.

More formally, step 2 can be done by searching for a permutation that has multiple circuits computing it. We can then select two of them, say C_i and C_j , and then we have $I = C_1 C_2^{-1}$.

For the pair replacements, we will classify each type of overlap from the right gate onto the left gate. For instance, 123;321 we can say that the active pin of the left gate's active pin shares a wire with the second control pin of the right gate, the left gate's first control pin shares a wire with the right gate's first control pin, and the left gate's second control pin shares a wire with the right gate's active pin. For pairs of gates, there are 33 different ways to have an overlap (excluding the case where the two gates don't overlap at all). This is because there are 6 ways to have an overlap on every single wire of the left gate, 18 ways to have 2 overlaps on the left gate, and 9 ways to have an overlap on a single wire. This means we can easily identify each of the overlaps with an id. We can now define how to make pair replacements.

Pair Gate Replacements

0. Choose how many pairs to be replaced X .
1. Pair up all gates in the circuit and store their overlap ids in a table T .
2. Create a random identity I by finding two friends from our rainbow table.
3. If I 's taxonomy matches one in T , say that of $g_1 g_2$, then rewire the first and second gate to match the pair to get $I' = g_1 g_2 B'$.
4. Remove the first two gates of I' that match the pair with the matching taxonomy to get B' .

5. Replace the pair with B'^{-1} as $I' = g_1 g_2 B'$ which means $g_2 g_1 = B'$.
6. Repeat until X pairs have been replaced.

The biggest advantage to using the single gate replacements is the simplicity. It allows us to hide a single gate amongst 4 to 8 gates. However, the state before these 4 to 8 and after them will remain the same. Thus, by replacing a pair of gates, we get to remove the seam between those two gates. So while it is harder to replace a pair, as the best we can do is check a random identity and see if it matches, it is more effective in blurring. Thus, the first question is whether we can get enough randomness with only single gate replacements, similar to how we chose to use random shooting over random walking.

Figures 24 and 25 are two heatmaps of tests that use random shooting, ancilla replacements, single gate replacements, and 5 and 30 rounds of asymmetric butterfly, respectively.

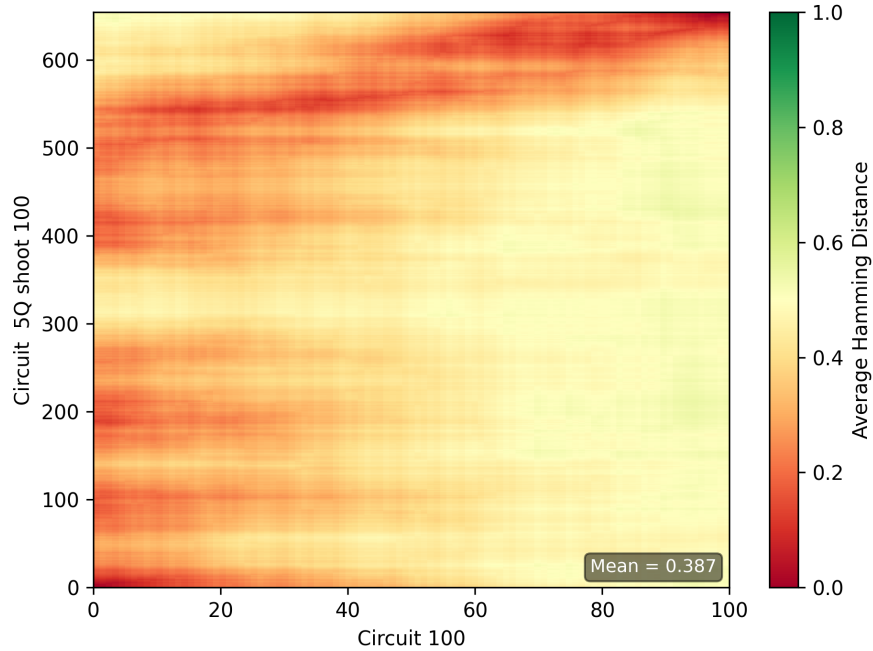


Figure 25: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, single gate replacements, combined in 5 rounds of asymmetric butterfly

Figure 24: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, single gate replacements, combined in 5 rounds of asymmetric butterfly.

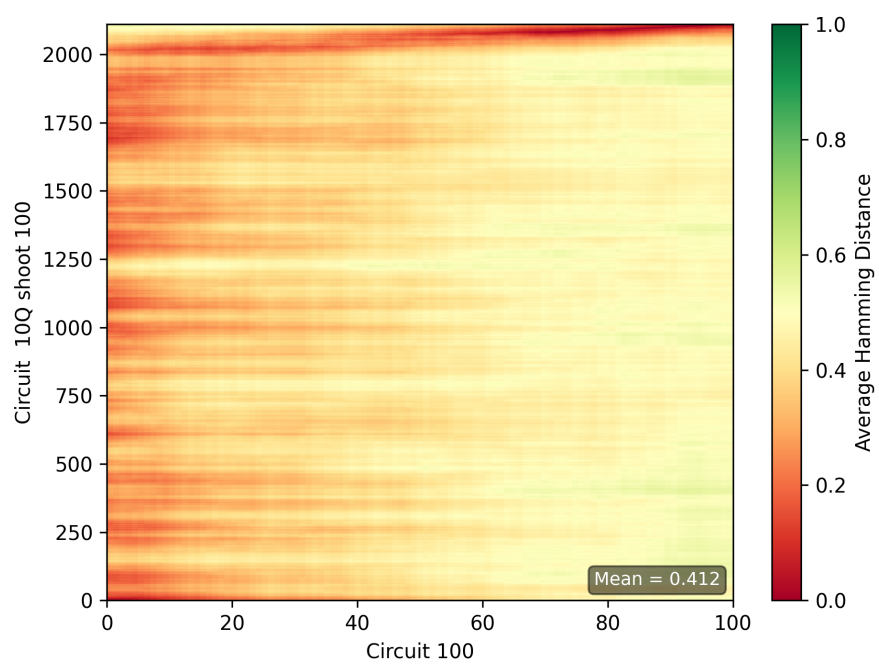


Figure 26: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, single gate replacements, combined in 30 rounds of asymmetric butterfly

Figure 25: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, single gate replacements, combined in 30 rounds of asymmetric butterfly.

While we get incompressibility from the above results, we see that there remains a “red beam” in the top right corner. In addition, for much of the obfuscated circuit, we see that it doesn’t move much, indicated by the red lines on the left until near the top of the heatmap. We call this a red boomerang. We suspected that this was a result of the structure of the random 100 gate circuit, but upon testing with other random circuits, the red boomerang remained. Thus, it would appear that we need to test with pair gate replacements. The heatmap below in Figure 26 shows the same test, but with only pair gate replacements.

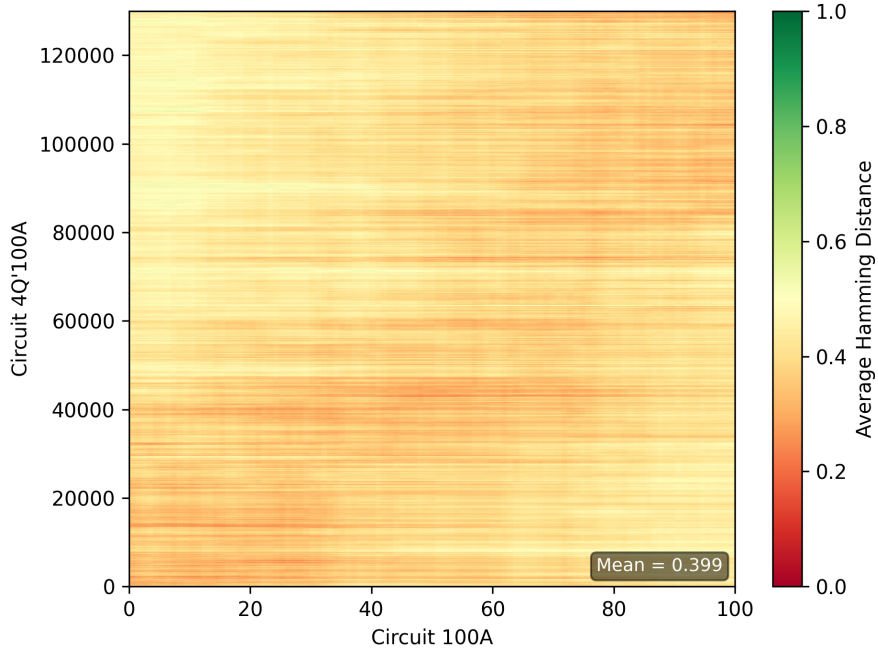


Figure 27: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, pair gate replacements, combined in 4 rounds of asymmetric butterfly

Figure 26: Heatmap showing a 100 gate circuit after random shooting, ancilla replacements, pair gate replacements, combined in 4 rounds of asymmetric butterfly.

We quickly note that we have tested this a number of times and got similar results each time. We see no “red beam” anymore and the randomness is much more spread out. There is still a very spread out red “line” on $y = x$ and so

additional blurring is needed. This is the only way we have found so far that has managed to get the blurring and incompressibility that we desire. We also believe that it would be good to use both single and pair gate replacements, rather than just one.

Methods on Pair Replacements

Above, we discussed a partitioned pair gate replacement. However, this doesn't actually provide much mixing outside of the replacements themselves. The gates adjacent to each other don't interact in any meaningful way, unlike how adjacent identities in the butterfly will attempt to randomize via shooting and compression. Thus, the methods that we discuss here will attempt to further introduce randomization amongst gates through the use of pair gate replacements. We note that all of these methods incorporate shooting somewhere in the algorithm as well. The point of these methods is to get away from the large blowup incurred from the butterfly methods, while still effectively mixing our circuits.

There are a number of methods that we have tried with motivation of pair gate replacements. We note that pair gate replacements are actually very similar to the ancilla replacements that we discussed before, as a pair could span anywhere from 3 to 6 wires. Thus, all we need to do is make a replacement on 6 or 7 wires in order to use ancilla wires.

Replace Pairs Sequentially

In this method, we make pair replacements in a sequential manner before doing compression. For short, this method is called RCS. RCS is very similar to the method described above. However, we no longer partition our circuit into pairs. Instead, we do the following

1. We take g_1g_2 as a pair and replace it with $u_1u_2 \dots u_l$
2. We then treat u_lg_3 as our next pair and replace it with $w_1w_2 \dots w_k$
3. Our next pair is w_kg_4 and do the same
4. Repeat until we have reached the end of the circuit.

Notice that our end result is a sequence of gates that never return to the original functionality of $g_i g_{i+1}$ because we replace it with $u_1u_2 \dots u_l$, but then use u_l in a later pair swap. Notice that it is possible that g_1 and g_2 do not collide at all. In fact, they may not even share any of the same wires. We experimented with some ideas on forcing collisions and while we didn't see any new results, we will state the ideas below for completion.

1. We take $g_i g_{i+1}$ as a pair
2. If g_i and g_{i+1} share any wires, then replace with $u_1u_2 \dots u_l$ as before. The next pair to consider is $u_l g_{i+2}$

3. Else, we shoot g_{i+1} to the right
4. If g_{i+1} eventually collides with some gate h_j , then we treat $h_j g_{i+1}$ as a pair and replace it with $w_1 w_2 \dots w_k$. The next pair to consider is $g_i g_{i+2}$
5. If g_{i+1} does not collide with any gate to the left, then make it a single gate replacement. The next pair to consider is $g_i g_{i+2}$

Notice that in the experiment above, we don't fully remove the $u_1 u_2 \dots u_l$ in the cases that we must shoot left. It remains unclear whether this is a huge loss or not and we did not explore this much further as little to nothing was lost when switching to the simpler method discussed at first. Below, Figure 27 is the result of RCS after 3 rounds.

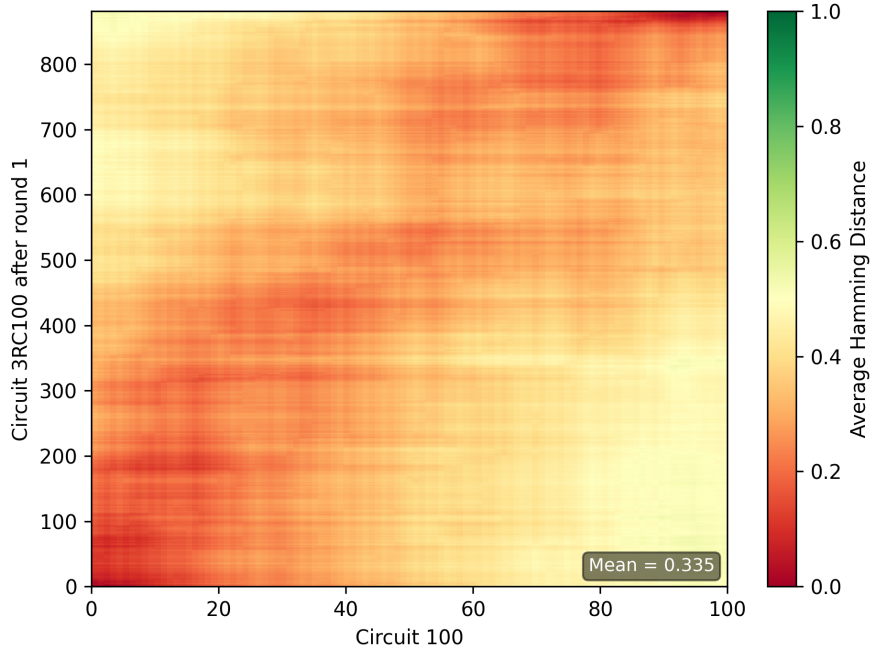


Figure 28: Heatmap showing a 100 gate circuit after 3 rounds of RCS

Figure 27: Heatmap showing a 100 gate circuit after 3 rounds of RCS.

This heatmap is quite similar to the heatmap taken from our partitioned version of pair replacements. Of course there is more correlation overall, but the gate blowup is much smaller. In fact, we can take this to 50 rounds and still be only at a fraction of the number of gates that the pair gate replacements with asymmetric butterfly.

For comparisons sake to the next method, Figure 28 depicts heatmap of a test done on 16 wires.

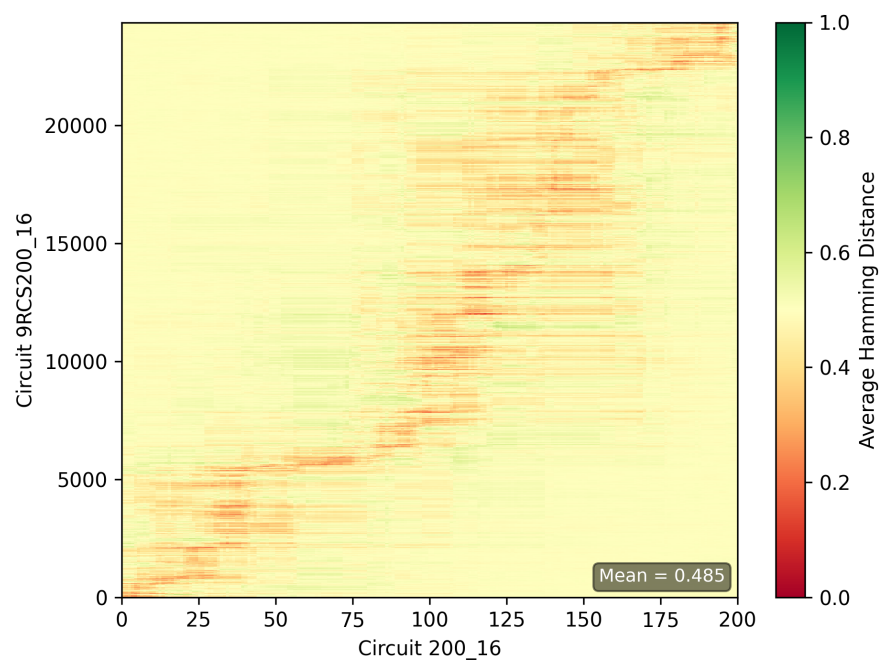


Figure 29: Heatmap showing a 100 gate circuit after 9 rounds of RCS on 16 wires

Figure 28: Heatmap showing a 100 gate circuit after 9 rounds of RCS on 16 wires.

We will return to this method after discussing the other two methods.

Replace Pairs by Distance

One problem with the above method is that we replace pairs in a very standard manner, without caring for the kinds of pairs that we replace. For instance, suppose a circuit is extremely randomized in the first half and completely untouched in the second half. Then ideally, we would further randomize the second half in order to get complete randomness. The sequential method constantly gets close to the original states, and we wish to stay away from these points until the end of the circuit. This is the idea of the replace pairs by distance method (RCD).

This method records the notion of “distance” in between each gate. These distances measure how far we have gone from the original states of our original circuit. The idea is to track how far the circuit is from the **original circuit** after each gate. We call this the *distance*. As each gate can only flip a single bit, then a single gate can only affect the distance by ± 1 . So suppose we have a circuit $g_1g_2g_3$. Then there are 4 initial states for us to record: before g_1 , between g_1 and g_2 , between g_2 and g_3 , and after g_3 . As this is the original circuit, our starting distances is thus $[0, 0, 0, 0]$. Now suppose we replace g_1g_2 with $u_1u_2u_3u_4$ (of course with equivalent functionality). Then the circuit becomes $u_1u_2u_3u_4g_3$. In addition, the gates within $u_1u_2u_3u_4$ will differ from g_1g_2 . So suppose and input state x_0 is changed by g_1 to become x_1 . Then x_0 may never be x_1 after any u_i . This means that the u_i give us some *distance* from the original circuit. Our new distances become $[0, 1, 2, 1, 0, 0]$. We note that x_0 after both g_1g_2 and $u_1u_2u_3u_4$ is equal as the circuits themselves are equivalent. We quickly note that this creates a small “hill” in the distances. Now let us replace u_4g_3 with $w_1w_2 \dots w_8$. The distances will then become $[0, 1, 2, 1, 1, 2, 3, 4, 3, 2, 1, 0]$. The beginning and end are ascending and descending respectively. However, notice that the “second” distance in our list of distances can never be greater than 1. In other words, we can not have a distance like $[0, 2, 3, 2, 0]$. This means that u_1u_2 is already as “steep” as possible and can not be improved further. We can therefore ignore these “edges” and only need to consider the middle section $[1, 1, 2, 3]$. Finally, notice that all the 0s have been eliminated. If we continue with this process, we can work to eliminate all the 1s, and then the 2s, etc. Ideally, we would be able to continue until we have removed all 30s, but practically this becomes very hard due to gate blowup. Thus, our tests only go as far as removing all 10s.

Figure 29: Heatmap showing a 100 gate circuit after 5 rounds of RCD on 16 wires.

Figure 29 looks extremely similar to that of the RCS method. With further rounds, we only get a lighter red line, but can’t seem to escape the red line that reveals our correlation. In addition, we clearly notice that the line is much

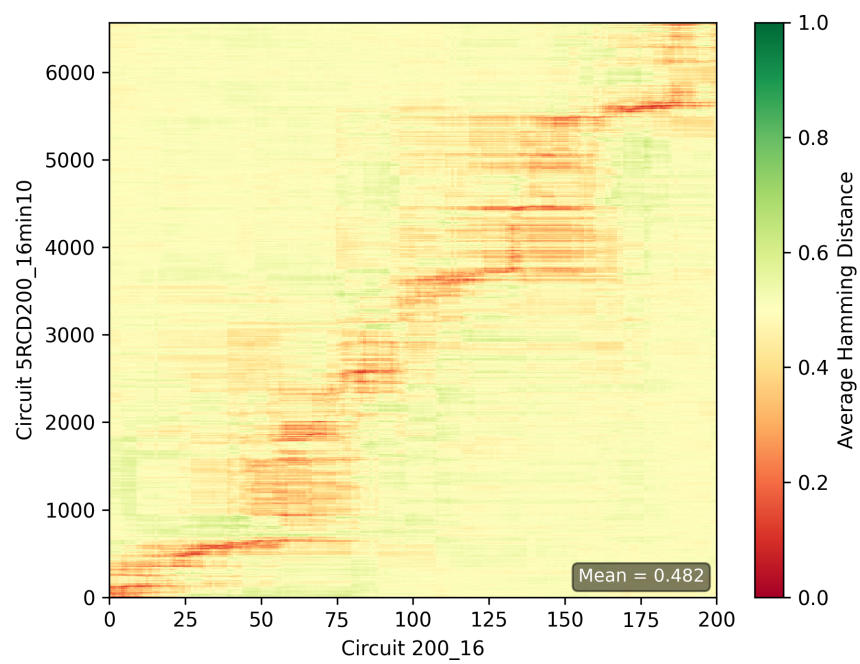


Figure 30: Heatmap showing a 100 gate circuit after 5 rounds of RCD on 16 wires

clearer in the 16 wire version than the 64 wire version. This reveals that our randomization methods are merely fixing the blurring that we have from our shooting. On 16 wires, shooting is much less effective as gates will collide much more often. Thus, it is clear we need some further form of randomization. That is not to say that the above methods are useless though, as we have still achieved incompressibility. At this point, we are most concerned with the heatmap attacks.

More on RCS and RCD

We earlier discussed that our random circuits needed a sufficient number of gates to actually appear random. Figures 30 and 31 below compare circuits on 128 wires. Figure 30 uses 100 gates and still looks very blurry, while Figure 31 uses 1000 gates and does not. Thus, these tests also need enough gates for the number of wires.

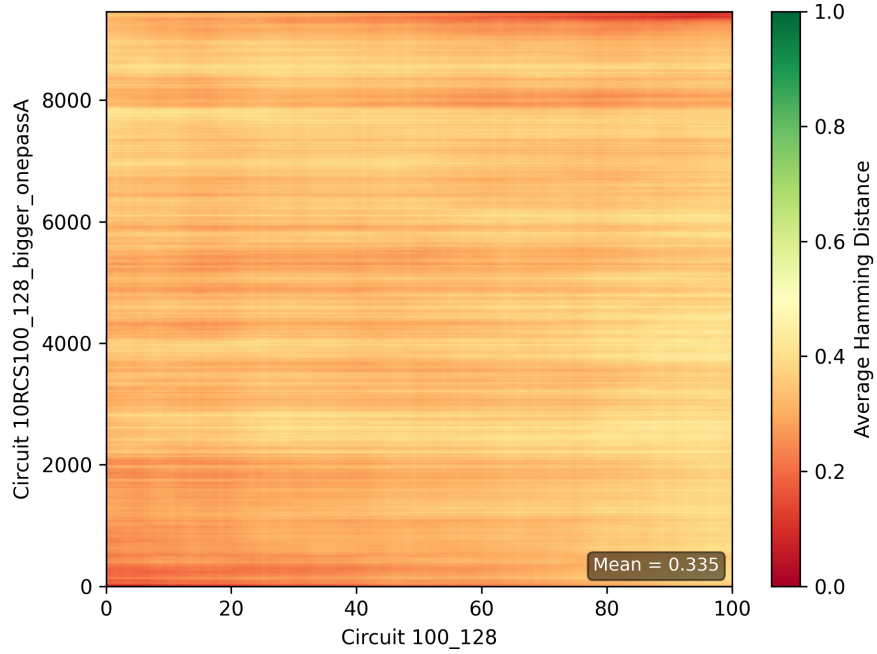


Figure 31: Heatmap showing a 100 gate circuit after 10 rounds of RCS on 128 wires

Figure 30: Heatmap showing a 100 gate circuit after 10 rounds of RCS on 128 wires.

Figure 31: Heatmap showing a 1000 gate circuit after 3 rounds of RCS on 128 wires.

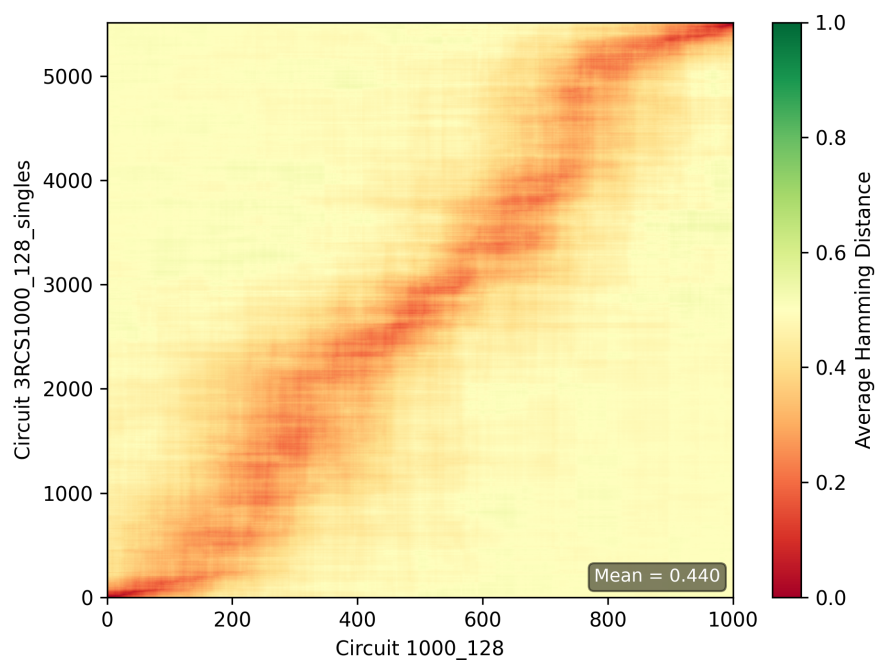


Figure 32: Heatmap showing a 1000 gate circuit after 3 rounds of RCS on 128 wires

One thing that we have retained throughout all of our experiments, is maintaining functionality of the circuit. As a result, we have always gotten a strong diagonal on our heatmaps due to the functionality being equivalent, causing there to be guaranteed strong correlation in the corners along $y = x$. As discussed earlier when we detailed our measures of success, our results may only be a result of working on 100 gates on 64 wires. As we saw, this isn't enough to get a full amount of randomness on 64 wires. However, one idea that comes from this is given our starting number of gates, to simply increase the number of wires that we work on in hopes that we can force a heatmap that blurs in the way we saw previously. However, this would continue to limit us, as even those blurred heatmaps showed a lot of correlation. This correlation is, in a way, expected as for equivalent circuits, we will always have correlation in the corners along $y = x$. Thus, one question to ask is if we really need to maintain equivalence?

Generation Mixing

Generation mixing began as a variation on RCD. RCD tries to work on the parts of the circuit which remain closest to the original computation. Instead of repeatedly measuring that distance on many input states, generation mixing gives each gate a cheap score which records how deeply it has been rewritten. It also keeps the useful idea from RCS that one replacement should overlap the next, rather than leaving a sequence of isolated replacement blocks.

A direct way to use the score is to select a gate at the lowest useful generation, shoot it through the gates with which it commutes, replace the window at the resulting collision, and continue from a gate made by that replacement. We can also obtain overlap without choosing the least rewritten gate each time. The method below makes fresh database-replacement attempts under a size schedule and moves the products of each successful replacement outward. Repeated sampling and this outward motion make the replacements overlap.

This is the point at which the frozen table is most useful. Apart from the occasional twist described below, every round attempts the following database move:

1. Choose either a contiguous window or a convex window. As described in *Sampling Subcircuits*, the gates of a convex window can be brought together by commuting them past gates with which they do not collide.
2. Compute the exact polynomial functionality of the selected window and canonicalize it to obtain its database key. We first check the curated table and then fall back to the general table. The identical spelling is discarded; a successful lookup must actually change the window.
3. Choose a different circuit with the same functionality. A **MIX** move chooses randomly among the non-growing spellings when any are available. If every available spelling is larger, it may instead choose a random

larger one and pay the necessary growth. A **COMP** move never grows the window and chooses among its shortest available spellings. Therefore, a COMP move shrinks when it can, but it may also make an equal-length re-spelling. COMP normally begins with a larger window and tries shorter prefixes when the full window has no usable replacement, while MIX samples one relatively short window without walking through all of its prefixes.

4. Map the canonical circuit back onto the physical wires, verify that it has the same functionality as the outgoing window, and splice it into the circuit. The two boundary states are unchanged, but the gates and intermediate states inside the window may be completely different.
5. Point the new gates outward from the center of the replacement and move each one through a fixed fraction of the gates across which it can legally commute. The new gates therefore do not remain together as an obvious replacement block. A later sampled window is likely to contain gates made by several different earlier replacements.

If a lookup fails, that round simply makes no replacement. It does not fall through to a different kind of move. This keeps generation mixing focused on repeated database re-spelling rather than turning it into the fragmentation walk which comes later.

For a small example, suppose part of the circuit is

$$g_1 g_2 g_3 g_4 g_5.$$

We may first replace $g_2 g_3$ by an equivalent circuit $u_1 u_2 u_3 u_4$, giving

$$g_1 u_1 u_2 u_3 u_4 g_4 g_5.$$

After the u_i are moved through their available commuting ranges, a later window may contain, for example, $u_3 u_4 g_4$. Replacing that window folds part of the first replacement together with previously untouched material. We no longer have a clean boundary at which we can separate the first replacement from the rest of the circuit. Repeating this process gives us the cascading behavior that motivated RCS.

We can record generations in order to measure this process. Input gates begin at generation 0. After a database replacement, every new gate receives one more than the upper median generation of the window which it replaced. The gates made by one replacement also share a **litter** label, which lets us recognize gates which were born together. These are coarse accounting tools rather than exact ancestry statements. In the method above, they measure the rewriting without selecting its windows or deciding when to stop. The size schedule described next controls how much mixing we do.

Expansion, Holding, and Compression A generation-mixing schedule can have three size periods: expansion, holding, and compression. Expansion gives us more alternative spellings and more room for the new gates to move into other neighborhoods. Holding keeps that space while many overlapping replacements re-spell the circuit. Earlier recipes then compressed part of the way back down.

The complete mixing method uses only the first two periods here. We grow toward twice the incoming gate count over three work units, then hold that size for another 27. One work unit represents roughly one mixing round per gate, counting the circuit's changing size as we go. There is no shrinking period at this stage: we leave final compression until after splitting and crossing.

The schedule controls the balance between MIX and COMP on every round. During expansion it favors MIX, during the hold it balances the two so that growth and compression roughly cancel, and during the final period it favors COMP. The schedule measures work relative to the current circuit size, so a larger circuit receives proportionally more replacement attempts. Thus, the chosen peak size controls how much room the mixer has, while the length of the hold controls how long it continues re-encoding at that scale.

Changing the Internal Wire Frame A database replacement changes the inside of a short window while preserving the states at its two ends. Generation mixing also occasionally changes the physical wire frame throughout a much longer window. Let W be that window and let P swap two wires. Since P is its own inverse,

$$P(PWP)P = W.$$

The middle copy PWP is obtained by relabeling the two wires throughout W . The opening and closing copies of P move into this swapped frame and then move back out of it. The complete circuit therefore has the same functionality, but its intermediate states use a different physical representation throughout the window.

We do not insert a bare, easily recognized swap and merely hope that later database moves hide it. Each boundary is instead synthesized directly as an all-r57 word which can absorb a few of the real neighboring gates. The new boundary gates are then moved outward in the same way as database products, and later database rounds may re-spell them again. This is the part of the method which borrows the wire-frame idea developed in the later SAMF section. This twist uses the swap part only; it does not add the bit flips of the gate-by-gate SAMF construction.

At the end of generation mixing, every gate is given one final random position within the full interval through which it can commute. This changes no gates and preserves their legal order, but removes positioning left over from the final replacement round. Generation mixing therefore leaves us with an equivalent

circuit whose database replacements are in the r57 vocabulary. Non-r57 gates from preprocessing can remain. The circuit has undergone many overlapping local re-spellings and changes of wire frame over long intervals.

Pre-Interleaving

In order to play around with the idea of not maintaining functionality, we must first establish that we can not just destroy all semblance of equivalence, as in the end, an obfuscated circuit should still be able to used for its original purpose. Despite this goal, we can still create a random circuit via pre-interleaving.

The idea of pre-interleaving is to create a circuit that is completely random overall on $2n$ wires, but on the first n wires maintains functionality. A simple way to do this is with the following:

1. Let C be a random circuit on n wires with m gates
2. Generate a random C' on n wires and m gates and then shift to wires to lie on $n..2n$ as opposed to $0..n$
3. Interleave C and C' together to get a new circuit D

From this, we have that C' gives us a completely random circuit D , but as we shifted all of its gates to commute with the gates of C , then the functionality of D on the first 64 wires will depend solely on C . In other words, given D , we can just limit our view to the first 64 wires in order to use it as intended. The hope here is to not hide the fact that we are using only half the wires, nor is it to hope that we can hide which 64 wires we have chosen. Instead, we can combine this interleaving idea with our previous RCS/RCD methods. As we have interleaved our gates together, then each adjacent pair will be noncolliding. When we replace them with our identities, we hope to enforce some collisions and, in a way, fix these gates together such that the interleaving can not be undone. This is very similar to our hope of fixing our blurring methods from earlier. In addition, by allowing a completely random C' to help create a D , a heatmap of D and D' would, in theory, look completely random.

Unfortunately, we did not see great results from this method. The heatmap in Figure 32 reveals that we can still easily correlate between the original circuit and the new circuit. We tested a number of heatmaps, and they all revealed the underlying correlation. We tried first to take the inputs only on the first 64 wires. If things worked as we thought they would, then the additional C' would incur great randomness on these 64 inputs. After, we took a heatmap on 128 wires, shown in Figure 33, which again, should be quite difficult to relate to the original circuit due to the circuits not even being correlated.

Figure 32: Heatmap showing a 1000 gate circuit, interleaved with a second 1000 gate circuit, using inputs on 64 bits.

Figure 33: Heatmap showing a 1000 gate circuit, interleaved with a second 1000 gate circuit, using inputs on 128 bits.

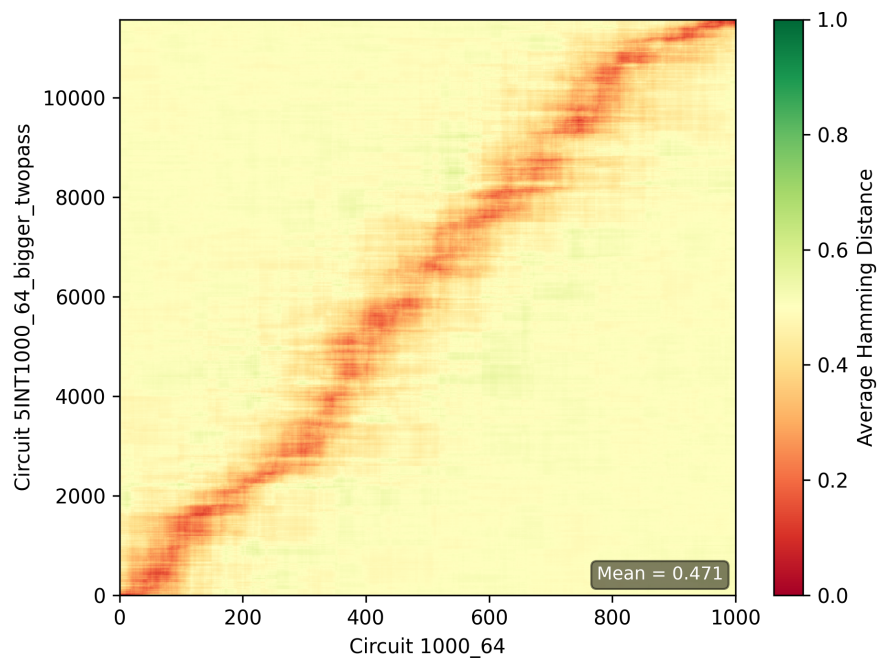


Figure 33: Heatmap showing a 1000 gate circuit, interleaved with a second 1000 gate circuit, using inputs on 64 bits

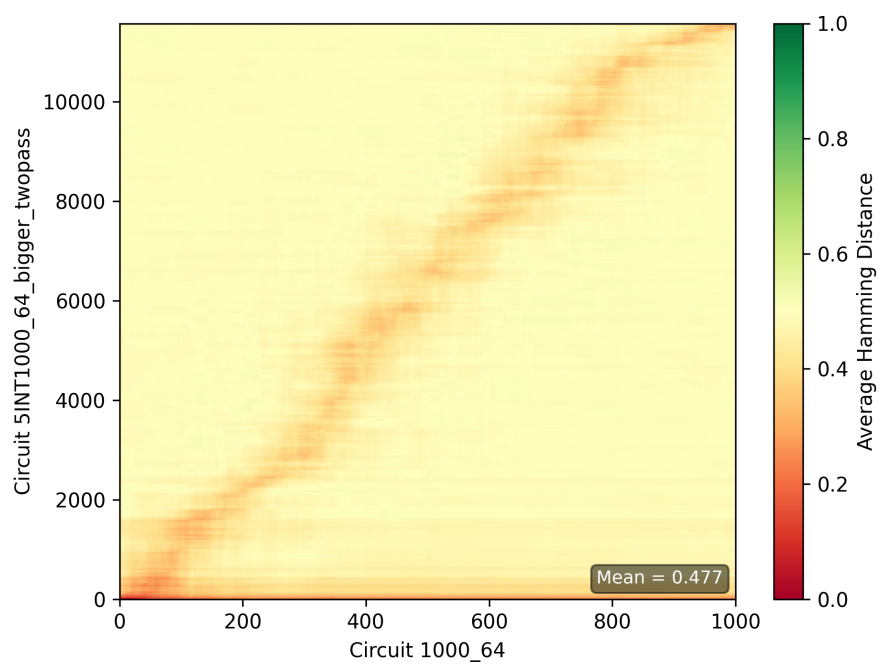


Figure 34: Heatmap showing a 1000 gate circuit, interleaved with a second 1000 gate circuit, using inputs on 128 bits

In a way, this might not be too unexpected. The butterfly methods that we worked on before, were initially tested on identities over 5 wires. Our results in the end, would constantly look correlated no matter how much we increased the wires, but that was largely due to our resulting circuit being heavily condensed on the original 5 wires. From this, it seems that adding gates on new wires is ineffective in randomizing a circuit. So while we leave the interleaving method largely untested from these results, we still find promise in this idea of “leaving functionality behind”. Merging in more clever ways will eventually lead us to the later gadgetization ideas.

SAMFs: Wire Shuffles and Bit Flips

The following construction develops wire shuffles and bit flips as a mixing method. The swap part also gives us the change of wire frame used in generation mixing.

A **SAMF**, short for *swap and maybe flip*, is a reversible circuit which swaps two selected wires and may also flip the value on either one. As we add SAMFs, we keep track of the accumulated wire permutation and negations, relabel the later gates into that changing frame, and undo the final accumulated transformation at the end. The discussion below first develops this idea as wire shuffles and then adds the bit flips.

We take a different approach here. We recall that we have already achieved the goal of incompressibility and only need to beat the heatmap attacks (at least for now). The heatmaps simply record hamming distances between states as they evolve throughout the two circuits. Thus, one simple way to force this to differ greatly, is to shuffle the wires that we are working on. If we are to do this, then we have artificially beaten the heatmap attacks, as two different wire shuffles on the same circuit C , would result in C having a higher hamming distance than if we took a heatmap of C with itself.

Let us think about how we can achieve these shuffles. One way that utilizes a similar thought to our butterfly method, is to do the following.

Butterfly-like

1. Start with some circuit $C = g_1 g_2 g_3 \dots g_m$
2. For g_i , replace it with some shuffle $B_{w_i} g'_i B_{w_i}^*$, where g'_i has been shifted accordingly with B_{w_i} so that functionality is maintained.

The above would yield something like $B_{w_1} g'_1 B_{w_1}^* B_{w_2} g'_2 B_{w_2}^* \dots$. One of the advantages of using our wire shuffles, is that they form a subgroup of all circuits. This can easily be verified. The identity is the trivial shuffle where nothing gets shuffled. Every shuffle can be shuffled in reverse order and so every shuffle has an inverse. Finally, if C_1 and C_2 are shuffles, then of course $C_1 C_2$ is also a shuffle.

In addition, it is much easier to find equivalent wire shuffles than it is to find equivalent circuits. The main reason for this is that we do not need to work in the world of circuits. Instead, we can find two different shuffles S_1 and S_2 that compute the same permutation, and then convert each of them into a circuit. In addition, given a random permutation that is known to be a shuffle, we can find all the transpositions that make up the shuffle and then convert that into a circuit. Now consider the task of finding a circuit corresponding to a particular completely random circuit. On 16 wires, there are already 2^{16} different permutations, and it wouldn't even be known how many gates would be needed to compute the desired permutation.

In mixing our shuffles, we can actually find a random, yet equivalent, circuit representation of $B_{w_1}^* B_{w_2}$. Thus, we would ideally never return to the original states until we reach the very end of the circuit. From what we see below in Figure 34, heatmaps reveal that we can not actually get far enough from the original functionality for this to be the case. The many red horizontal lines show that we are constantly returning back to the original functionality, or at least getting close enough to it to reveal correlation.

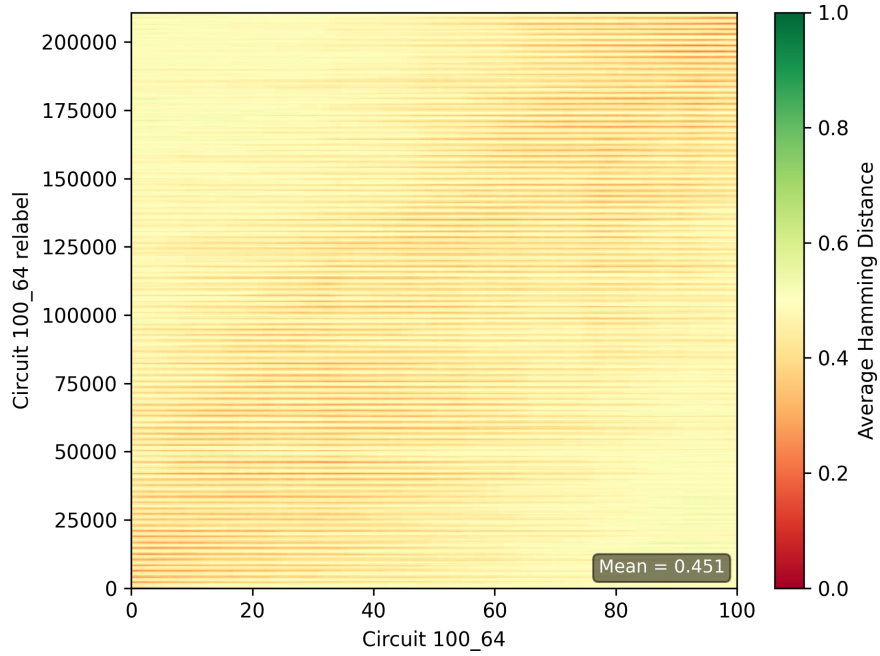


Figure 35: Heatmap showing a 100 gate circuit on 64 wires after a butterfly-like shuffle of wires

Figure 34: Heatmap showing a 100 gate circuit on 64 wires after a butterfly-like shuffle of wires.

Thus, we need to carry our wire shuffles throughout the entirety of the circuit. Let us take this in the simplest way possible.

Singular Shuffle

1. Start with a circuit $C = g_1 g_2 g_3 \dots g_m$
2. Insert a shuffle at the very beginning and at the very end, shifting g_i to g'_i such that $g_1 g_2 g_3 \dots g_m = B_w g'_1 g'_2 g'_3 \dots g'_m B_w^*$

Of course, as before, we can get two random circuits that compute B_w so that we are not merely reversing the same circuit. Thankfully, our sanity checks pass and our heatmap now looks extremely random, as shown in Figure 35.

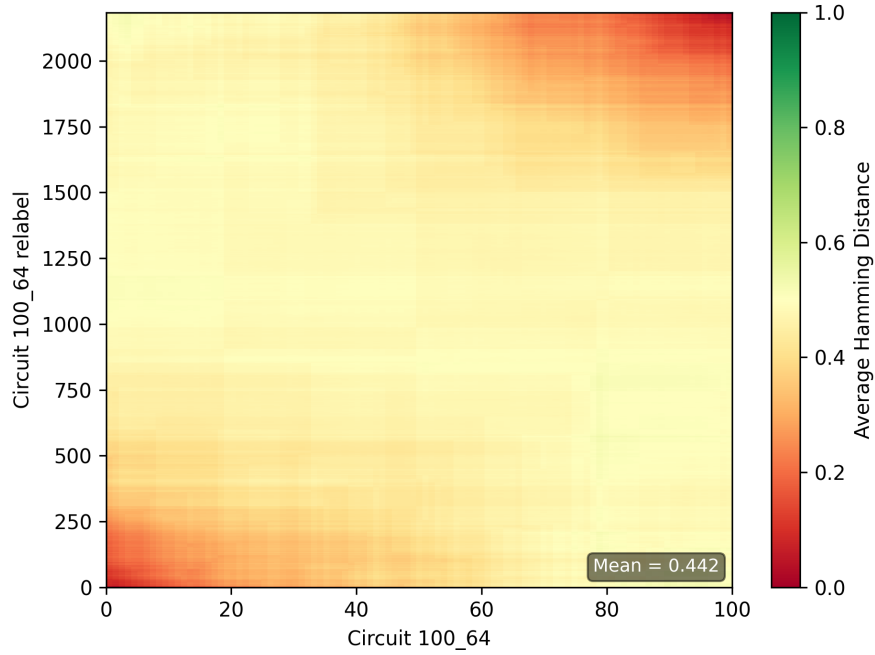


Figure 36: Heatmap showing a 100 gate circuit on 64 wires after a simple shuffle of wires

Figure 35: Heatmap showing a 100 gate circuit on 64 wires after a simple shuffle of wires.

The main reason that this works so much better, is that in no point in the circuit, are we returning to the original functionality. We remain far from it with our first shuffle and do not return until the very end. Of course, this does not hide anything. An attacker can easily look at our circuit and then see that our initial gates are merely a shuffle amongst the wires, undo the shuffle, and

then recover the original circuit. Thus, we need to do something more than just a simple singular shuffle. Let us do the following instead.

Gate-by-gate Shuffles

1. Start with a circuit $C = g_1 g_2 g_3 \dots g_m$
2. For each gate, insert a B_{w_i} and shift g_i as needed to maintain functionality to get

$$B_{w_1} g'_1 B_{w_2} g''_2 B_{w_3} g'''_3 \dots B_{w_m} g^{(m)'}_m B^*,$$

where B^* is the inverse of the composition of all B_{w_i} .

This alone makes it much harder to find our original gates g_i . If we combine this now with our previous mixing methods, such as RCS, then we can ensure that this is also incompressible. This also gives us some randomization amongst the swap gates to hide the fact that they are doing swaps or even hiding the original gates into these swaps, making it even harder for an attacker to determine which gates are part of swaps and which gates are from the original.

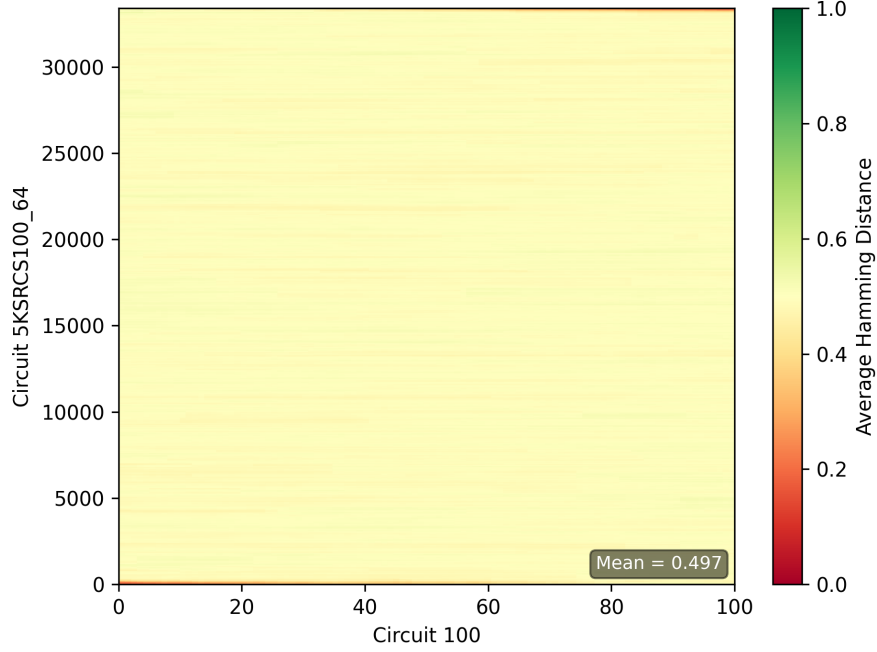


Figure 37: Heatmap showing a 100 gate circuit on 64 wires after gate-by-gate shuffles

Figure 36: Heatmap showing a 100 gate circuit on 64 wires after gate-by-gate shuffles.

As seen in Figure 36, the heatmap remains close to the ideal that we are looking for. One thing to note is that the blowup in the number of gates is much greater. As we are limited to gate r57, in order to compute a swap of two wires, it requires at minimum, 6 gates and 3 wires. The algorithm above utilizes a random choice of a swap using 3 wires and between 6 and 20 gates, just for some additional randomness. In order to compute a completely random shuffle on the wires, we can then use a Knuth shuffle. This requires n swaps in total, which gives us a total of $6n$ blowup in the total number of gates. Thus, one thing that one could try is to instead only add wire shuffles to some gates, rather than before every single gate. Then repeat it in combination with RCS and compression. In other words,

1. Start with a circuit $C = g_1 g_2 g_3 \dots g_m$
2. Insert x number of shuffles randomly in between the gates of C , as well as before g_1 and after g_m .
3. Mutate the shuffled C via RCS
4. Compress
5. Repeat

The shuffle before g_1 is meant to ensure that the entire circuit is shuffled, at least a little. The shuffle after g_m is meant to undo everything and maintain functionality. One advantage to this relies on the fact that adding shuffles without RCS remains compressible. For instance, a gate-to-gate shuffle insertion can blow up a 100 gate circuit to a 60,000 gate circuit, on 64 wires. This compresses down to around 31,000 gates. Thus, we believe that we can add many shuffles and compress them along the way, such that we have both many shuffles as well as minimal gate blowup. However, this remains untested.

As discussed in *How We Measure Success*, an ordinary heatmap compares physical states directly. A wire permutation can therefore make this map look random without hiding the computation. We use the Hamming-weight heatmap here because a pure wire permutation preserves Hamming weight. This lets us test whether the shuffles above have only moved the same intermediate state to different wires. We compute it as follows.

To compute heatmap between C and C' on n wires:

1. Let i be the i th gate of C and j be the j th gate of C' .
2. Choose a shared set S of random input states in $\{0, 1\}^n$.
3. For each $x \in S$, compute $y = C_i[x]$ and $y' = C'_j[x]$.
4. Average $|H(y) - H(y')|$ over S , where H is Hamming weight.
5. Repeat for all pairs i, j .

Let us check on a circuit that has been shuffled using this new type of heatmap. We will test on 32 wires as to keep the wire blowup minimal, however, the results can be extended to a larger number of wires as well. We will test in the following manner.

1. For a given C , shuffle/shoot the circuit
2. From above, shoot/shuffle
3. Compress

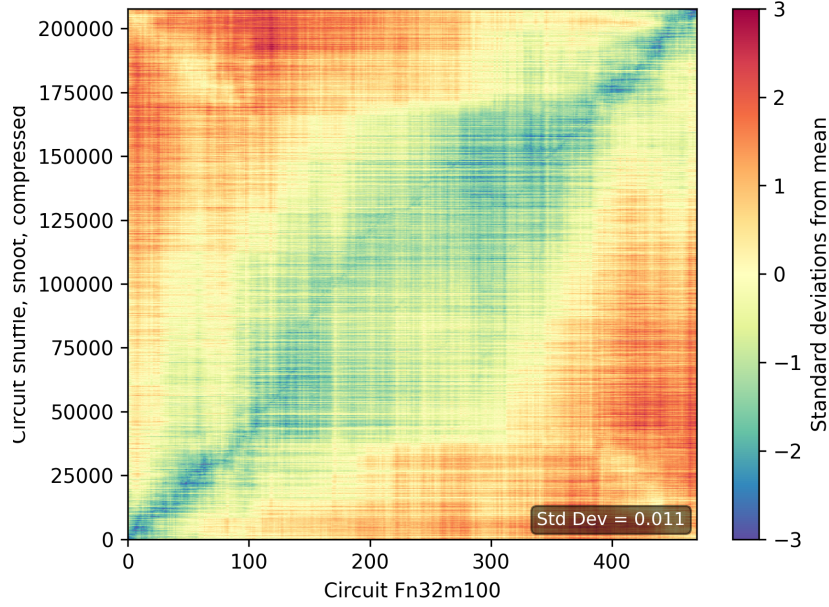


Figure 38: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gateshuffles, shooting, and compression

Figure 37: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gateshuffles, shooting, and compression.

Figure 38: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gate shooting, shuffles, and compression.

As we can see in Figures 37 and 38, whether we shoot or shuffle first, we can still clearly see the structure of the original circuit with shuffles. Thus, we need to destroy this structure in some way.

One way that we can introduce some functional changes to the circuit, is by introducing bit flips. We note that the set of all circuits which swap as well as flip bits, remains a subgroup of all circuits. Thus, we can do all of our methods

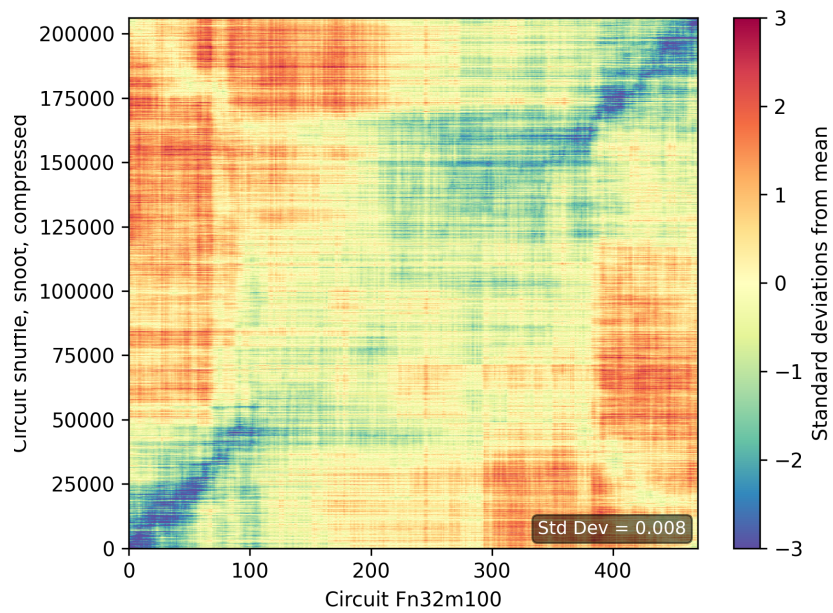


Figure 39: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gate shooting, shuffles, and compression

above, but now allow our swaps to also flip a single bit.

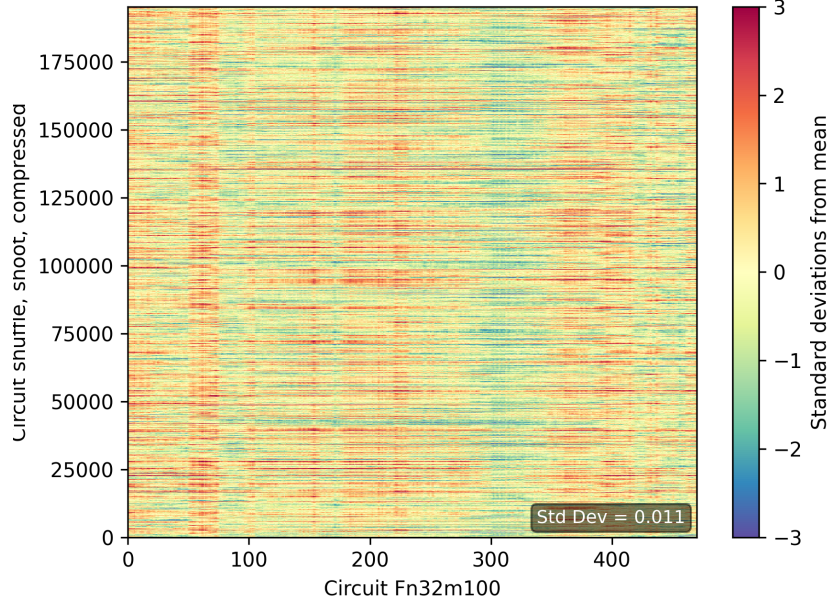


Figure 40: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gate shuffles & bit-flips, shooting, and compression

Figure 39: Hamming weight heatmap showing a 100 gate circuit on 32 wires after gate-by-gate shuffles & bit-flips, shooting, and compression.

From Figure 39, we can see that this heatmap no longer shows the earlier correlation. We still need to test whether other observations recover information from the original circuit.

Beyond Heatmaps & Compression

We recall that a flat global heatmap only rules out the statistic and input distribution that we tested. We therefore repeat the same comparisons on pieces of the circuit and on deliberately chosen input families. These tests are more targeted, but they still do not rule out the other attacks listed in *How We Measure Success*.

One variation of the heatmap that we showed briefly, was to instead take a heatmap on partial circuits. For instance, a 1000 gate circuit can be split into 100 gate “pieces”, to which we can take a heatmap on each piece. This can either be done by treating each piece as its own isolated circuit, but also we can

combine pieces together to form a partial circuit. The former method allows us to test whether each part of the larger circuit is truly random. Suppose a circuit is extremely correlated, except at the very beginning. Then the randomness incurred at the beginning of the circuit would propagate throughout the entire circuit. This method allows us to get around this. On the other hand, by attaching pieces together, we can measure how the progression of the circuit as we reach the final obfuscated version. Indeed, we can measure how the mean (from the heatmaps) of the circuit changes as we add more gates in order to see if it acts in line with a completely random circuit.

We can also choose our inputs more carefully, instead of solely relying on random inputs. It is possible that some inputs will reveal more correlations than others. This is because even a single bit flip at the beginning of the circuit can cause many bit flips throughout the entire circuit. In other words, some inputs may look more random than others and there may be inputs that reveal much of the structure of the original circuit. We have three ways of going about this. First, it is to take inputs with only small hamming weight. The second is to take a random input x_0 , and then only take x_i such that the hamming distance between x_i and x_0 is small. Thirdly, we can take a random input x_0 and then fix every single bit except for bit k . We can then flip k to see how this single flip propagates throughout the entire circuit, and then do the same for every bit k of x_0 .

There are also methods for us to probe the circuit in order to attempt to determine what the original shuffles are. This remains largely untested and it remains unclear whether we are able to retrieve the original shuffle even after our randomization methods. In addition, even with the original shuffle, we may not be able to undo them as the original shuffle has been mixed throughout the circuit. In other words, finding the shuffle means having to reverse engineer it even through the noise we have added in compression and the pair replacement methods, and then undoing it means we have to deal with the noise as well.

Differential Attacks and Algebraic Degree

We recall that the differential test writes every wire at a circuit prefix as a polynomial in the original inputs and looks at its algebraic degree. The final degree is fixed by the original functionality, so our goal is only to keep the intermediate states at high degree. For more on differential attacks, see Bard's *Algebraic Cryptanalysis*.

It is natural to first ask whether a random circuit already gives us this property. At least in the regime we care about, it does not. While degree does increase with gate count and a random circuit will eventually saturate every wire, it increases slowly and unevenly in the low gate regime. For instance, on 16 wires, a random 32 gate circuit can have three wires at degree 1 while other wires are already at degree 8. Thus, heatmap randomness and degree are not the same measure, and a circuit can pass the first while failing the second on only

a handful of wires. Since this is enough for an attacker, we must construct high degree on every wire deliberately.

We use the following construction. Reuse the last active wire as a control wire, introduce a fresh wire as the second control wire, and recycle a previously used wire as the new active wire. Each gate then folds the previous output polynomial into the next and gains one degree of interaction, so the maximum degree grows by exactly one per gate, reaching $m + 1$ after m gates, until it saturates at the permutation ceiling of $n - 1$. Below is the beginning of such a circuit.

Gate	Output	Degree
[1,0,2]	$x_1 = x_0x_2 + x_1 + x_2 + 1$	2
[2,1,3]	$x_2 = x_0x_2x_3 + x_1x_3 + x_2x_3 + x_2 + 1$	3
[3,2,4]	$x_3 = x_0x_2x_3x_4 + x_1x_3x_4 + \dots$	4
[4,3,5]	$x_4 = x_0x_2x_3x_4x_5 + x_1x_3x_4x_5 + \dots$	5

Of course this only guarantees high degree on the wire that was most recently active. To get every wire high, the active wire cycles through all wires repeatedly, so that after a full pass every wire carries its own rung of the ladder.

Gadgetizing

We begin with a paired construction, where each logical value is the XOR of two physical wires. Its gadgets let us compute while keeping the values shared. We will then see where the linearity of this sharing hurts us, and how it motivates the nonlinear constructions which follow.

We will call integrating our original circuit so that it carries high algebraic degree *gadgetizing* it. This is in reference to our heavy use of “gadgets” in order to do secret-preserving computations, secret-preserving swaps, and more.

The idea is to add auxiliary wires which already have high algebraic degree, and then use them in the computation of our circuit, while ensuring that by the end we have returned to the original functionality. This is quite reminiscent of our SAMF gates (swap and maybe flip), where we swap wires and flip the value of many wires, changing the functionality throughout the circuit, but then are able to undo everything and return to the original circuit at the end. Instead of just simply swapping and flipping though, we want something that will actually change the structure of the circuit and incur a higher algebraic degree.

To do this we use an extra n auxiliary wires, so the gadgetized circuit lies on $2n$ wires. The first n outputs will have the functionality of the original circuit, while the latter n outputs are allowed to contain random values. The goal is therefore to integrate the auxiliary computation into the original computation without changing what is eventually returned on the first n wires.

Suppose r_i is an auxiliary value with high algebraic degree and w_i is a computation value from our original circuit. One way to imbue the physical representation of w_i with high algebraic degree is to form $w_i \oplus r_i$. Of course, if we replace w_i by this value directly we have lost the ability to read w_i without later removing r_i , and simply removing the same mask at every use would expose the original computation in the middle of the circuit anyway. Instead we represent each computation value w_i by a *secret* value s_i and an auxiliary value r_i with

$$w_i = s_i \oplus r_i.$$

Here w_i is the value the original circuit uses, while s_i and r_i are the two physical values the gadgetized circuit actually holds. We call the pair (s_i, r_i) a *pairing*, and w_i the *virtual value* it carries. The gadgetizer only needs to remember which physical wires currently hold s_i and r_i , and that bookkeeping never appears in the output.

If our original circuit is $C = g_1 g_2 \cdots g_m$, we would like to replace each gate g_k by a gadget whose effect on the physical values is the same as g_k 's effect on the virtual ones. We do not want to decode w_i , apply the gate, and re-encode it, since that exposes w_i . The gadget must compute the r57 gate *homomorphically*, with the values staying encoded throughout.

The gadgets

An early version of this used a single 12 gate gadget, $[[0,3,5], [0,3,6], [0,4,5], [0,4,6], [1,0,2], [0,2,1], [2,0,1], [1,2,0], [0,2,1], [2,1,0], [0,3,4], [0,4,3]]$, which did both the secret computation and a secret swap of the auxiliary pair tied to a particular computation wire. This cost 12 gates and led to a 13x blowup in the total number of gates. We now split those jobs apart. The following table lists the gadgets we use.

Gadget	Purpose	Gate sequence ([active, ctrl1, ctrl2])
SG	6-gate homomorphic gadget for a secret- shared r57	[4,5,6] [0,4,6] [0,5,4] [4,5,6] [0,6,3] [0,3,5]
RG1	Swap virtual values between two pairs	[1,2,3] [0,3,2] [3,1,0] [2,0,1] [0,3,2] [1,2,3]

Gadget	Purpose	Gate sequence ([active, ctrl1, ctrl2])
RG2	Re-pair two pairs (break pairings, keep virtual values)	[0,3,2] [1,0,2] [2,0,3] [2,3,0] [1,3,2] [3,0,2]
RG3	Refresh a pair's shares against two random wires	[share_i, r1, r2] [pad_i, r1, r2]

SG is the gadget used for each original gate. Given an original gate $[a, b, c]$, it acts on the five wires s_a, s_b, r_b, s_c, r_c , which are local slots 0, 5, 6, 3, 4, respectively, in the sequence above. The auxiliary wire r_a of the active value does not appear in the gadget at all and is left untouched, which is what lets the result decode. After the six gates,

$$w'_a = s'_a \oplus r_a = w_a \oplus (w_b \vee \neg w_c),$$

and both control pairings still carry w_b and w_c unchanged. So six physical gates perform exactly one original gate on the encoded state, without ever exposing w_a , w_b , or w_c on any individual wire. Every original gate is replaced by one SG, and in principle SG alone would reconstruct the entire original functionality. However, a circuit made only from a repeating six gate period would leave an obvious trail, and so we also use the *RGi* gadgets below.

For **RG1**, assume we have virtual values s_1, s_2 where s_1 is carried by wires w_0, w_1 and s_2 is carried by w_2, w_3 . That is:

$$w_0 + w_1 = s_1$$

$$w_2 + w_3 = s_2$$

From the gate sequence above we have:

$$w'_0 = w_0 + w_1 + w_2 + w_0w_1 + w_1w_2 + w_0w_3 + w_2w_3 + w_0w_1w_2 + w_0w_1w_3$$

$$w'_1 = w_0 + w_1 + w_3 + w_0w_1 + w_1w_2 + w_0w_3 + w_2w_3 + w_0w_1w_2 + w_0w_1w_3$$

$$w'_2 = 1 + w_0 + w_0w_1 + w_1w_2 + w_0w_3 + w_2w_3 + w_0w_2w_3 + w_1w_2w_3$$

$$w'_3 = 1 + w_1 + w_0w_1 + w_1w_2 + w_0w_3 + w_2w_3 + w_0w_2w_3 + w_1w_2w_3$$

which means that:

$$w'_0 + w'_1 = s_2$$

$$w'_2 + w'_3 = s_1$$

This swaps the secret values between the two pairs, and does not swap the auxiliary pairings.

For **RG2** we have:

$$w'_0 = 1 + w_0 + w_2 + w_2w_3$$

$$w'_1 = 1 + w_0 + w_1 + w_2 + w_3 + w_0w_2 + w_0w_3 + w_2w_3$$

$$w'_2 = 1 + w_0 + w_3 + w_2w_3$$

$$w'_3 = 1 + w_2 + w_3 + w_0w_2 + w_0w_3 + w_2w_3$$

which means that:

$$w'_0 + w'_2 = s_2$$

$$w'_1 + w'_3 = s_1$$

This also swaps the secret values between the two pairs, but it swaps the auxiliary pairings along with them, so virtual value i now lives on (r_i, s_j) and virtual value j on (s_i, r_j) . The XOR of each new pair is still the correct computation value, but the physical relationship between computation and auxiliary wires

has changed, which makes the pairing structure much less stable across the circuit.

For **RG3**, assume the virtual value s_1 is carried by wires w_0, w_1 , i.e.

$$w_0 + w_1 = s_1$$

and let r_1, r_2 be two further (arbitrary) wires. Writing $m = r_1 \vee \neg r_2 = 1 + r_2 + r_1 r_2$, we have:

$$w'_0 = 1 + w_0 + r_2 + r_1 r_2$$

$$w'_1 = 1 + w_1 + r_2 + r_1 r_2$$

$$r'_1 = r_1$$

$$r'_2 = r_2$$

which means that:

$$w'_0 + w'_1 = s_1$$

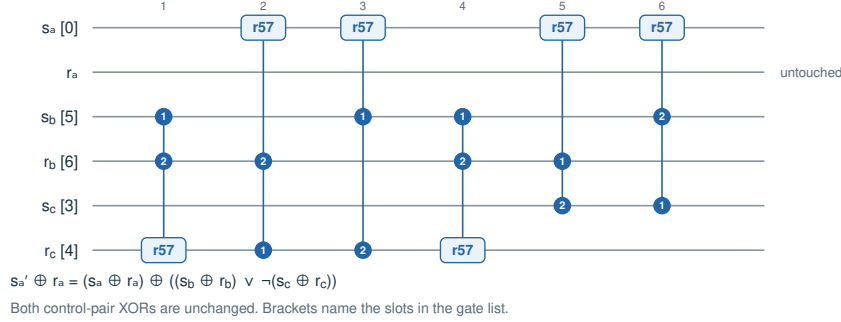
The same mask m is added to both shares, so it cancels in the sum: the secret stays on its own pair and the share/pad pairing is unchanged. RG3 therefore re-randomizes the two shares of a single pair against two unrelated wires, without moving the secret value or swapping any pairings. It is also by far the cheapest of the three, at 2 gates rather than 6.

The diagram below shows the six gates of SG and the two gates of RG3. The target's pad r_a stays untouched during SG, while RG3 puts the same mask on both wires of a pair.

Notice that within our *RGis*, we no longer need to consider freed versus paired wires. In other words, we do not need to consider whether an auxiliary wire r_i is currently paired with some computation wire w_i . In the past, we would attempt to update r_i , which meant that we needed to ensure that r_i wasn't paired in order to update it. In other words, we could only update the “free” wire, and we had to schedule which wire that would be. This is no longer a problem. It also means that we can get away with only an additional n auxiliary wires, rather than the $n + 1$ the older design needed to keep one free.

The paired gadgets: SG and RG3

SG: one logical r57, six physical r57 gates



RG3: refresh both shares

$$m = r_1 \vee \neg r_2$$

$$(s \oplus m) \oplus (r \oplus m) = s \oplus r$$

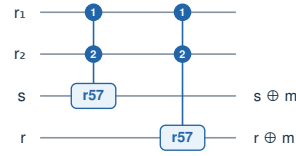


Figure 41: The six-gate SG and two-gate RG3 circuits

Putting the gadgets together

Recall that when integrating SAMFs into our earlier mixer, we didn't need to have a full wire shuffle every single time. While this indeed provides us with the most randomness, it also incurred a huge gate blowup. Similarly, we do not need to use every single *RGi* at every point in the integration process. The gadgetizer is thus broken up into five blocks: randomization, pairing, gadgets, unpairing, randomization. The final randomization block is there to make the gadgetized circuit look more symmetric. In addition, the randomization blocks add padding which is meant to make it harder for an adversary, after our mixing, to determine the original rewiring. It is not clear whether this is actually needed or not, as the mixing afterwards may already hide this information.

Before any of it runs we also shoot the original gates left and right past non-colliding neighbors. This changes the visible gate order while preserving functionality, so two gadgetizations of the same input do not begin from the same ordering.

The **randomization blocks** are what supply the algebraic degree. Every gate's active wire is drawn round-robin from the auxiliary half, each auxiliary wire being targeted once per round before any wire is targeted twice, while both control wires are drawn uniformly from all $2n$ wires. Restricting the active wire to the auxiliary half is what makes these blocks safe: they cannot disturb the lower n outputs at all. Drawing controls from all $2n$ wires is what makes them useful, since it lets the auxiliary values become complicated functions of both

the original and the auxiliary inputs rather than of the auxiliary inputs alone. We note that this follows the cycling schedule of the degree-raising construction from the previous section but does not chain the previous active wire as a control; the controls are simply random.

The **pairing block** is where a computation wire and an auxiliary wire get bound together into a pair. We do this with a block W_i , which takes four wires (q_0, q_1, q_2, q_3) and produces $(q_2, q_3, q_0 \oplus q_1, q_1)$. The following 11 gates compute this: $[[0, 3, 2], [3, 2, 1], [1, 3, 2], [2, 0, 1], [2, 1, 0], [0, 1, 2], [0, 2, 1], [1, 0, 3], [3, 0, 1], [3, 1, 0], [1, 3, 0]]$. It was found by a meet-in-the-middle search over r57 gates, and since every r57 gate is its own inverse, reversing the sequence gives W_i^{-1} , which is what the unpairing block uses.

To see what this does, call W_i with q_0 the wire currently holding the computation value, q_1 the wire holding its auxiliary value, and q_2, q_3 two randomly chosen destination wires. Afterwards q_2 holds $q_0 \oplus q_1$ and q_3 holds q_1 , so the pair (q_2, q_3) XORs back to the original computation value: the value has been encoded, and it has been moved. Meanwhile the old contents of q_2 and q_3 have landed on q_0 and q_1 rather than being destroyed.

The reason we use one block for this rather than separate XORs and swaps is exactly that it does both at once. It pairs a single computation wire with a single auxiliary wire while simultaneously relocating the other wires involved, so that after the pairing block the computation wires are no longer sitting on the first n wires and the auxiliary wires on the latter n . They are mixed throughout all $2n$. Of course, this can be done with separate XORs and SAMFs, but for three relocations we would need 6×3 gates. It is not clear though whether these W_i are better than simply using our SAMFs.

We need to choose these destinations carefully. Since W_i relocates wires, a fixed choice of destinations can have one W_i overwrite a wire which a later W_i still needs. We therefore keep a live record of where every unencoded computation value, every auxiliary value, and every pair currently lives, and choose each W_i 's destinations against that live map. The unpairing block does the same in reverse, decoding values onto output wires in increasing order so that a finished output is never touched again. In the special case where a pair's share or pad already sits on its output wire, we skip W_i^{-1} entirely and just XOR the other half in, which takes 4 gates instead of 11.

In this paired gadgetizer's **gadgets block**, every gate of the original circuit becomes one SG, and every f original gates we insert one RGi drawn uniformly from $\{RG1, RG2, RG3\}$. Here f is the refresh spacing. We use $f = 2$ in the benchmarks below, meaning two original gates are simulated and then one rerandomization gadget is inserted. The pairs (i, j) that $RG1$ and $RG2$ act on, and the single index i that $RG3$ acts on, are drawn from shuffled queues, so that every pair gets its turn before any pair repeats. This spreads rerandomization across the whole state rather than hitting the same few values over and over.

We note that the gadgetizer’s internal pairing map changes after RG1 and RG2, while the represented computation does not. The next SG reads the updated map and continues the original computation on the newly represented values. Once gadgetization is over, none of this map is saved. The only output is a sequence of r57 gates, so an adversary cannot simply look up which physical wires formed which pair.

Some gate blowup benchmarks

The randomization portion costs $\max(2n \lfloor \ln n \rfloor, 64)$ gates per bookend, and since there is one on each side, this contributes about $4n \lfloor \ln n \rfloor$ gates in total. Of course, this is only our estimate on the number of gates necessary in order to get enough randomness for our mixing. We note the floor: at $n = 128$ this is 1024 per bookend, not 1243.

The pairing is done with the W_i block, which costs exactly 11 gates per value, so $11n$ per side. The unpairing side is at most $11n$, since the 4 gate special case above fires occasionally, so $22n$ for both sides is a good approximation rather than an identity.

Each SG costs 6 gates, so if there are m original computation gates this yields a $6m$ blowup. $RG1$ and $RG2$ take 6 gates each while $RG3$ takes 2, so an RGi costs $\frac{6+6+2}{3} = \frac{14}{3}$ gates on average. We insert one every f original gates, so the total gate blowup is approximately

$$4n \lfloor \ln n \rfloor + 22n + \left(6 + \frac{14}{3f}\right)m.$$

At $f = 2$ the coefficient on m is $\frac{25}{3} \approx 8.33$; at $f = 3$ it is $\frac{68}{9} \approx 7.56$.

As a sanity check, on $n = 256$ with $m = 6768$ and $f = 2$, the formula predicts $5120 + 2816 + 2816 + 40608 + 15792 = 67,152$ gates. Two recorded runs produced 67,104 and 67,184 gates on 512 wires, both within half a standard deviation of the RGi draw.

Where the linearity hurts

Everything above is an improvement on the 12 gate gadget, and it does what we asked of it: the auxiliary wires carry high algebraic degree, and that degree is present on every carrier wire throughout the body of the circuit. However, the sharing itself introduces a weakness at the center of the construction, and it is not a mixing deficiency that additional rounds can fix.

A logical value v lives on two carrier wires as $v = c_0 \oplus c_1$. Since the decode is an XOR, it is *linear*, and so at every instant every logical value of the original circuit C is a degree-1 affine function of the gadgetized circuit’s wires. Thus, an adversary who fits affine functions can read C ’s intermediate state out of the gadgetized circuit at any point, regardless of how algebraically complex that

state is as a function of the input. Although the previous section raised the degree of the individual wires, the linear decode still exposes the low degree logical state.

We recall that the affine reconstruction heatmap does not compare physical wires directly. At each pair of prefixes, it asks whether the logical state of C can be recovered as an affine function of the gadgetized circuit’s current wires under the declared auxiliary-input policy. The paired decode $v = c_0 \oplus c_1$ gives the attacker exactly such a relation, regardless of the carriers’ degree in the original inputs. We now apply that test to this gadget.

Under the linear sharing, affine heatmap reveals **diagonal**. It is the same diagonal we have been fighting since the very beginning of this document, except now it is drawn in a metric that our shuffles and bit-flips do not touch. This means that with SAMFs and our linear gadgetization, we are effectively not hiding our circuit at all.

The difficulty we found in removing this diagonal motivates the next stage of our construction. No matter the expansion/compression method, we were not able to get rid of this diagonal. The reason is structural: mixing re-randomizes *which* wires carry the sharing, but the value stays affine in the new wires, and so the affine adversary can simply fit it again. Wire permutations, NOTs, CNOTs, and SAMFs are all affine changes of basis, while a degree-1 predictor is invariant to that entire class. Similarly, every re-randomization in our gadgetizer, including RG1, RG2, RG3, and both randomization bookends, is affine or affine-preserving on the carrier pair. Thus, these methods cannot fix this leak by construction.

The secret share itself unable to help us here. An affine adversary obtains every degree-1 term at no cost by XORing it into its predictor, and so the second carrier does not add any cost for the adversary. In other words, the secret-sharing scheme above gives us no protection against a degree-1 attacker.

This also gives us a constraint on how we can fix the problem, as we cannot simply remove the XOR pair and replace it with something nonlinear. A pure two-wire product decode is unbalanced: over the four states of (w_p, w_q) , the value 1 has exactly one preimage and 0 has three. No reversible gadget can conditionally flip between classes of unequal size, since a bijection cannot map a 3-element class onto a 1-element class, and this holds for every choice of constants, with any ancillae or garbage. Thus, the linear part is structurally necessary because it re-symmetrizes the classes.

Thus, the diagonal cannot be removed by tuning the mixer. Since the leak comes from the linear decode, we need an encoding whose decode is itself nonlinear. We discuss this construction in the next section.

Nonlinear Product-Share Gadgetization

The first nonlinear construction replaces the paired XOR decode with a carrier masked by products of band values. We describe its decode, Gray fold,

and measurements before moving on to the attack which motivates embedded masking.

Let the circuit entering gadgetization have q logical wires. The nonlinear gadgetization uses $2q$ physical wires: one carrier for each logical value and one band wire for each carrier. Thus, this step doubles the width of the circuit entering gadgetization. The carriers hold the encoded computation, while the band wires supply the nonlinear product terms used to mask it. During gadgetization, a ledger records where every carrier and band variable currently lives, which mask terms belong to each logical value, and each value's constant, chosen when the gadget is constructed.

The product-share decode

At an ordinary gadget boundary, each logical value V_i is represented as

$$V_i = C_i \oplus M_i \oplus \kappa_i,$$

where C_i is the value's current carrier, κ_i is a constant recorded in the ledger, and

$$M_i = \bigoplus_j \prod_l (B_{bjl} \oplus a_{jl})$$

is its product mask. Each B_b names a band variable, while each a_{jl} is a fixed bit which chooses whether that band variable is used normally or negated. The mask slots name band variables rather than physical wires. When a gate is emitted, the ledger resolves each variable to the wire which currently holds it. The offsets and κ_i never need their own physical wires.

The nonlinear band fill

The second half of the physical wires initially contains incoming junk $z_1, \dots, z_q$. We turn these wires into band variables $B_1, \dots, B_q$ in order. Each band variable has the form

$$B_j = z_j \oplus x_{p_j} \oplus L_j \oplus P_{j1} \oplus P_{j2} \oplus \delta_j,$$

where x_{p_j} is a randomly chosen pivot data wire. In the construction used here, L_j is the XOR of between one and seven additional data wires, limited by the number available, and P_{j1} and P_{j2} are two-literal products. The two sources within one product are distinct, either literal may be negated, and a source may be a data wire or an earlier band variable. The fixed bit δ_j accounts for a possible complement introduced by the chosen g57 spelling of those products. It is determined when the gadget is generated and can be absorbed into the auxiliary junk value for purposes of the examples below.

The pivot appears exactly once, as the linear term x_{p_j} . It cannot appear in L_j , in either product directly, or inside the transitive support of an earlier band variable used by a product. Therefore, after fixing every other source, flipping the pivot always flips B_j . This makes the individual band value balanced even though the rest of its expression is nonlinear. The band variables are constructed in order, so B_1 can use only data wires, B_2 may also use B_1 , and in general B_j may use any earlier B_r which passes the pivot-support check. Different band variables may choose the same pivot.

These choices are random when the gadget is generated, not while it is being evaluated. Each B_j is computed in place by XORing its selected terms into the wire which initially holds z_j . Once filled, it is a fixed physical bit until an explicit retire-and-refill step changes it. In particular, later source-gate updates to the logical values do not retroactively change the band or require every mask using it to be updated.

Mask injection

After filling the band, we draw four product terms for each logical value: three products of two band literals and one product of three band literals. A literal may be either B_j or $\neg B_j$. XORing the four terms assigned to V_i gives its mask M_i . In particular, the three-literal term gives the decode degree 3 in the band variables; it is not restricted to pairwise products.

The circuit injects the four terms into V_i 's carrier one at a time. If the carrier initially contains V_i , it contains $V_i \oplus M_i \oplus \kappa_i$ after the injection. The full value of M_i is never collected on another wire: the ledger only records its four terms and resolves their current wire locations whenever it emits a gate. After the band injection, the first q wires hold the masked carriers and the latter q wires hold the band values.

Gray Folding Source Gates

Suppose a source gate updates a logical target as

$$V_t \oplus = f(V_{i_1}, \dots, V_{i_r}).$$

We write f in algebraic normal form and substitute the complete decode of every control. Each control contributes its carrier, its individual product mask terms, and its ledger constant. Distributing the expression produces a collection of conjunction fragments, each of which is XORed into the target carrier. A product mask is therefore used only through its band literals as part of a fragment's controls; neither the mask nor the decoded logical value is first reconstructed on a physical wire.

For a two-control gate, write the complete control expressions as

$$S_b = L_b \oplus M_b \quad \text{and} \quad S_c = L_c \oplus M_c,$$

where L_b and L_c contain the carrier literal and any ledger constant, while M_b and M_c are the XORs of their product-mask terms. The product part of the source gate is then

$$S_b S_c = L_b L_c \oplus L_b M_c \oplus M_b L_c \oplus M_b M_c.$$

A direct expansion would form these contributions by repeatedly pairing the individual mask terms. The Gray fold instead gathers each complete mask sum once and reuses it. It borrows two *dirty* wires u and z , whose incoming values u_0 and z_0 are unknown and are not cleared. Call the four working states

$$\begin{aligned} A &= (u_0, z_0), & B &= (u_0 \oplus M_b, z_0), \\ C &= (u_0 \oplus M_b, z_0 \oplus M_c), & D &= (u_0, z_0 \oplus M_c). \end{aligned}$$

The circuit moves through them as

$$A \xrightarrow{\text{gather } M_b} B \xrightarrow{\text{gather } M_c} C \xrightarrow{\text{strip } M_b} D \xrightarrow{\text{strip } M_c} A.$$

This is a Gray-code order because only one gathered mask changes between two adjacent states. While moving through the cycle, the circuit XORs several products into the target carrier. One valid placement of those products works as follows.

Recovering $L_b M_c$. In state A , the z accumulator contains z_0 ; in state D , it contains $z_0 \oplus M_c$. XORing $L_b z$ into the target once in each of these states contributes

$$L_b z_A \oplus L_b z_D = L_b z_0 \oplus L_b (z_0 \oplus M_c) = L_b M_c.$$

The two copies of $L_b z_0$ cancel because addition is XOR.

Recovering $M_b L_c$. In state A , the u accumulator contains u_0 ; in state B , it contains $u_0 \oplus M_b$. XORing $u L_c$ into the target in both states contributes

$$u_A L_c \oplus u_B L_c = u_0 L_c \oplus (u_0 \oplus M_b) L_c = M_b L_c.$$

Here the two copies of $u_0 L_c$ cancel.

Recovering $M_b M_c$. The circuit XORs uz into the target once in every state. The combined contribution is

$$\begin{aligned}
u_A z_A \oplus u_B z_B \oplus u_C z_C \oplus u_D z_D &= u_0 z_0 \oplus (u_0 \oplus M_b) z_0 \\
&\quad \oplus (u_0 \oplus M_b)(z_0 \oplus M_c) \oplus u_0(z_0 \oplus M_c) \\
&= M_b M_c.
\end{aligned}$$

After expansion, $u_0 z_0$ appears four times, while $u_0 M_c$ and $M_b z_0$ each appear twice. All of them cancel, leaving only $M_b M_c$. Finally, $L_b L_c$ does not use either accumulator, so it is XORed into the target once at any state. All four required terms have now been added to the same target carrier; the circuit never has to place their intermediate XOR on another wire. We may randomize which equivalent bare and gathered states receive the $L_b z$ and $u L_c$ contributions, but the same cancellation argument applies.

Thus, *folding* refers to obtaining the complete product from differences between these dirty states rather than reconstructing either mask on a clean wire. The final two transitions strip the gathered masks and restore both borrowed wires exactly. A degree-3 mask term needs a temporary dirty helper while it is gathered, and that helper is also restored. Fixed complements from the g57 spellings are absorbed into the fold’s constant atoms, which are taken from the ledger.

The dirty accumulators are restored, but an observer can still compare their values before and after a gather. The unknown starting value then cancels and reveals the gathered mask. We explain this attack after the construction, in Linear Correlators Across the Entire Circuit. Thus, the Gray fold solves the problem of wide gates while leaving a problem for an observer who combines different times.

We also tested a version without the Gray fold. It expands the products one term at a time and uses dirty helpers to break only the three- and four-control fragments into two-control gates. The borrowed-wire circuits below show these replacements. In a matched 64-wire experiment, this made 92.22% of all gates reachable by the frozen regular table, versus 97.37% with the Gray fold, while never gathering a complete mask on one accumulator. Breaking down every wide fragment reached only 92.37% at substantially higher cost. Applying our fragment compressor before database mixing reduced the selective version to 77.96% reachability. These measurements compare variants of the product-share construction; they do not measure embedded masking.

That product-share construction uses this Gray fold for suitable two-control source gates. Other gate shapes use the same full-decode substitution but fall back to the direct fragments or the dirty-helper construction. In either case, constant contributions update κ_t , and the target’s mask does not have to change merely because its logical value changed.

The purpose of this Gray folding is that it uses fold gates with at most two controls, which keeps them close enough to r57 for the frozen database to replace them during the later mixing. It does not make the complete circuit r57, nor

does it need to. It only needs to keep the fold from creating a large class of gates which the database cannot reach.

We can see the difference by comparing the Gray fold with the earlier direct expansion on the same $n = 128$ sandwich and source circuit. The direct expansion emits one gate for every cross-product of mask atoms, so its control count grows with both the source-gate arity and the mask degree. The Gray fold replaces those wide gates with the gather, read, and restore cycle described above.

	direct wide fold	Gray fold
gadget gates	339,786	808,618
fold gates above two controls	153,421	0
gates reachable by the frozen database	31.55%	95.47%

The later product-share variant measured in that campaign was cheaper and slightly more reachable than the Gray column above, at 692,653 gates with 97.53% of the circuit inside the database’s reach. The important number is reachability, not the raw database match rate: a gate outside the eligible width can never be re-encoded regardless of how many replacements the table stores.

Thus, the Gray fold makes the product-share encoding easier for the database to reach. The nonlinear encoding removed the affine diagonal in the local-state tests we ran, while the fold kept it from producing wide fragments which would pass through the mixing stage unchanged. Neither result rules out a recovery which combines different times.

A four-wire example

Let us walk through a simplified example. Suppose the source computation begins with four logical wires x_1, x_2, x_3, x_4 . Gadgetizing this computation gives us eight physical wires, $C_1, \dots, C_8$. Before the band fill, they contain

$$C_1 = x_1, \quad C_2 = x_2, \quad C_3 = x_3, \quad C_4 = x_4$$

and

$$C_5 = z_1, \quad C_6 = z_2, \quad C_7 = z_3, \quad C_8 = z_4,$$

where the z_i are the incoming auxiliary junk values. The fill targets C_5, C_6, C_7, C_8 in order, turning them into B_1, B_2, B_3, B_4 . One valid draw is

$$\begin{aligned} B_1 &= z_1 \oplus x_2 \oplus x_4 \oplus x_1 x_3 \oplus x_1 x_4, & \text{pivot } x_2, \\ B_2 &= z_2 \oplus x_2 \oplus x_3 \oplus x_1 x_4 \oplus x_3 x_4, & \text{pivot } x_2, \\ B_3 &= z_3 \oplus x_4 \oplus x_1 \oplus x_1 x_2 \oplus x_2 x_3, & \text{pivot } x_4, \\ B_4 &= z_4 \oplus x_1 \oplus x_4 \oplus x_2 x_3 \oplus x_2 x_4, & \text{pivot } x_1. \end{aligned}$$

For example, B_1 starts with the junk value z_1 . The construction XORs in pivot x_2 , the extra linear source x_4 , and the two embedded terms x_1x_3 and x_1x_4 . Nothing besides the pivot term depends on x_2 . Likewise, the remaining expressions exclude their stated pivots from every other term. The first two bands deliberately show that pivots do not have to be unique: both may use x_2 .

This particular four-wire draw uses only data wires because each displayed band ends up depending on all four data inputs. At larger widths, an eligible earlier band may be used as one source of a later product, and this is how the band fill can rise above degree 2. For example, suppose an earlier band contains an embedded term x_ax_b and a later fill includes the product B_rx_c . Expanding just that product gives

$$B_rx_c = (\cdots \oplus x_ax_b \oplus \cdots)x_c = \cdots \oplus x_ax_bx_c \oplus \cdots.$$

For distinct x_a, x_b, x_c , the later band can therefore contain a degree-3 monomial in the original data inputs, provided that another contribution does not cancel it. Further eligible uses of earlier bands can raise the degree again, even though each individual fill gate uses only two source literals. After all four fills in our concrete example, the physical wires contain

$$x_1, x_2, x_3, x_4, B_1, B_2, B_3, B_4.$$

We next draw product mask terms for each logical value. A band literal may be either B_j or $\neg B_j$. For instance, one possible toy mask for the first value is

$$\begin{aligned} M_{11} &= B_1B_2, & M_{12} &= (\neg B_1)B_3, \\ M_{13} &= B_2(\neg B_4), & M_{14} &= B_1(\neg B_3)B_4, \end{aligned}$$

and

$$M_1 = M_{11} \oplus M_{12} \oplus M_{13} \oplus M_{14}.$$

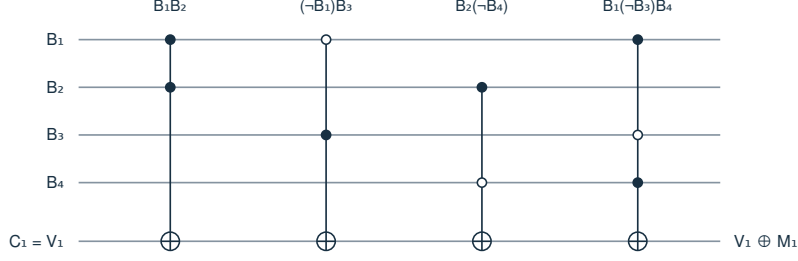
Here is the injection of these four terms. The cubic gate is drawn as one logical fragment; the dirty-helper construction below shows how to implement it using two-control gates.

The first three terms have degree 2 in the band variables. The last has degree 3. In particular, treating a negated literal as $\neg B_3 = 1 \oplus B_3$ gives

$$M_{14} = B_1(\neg B_3)B_4 = B_1B_4 \oplus B_1B_3B_4.$$

Thus, M_{14} contains the cubic monomial $B_1B_3B_4$. This degree is measured in the physical band variables. Since the B_j may themselves be nonlinear functions

Product-share mask injection



Each term goes straight into the carrier; M_1 is never gathered on a separate clean wire.
 Logical fragments shown here. The cubic fragment can be lowered with a dirty helper; κ_1 is omitted.

Figure 42: The four product terms injected into one carrier

of the incoming data, the mask can have still higher degree when it is expanded all the way back into the original inputs.

The other values receive their own independently drawn masks M_2, M_3, M_4 . Ignoring the ledger constants for one line, injecting the masks leaves the eight physical wires as

$$x_1 \oplus M_1, x_2 \oplus M_2, x_3 \oplus M_3, x_4 \oplus M_4, B_1, B_2, B_3, B_4.$$

More precisely, if V_i denotes the current logical value, then at every ordinary gadget boundary we maintain

$$V_i = C_i \oplus M_i \oplus \kappa_i,$$

where κ_i is the constant recorded in the ledger. Neither V_i nor M_i is reconstructed on a physical wire.

Now suppose the next source gate is

$$V_1 \oplus = V_2 \vee \neg V_3.$$

At the level of the decode invariant, its control function is

$$F = (C_2 \oplus M_2 \oplus \kappa_2) \vee \neg(C_3 \oplus M_3 \oplus \kappa_3).$$

Equivalently, over $GF(2)$,

$$F = 1 \oplus (C_3 \oplus M_3 \oplus \kappa_3) \oplus (C_2 \oplus M_2 \oplus \kappa_2)(C_3 \oplus M_3 \oplus \kappa_3).$$

The ledger constants are known when the gadget is generated, so we substitute them and simplify the expression. To display the nonconstant parts clearly, suppose for one line that $\kappa_2 = \kappa_3 = 0$. Algebraically, the required carrier update is then

$$\begin{aligned} C'_1 &= C_1 \oplus 1 \oplus C_3 \oplus M_3 \oplus C_2 C_3 \\ &\quad \oplus C_2 M_3 \oplus M_2 C_3 \oplus M_2 M_3. \end{aligned}$$

This shows each carrier, mask, and mixed contribution which must be folded. If

$$M_2 = \bigoplus_a M_{2a} \quad \text{and} \quad M_3 = \bigoplus_b M_{3b},$$

then the last three parts expand further as

$$\begin{aligned} C_2 M_3 &= \bigoplus_b C_2 M_{3b}, \\ M_2 C_3 &= \bigoplus_a M_{2a} C_3, \\ M_2 M_3 &= \bigoplus_{a,b} M_{2a} M_{3b}. \end{aligned}$$

Every M_{ia} in these equations is one specific product of two or three band literals. These equations describe the function to be folded, but the circuit does not first place M_2 , M_3 , or either decoded control on a clean wire. On a circuit with enough spare carrier wires, the Gray fold introduced above gathers the masks onto dirty accumulators, walks the four-state cycle, and emits the carrier-mask and mask-mask contributions while the unknown incoming values cancel. More precisely, a g57 spelling may leave a fixed residual, so the accumulators can hold $u_0 \oplus M_2 \oplus \delta_2$ and $z_0 \oplus M_3 \oplus \delta_3$. The fold compensates for these residuals through the controls' constant atoms and restores every borrowed wire exactly.

The four-wire example is meant to show the algebra of the fold. A literal four-carrier circuit does not leave enough spare carrier wires for both dirty accumulators and the helper used by degree-3 terms, so that tiny instance uses the direct expansion with dirty helpers. At the wider sizes where the Gray path is available, it computes exactly the expansion shown here without placing a control or complete mask on a clean wire.

The constant pieces, including the leading 1 above, are absorbed into κ_1 . Thus, the physical updates are arranged so that

$$C'_1 \oplus \kappa'_1 = C_1 \oplus \kappa_1 \oplus F,$$

while M_1 is unchanged. Decoding after the fold gives

$$\begin{aligned}
V'_1 &= C'_1 \oplus M_1 \oplus \kappa'_1 \\
&= C_1 \oplus M_1 \oplus \kappa_1 \oplus F \\
&= V_1 \oplus (V_2 \vee \neg V_3),
\end{aligned}$$

which is exactly the source gate. The controls were used only through their complete encoded expressions, so neither V_2 nor V_3 was ever revealed on an individual physical wire. We also did not update M_1 merely because the logical value changed: the same mask now hides the new value carried by C'_1 .

Re-masking is a separate maintenance step. Suppose we later replace the degree-2 term M_{12} by a fresh degree-2 term M'_{12} . At the level of the decode, the new carrier and mask are

$$C''_1 = C'_1 \oplus M_{12} \oplus M'_{12}, \quad M'_1 = M_1 \oplus M_{12} \oplus M'_{12}.$$

Therefore,

$$C''_1 \oplus M'_1 = C'_1 \oplus M_1,$$

so the logical value does not change. The circuit emits the fresh term first and strips the old term second, which means that the value is never left with less masking in between. Similarly, before retire-and-refill rewrites a band variable, every live mask which names that variable is re-sourced in this way.

Moving and refreshing the representation

The construction also changes where the pieces of the representation live. A carrier relocation swaps the physical locations of two logical carriers and updates the ledger. A band roll swaps a band variable with another physical wire and updates both the band-location map and the carrier map when necessary. The values and mask formulas do not change during either operation; only their locations do. Because mask slots name band variables rather than wire numbers, later folds, re-sources, and strips automatically use the new locations.

A retire-and-refill step changes the band value itself. Before rewriting a band variable, the construction finds every live mask term which names it and re-sources that term onto a fresh product. Only after the old variable has no remaining references does the refill change its wire. This mid-computation refill does not rebuild the initial fill from z_j . Instead, it XORs one fresh source and a small number of fresh two-source products into the current band wire, drawing those sources from other band variables and current carriers. The order is important: inject each replacement mask term first, remove the old term second, and refill the band variable only after all references are gone.

Leaving the gadget

At the end of the gadget, the circuit strips each recorded mask term from its carrier using the band variables' current locations and discharges the remaining ledger constants. It then applies a routing permutation over the whole physical wire space, placing the decoded logical values on their required output wires and the band variables on the auxiliary wires. Finally, another nonlinear fill is emitted over the auxiliary side so that those outputs contain fresh junk rather than an exposed copy of the working band. The required outputs therefore compute the same function as the source circuit, while the other outputs remain unrestricted junk. Of course, further mixing with our database replacements will be needed to ensure these remain hidden.

Linear Correlators Across the Entire Circuit

The nonlinear gadgetization above removes the simple XOR decode which hurt our paired construction. However, we recall what the last linear correlator heatmap actually checks. Each cell tries to recover a source value from the wires at one point in the gadgetized circuit. Even when the heatmap covers the entire circuit, each recovery is local to one state. It does not combine the wire values at different points in the computation.

An observer of our circuit does not have this restriction. They can run the published gate list and keep the intermediate values. Thus, we should allow the correlator to use observations from across the entire circuit. Suppose $Z_w^{(j)}(x)$ is the value of physical wire w after gate j , and $v(x)$ is one of the source circuit's intermediate values. We now ask whether

$$v(x) = \kappa \oplus \bigoplus_{j,w} \alpha_{j,w} Z_w^{(j)}(x)$$

for fixed coefficients $\kappa, \alpha_{j,w} \in GF(2)$. The coefficients may select different wires at different times, but they must work across the inputs. In other words, we have changed the information available to the attacker, while still only asking for a linear combination of that information.

This is particularly problematic for the Gray fold. Recall that its dirty accumulator starts at u_0 and then becomes $u_0 \oplus M_b \oplus \delta$, where δ is the fixed complement from its gate spelling. Neither state needs to contain the mask by itself. However,

$$u_{\text{before}} \oplus u_{\text{after}} = M_b \oplus \delta.$$

The unknown dirty value cancels. Combining these two observations with the entry carrier C_b therefore gives

$$V_b = C_b \oplus u_{\text{before}} \oplus u_{\text{after}} \oplus \delta \oplus \kappa_b.$$

This is an affine equation in three observations, even though M_b is a nonlinear function of the band. Splitting the gather into smaller pieces does not remove the problem: the observer can XOR together the changes from all the pieces. The same concern applies when individual product masks appear as gate firing bits. Once a nonlinear product is itself an observation, the attacker can use it as one term in a linear combination.

Recording the trace

We do not need to store the complete state after every gate to perform this attack. Let Δ_j be the firing bit of physical gate j . Since our gates only change one target,

$$\Delta_j = Z_{t_j}^{(j-1)} \oplus Z_{t_j}^{(j)},$$

and every later wire value can be recovered as

$$Z_w^{(r)} = Z_w^{(0)} \oplus \bigoplus_{\substack{j \leq r \\ t_j = w}} \Delta_j.$$

Thus, the initial wire values and all gate firing bits already span every intermediate wire value. We can use these as the features of our linear system. This is still a linear attack in the trace values; it does not claim that those values are linear in the original inputs.

We can display this attack as a cumulative heatmap. As we move through the gadgetized circuit, we add observations to one growing linear system. One version adds the full wire state at each sampled checkpoint. Another adds every gate's firing bit, giving the complete trace span. The last column therefore uses observations from across the circuit, rather than just its final state. Sampling only some checkpoints or gate deltas gives a restricted version of this attack.

As before, we fit the equations on one set of inputs and check them on separate inputs. We use three independent sets: fitting inputs to find an equation, validation inputs to reject spurious relations and choose a witness, and test inputs to measure that chosen witness without changing it. A long trace can have many more features than fitting samples. If the fitted span reaches the number of fitting samples, it can fit arbitrary labels on those samples. At that point, a failed test on new inputs is inconclusive: the procedure may have chosen a spurious equation instead of another one which generalizes. We cannot treat this as evidence that no affine equation exists.

Correlations without exact recovery

Exact recovery is only one kind of failure. A predictor which agrees with a source value on three quarters of the inputs fails an exact reconstruction test, but still reveals information. For a parity P of observed trace bits, we can write its signed agreement with v as

$$\rho = \mathbb{E}[(-1)^{v \oplus P}].$$

Choosing the better of P and its complement gives agreement $(1 + |\rho|)/2$. For instance, $|\rho| = 1/2$ gives agreement $3/4$. We should therefore look for biased linear correlators as well as exact linear equations.

To see why this matters for masking, suppose a carrier contains $v \oplus xy$, where x, y are independent uniform bits, independent of v . The product is zero on three quarters of the inputs, so the carrier already guesses v with agreement $3/4$. The decode is nonlinear, but the mask is biased. A wider product does not necessarily help: a product of three independent bits is zero seven eighths of the time.

The gadget gauntlet

We collect these tests in the **gadget gauntlet**. We begin with a chain of source `r57` gates, gadgetize it, optionally mix it, and compare the two executions on the same inputs. For each source gate $c \oplus = a \vee \neg b$, we record five target values:

$$a, \quad b, \quad c_{\text{old}}, \quad f = a \vee \neg b, \quad c_{\text{new}}.$$

On the gadgetized side, we record the initial wires and, for each physical gate, its firing bit and new target value. This includes values which exist only briefly. The main tests are:

Test	What the observer can combine
<code>a1</code>	One recorded value, or its complement, to recover a source value exactly.
<code>xrows</code>	Any affine combination of the wires at one physical-gate prefix. This is the local-state reconstruction test.
<code>xtrace</code>	Any affine combination of initial wires and gate firing bits across the complete recorded execution.
<code>w1</code>	Each individual trace feature, looking for statistical correlation.

Test	What the observer can combine
w2 , w3	Pairs and triples from selected trace features. These include XOR correlators and limited nonlinear combinations such as AND and OR.

The numbers in **w2** and **w3** count the features used, not the algebraic degree of every predictor. The statistical score is centered so that a predictor does not count as useful merely because the source firing bit is usually one. Namely, for the ± 1 versions of v and P , it measures

$$|\mathbb{E}[(-1)^{v \oplus P}] - \mathbb{E}[(-1)^v] \mathbb{E}[(-1)^P]|.$$

In the experiments below, we also run the same correlators against an independent random target. This gives us a noise baseline: a score can look nonzero merely because we have finitely many samples and try many predictors. A flag must exceed both the largest score against that random target for the same test and $6/\sqrt{N}$, where N is the number of correlation samples. The affine recovery tests check their equations on held-out samples as well. The unprotected circuit is included as a positive control, so that we can check that the attacks find the structure they are meant to find.

Of course, the gauntlet has limits. **w1** scans every recorded feature, but the usual **w2** and **w3** caps select 64 and 16 features respectively; they do not scan all pairs and triples of a large trace. **xtrace** also has a feature limit and explicitly reports when it is skipped. A skipped test is not a passed test. Small input domains can saturate the trace span, so the logical width, auxiliary-input policy, and sample counts must accompany the result.

The auditor knows the source values because it runs the source circuit. Finding a recovery equation demonstrates a relation with that computation; it is not already a blind reconstruction of an unknown circuit or a SAT preimage. We use it to find weaknesses in the gadget before asking the complete mixing method to hide it.

These attacks give us two concerns for the next construction. We want to compute without gathering a complete mask or temporarily making its decode linear. We also want the masks to be balanced and to change during the computation, so that simply observing a carrier or its change does not immediately give a useful correlator. This leads us to embedded masking.

embedded Masking

These concerns lead us to embedded masking. The idea is to leave embedded masks open on our computation wires and compute from inside them. We do

not gather a complete operand mask, and we do not first change the operand into a linear encoding just to read it. We also change the target mask during its update, so its before/after difference includes more than the source gate firing. These changes address particular relations found by the attacks above; we will return to the whole-trace test after describing the construction.

Let A be the circuit entering gadgetization, on q wires. We add a band of q wires, giving us $2q$ wires in total. In the complete mixing method, A is a sliced sandwich: a circuit on $2n$ wires which, when its n auxiliary inputs are zero, places $C(x)$ on one output half and allows the other half to contain junk. Thus, $q = 2n$ and the gadgetized circuit has $4n$ wires. The first q wires carry the masked computation, and the latter q wires hold the band values $B_1, \dots, B_q$. Their physical wire labels stay fixed.

Open embedded masks

Write V_w for a logical value and W_w for its current physical carrier. We maintain a collection $\mathcal{O}_w$ of masks which are *open* on that carrier:

$$V_w = W_w \oplus \bigoplus_{j \in \mathcal{O}_w} M_j(B).$$

We use masks of the form

$$M_j(B) = 1 \oplus B_y \oplus B_x B_y \oplus B_z.$$

Namely, we inject one r57 gate with band controls B_x, B_y , followed by the CNOT $W_w \oplus B_z$. Applying these gates again removes the mask as long as its band values have not changed. We call these open/close mask pairs locally geodesic identities, or LGIs.

The two gates on the left open one balanced mask. Repeating them on the right closes it. In this picture the three band values are unchanged between the open and close; we explain how to refresh them below.

The extra B_z matters. The r57 increment alone is one on three of its four inputs. Under independent uniform band values, a carrier under just this mask therefore remains correlated with the logical value. Adding a separate independent uniform bit makes the mask balanced while retaining its embedded term. This describes the mask as a function of its band variables; our actual band is derived from the input, so its variables are not assumed to be mutually independent.

We normally keep at least two masks open in the masked interior, with a rolling cap of three for ordinary mask placements. Temporary covers and replacement masks can exceed that cap. Before reading a thinly masked control, we add enough masks to reach the lower bound and keep those masks open afterward.

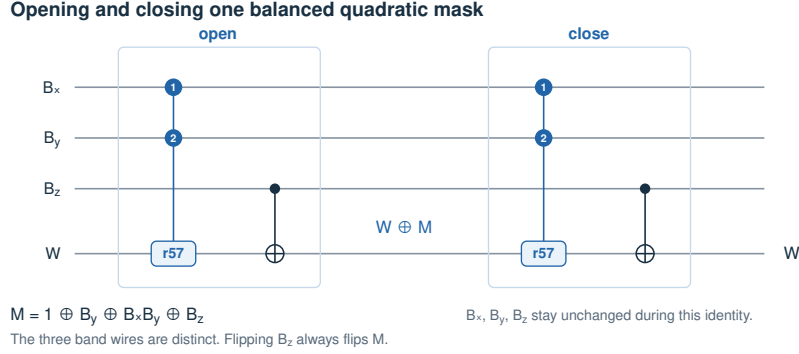


Figure 43: Opening and closing a balanced embedded mask

We try to choose their band wires disjointly from the other masks on that carrier. Small bands can force this condition to be relaxed.

Balancing alone does not settle the issue. A balancing wire is itself visible, and combining observations can cancel it. Keeping more than one mask open makes it harder for a single observed product to remove the remaining nonlinear part. This is why we check the mask depth throughout the computation, including around reads and band refreshes, rather than only counting how many masks we injected at the beginning.

Computing from inside the masks

Suppose the next logical gate is

$$V_t \oplus = 1 \oplus V_b \oplus V_a V_b.$$

Let $P_a = W_a \oplus M_a(B)$ and $P_b = W_b \oplus M_b(B)$ be the complete control decodes, where each M is the sum of that carrier's open masks. We expand the update

$$W_t \oplus = 1 \oplus P_b \oplus P_a P_b.$$

Each operand is embedded in the current physical variables, so their product has degree at most four. We emit its terms without first putting V_a or V_b on a physical wire. This is what we call **embedded fire**.

To see what changed, write $h(x, y) = 1 \oplus y \oplus xy$ for the r57 mask. The earlier read applied the reversed control pair because

$$h(x, y) \oplus h(y, x) = x \oplus y.$$

If a carrier held $v \oplus h(x, y)$, it now held $v \oplus x \oplus y$. The read was easier, but v had become affine in the current wires. Adding a balancing wire would still leave an affine expression, with that wire simply included in the XOR. Applying the reversed pair again after the read restores the mask, but cannot erase the observation. Later gate reorderings could also stretch this interval across the operand's idle time. embedded fire removes this linearization step and expands the still-embedded decode directly.

For the degree-three and degree-four terms, we borrow dirty band wires and restore their incoming values after each small block. In the gate lists below, $t \hat{=} a \& b$ means $t \oplus= ab$. For example,

```
t  $\hat{=}$  h & c
h  $\hat{=}$  a & b
t  $\hat{=}$  h & c
h  $\hat{=}$  a & b
```

adds abc to t and restores h , regardless of the initial value of h . For a degree-four term $abcd$, we use two dirty helpers h, k :

```
t  $\hat{=}$  h & k
h  $\hat{=}$  a & c
t  $\hat{=}$  h & k
k  $\hat{=}$  b & d
t  $\hat{=}$  h & k
h  $\hat{=}$  a & c
t  $\hat{=}$  h & k
k  $\hat{=}$  b & d
```

The four target updates visit all combinations of adding ac to h and bd to k . Their XOR is $(ac)(bd) = abcd$, and both helpers return to their incoming values. The cross-operand pairing matters: when ab and cd came from the two operand masks, we borrow ac and bd instead of collecting one operand's own mask term on a helper.

Thus, the fire uses gates with at most two controls and requires no additional clean scratch wires. Its surrounding slice guards can still contain three-control gates.

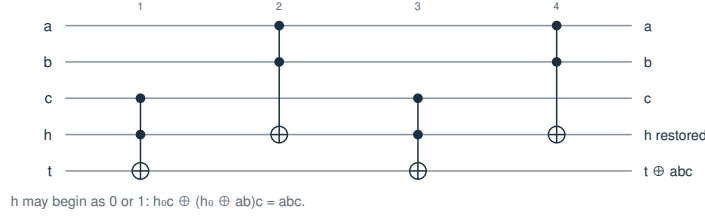
Both borrowed-wire constructions are shown below. All displayed source and helper wires are distinct. The intermediate helper values can change, but their incoming values are restored at the end of each complete unit.

Shuffling the computation

We treat each complete term block as a fire unit. Its internal order matters, but the complete units commute: each restores its helpers and adds its term to the same target. We divide the fire into two halves and independently shuffle the units within each half.

Dirty helpers for cubic and quartic terms

Cubic term: $t' = t \oplus abc$



Quartic term: $(ab)(cd)$, grouped across the operands as $(ac)(bd)$

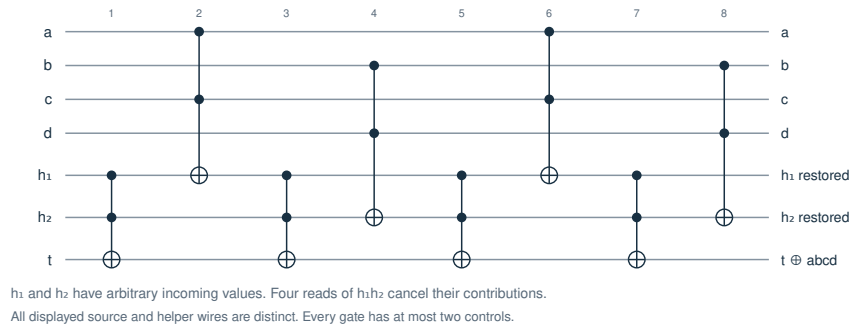


Figure 44: The four-gate cubic and eight-gate quartic dirty-helper circuits

We also open a temporary mask on the target before the fire and close it afterward. Between the two shuffled halves, we open a new persistent target mask. Thus, the order is

temporary cover ; first half ; persistent mask open ; second half ; cover close

Here T is the temporary cover and R is the new persistent mask. The two copies of T cancel, while R remains on the target.

Where space permits, these new masks avoid the band wires appearing in the operand polynomials. The temporary cover protects the target during the partial sum, while the persistent mask remains after the update. Consequently, the target's difference across this block includes both the logical firing increment and a new mask.

Filler masks are sampled throughout the computation, with target selection weighted by each wire's remaining mask budget. The logical gates are placed from a dependency-ready queue: reads remain after their earlier writes, and writes remain after earlier reads. Among gates whose dependencies have been satisfied, we take the one which entered the ready queue first. We randomize the surrounding masks, refreshes, and fire units; we do not randomly violate the logical dependencies. These are changes to gate ordering and mask placement, rather than the physical wire swaps of the earlier paired gadgetizer.

One quadratic fire, with a mask opened in the middle

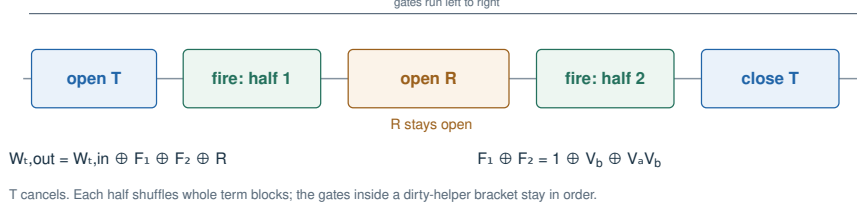


Figure 45: The target cover and persistent mask placed around the two fire halves

Band refresh and the two ports

During the compute, we spread refresh bursts across the mask and logical-gate placements. With m logical gates, we use $\lfloor m/8 \rfloor$ bursts of 16 updates each. A burst repeatedly updates one band wire by

$$B_j \oplus = \ell_a \ell_b.$$

Each ℓ is a wire value or its negation. We sample these literals from the band and the first n data wires, which initially hold the source input and change as the computation proceeds. Thus, these refreshes use both pools, whereas the dirty helpers inside a embedded fire are sampled only from the band. Before changing B_j , we open replacement masks where necessary, then close every mask which reads it. This preserves coverage without leaving a mask defined in terms of an obsolete band value. An optional repair refresh instead removes and reapplies the affected terms around the burst. The construction described here uses the replacement-and-close method.

The complete preprocessing has five parts:

opening guard ; band seed ; masked computation ; band reseed ; closing guard

The opening guard is a collection of gates targeting the first n data wires, each controlled by a positive literal from the outer band. Every such gate is dead when that band is zero. The outer band is the added masking register, separate from the sandwich's auxiliary register. We then seed each band wire with $x_a \oplus x_b$, choosing distinct wires from the original input prefix. After the masked computation, we close all remaining masks, leaving the sandwich output on its original q wires. A separately seeded collection of two-CNOT updates then reseeds the band from the first n data wires at that point. This is not an inverse fill: the band remains junk. Finally, the closing guard reads this band and changes only the sandwich's junk half.

We distribute the guards' slice controls across the band, shuffle those assignments, and shuffle the sampled gates. We sample the guards separately from

the masks used during computation, and make fresh choices for the initial band fill and the final refill. All these choices occur when we construct the circuit; evaluating its fixed gate list is deterministic.

For the sandwich which places the answer on the high half, the public promise is therefore

$$G(x, 0^n, 0^{2n}) = (\text{junk}, C(x), \text{junk}).$$

The circuit is reversible on all $4n$ wires, but we only constrain the answer block. The zero inputs specify the public slice: the input states on which the circuit must return $C(x)$ in that block. Neither junk output needs to be zero. The sandwich construction below explains why we separate the answer from the junk, and how the corresponding reverse slice behaves.

What the gauntlet still finds

The distinction between the local and whole-circuit attacks remains important for embedded masking. Removing the linearized read closes that particular local-state recovery. It does not show that every logical value has left the span of the complete trace.

We can see this directly before mixing. Each complete fire unit adds one polynomial term to the target and restores its helpers. Let J contain the physical gates which update that target inside the fire units, excluding the separate mask injections. Then the source firing bit satisfies

$$f = \bigoplus_{j \in J} \Delta_j.$$

The persistent target mask changes the XOR across the complete write block, but an observer who identifies its constituent gates can omit its updates from this sum. With public source inputs, recovering the logical firing bits also lets the observer follow the logical wire updates. Thus, hiding the before/after difference of one block is not enough to rule out every linear combination across its interior. Later mixing can change these intermediate values, so we must test the mixed circuit as well.

The following gauntlet experiments give a concrete example. Each circuit uses 128 logical wires, 128 band wires, and balanced embedded masks, with an ordinary open-mask cap of three. The source is a chain of 57 gates. Each source gate contributes the five target values defined above, so 16 gates give 80 targets and 64 gates give 320.

Source gates	Mixing moves	Physical gates	a1 hits	xrows hits	w1 / w2 / w3 flags	xtrace recoveries
16	0	5,546	0	0	0 / 0 / 0	77 / 80
16	20,000	24,537	0	0	0 / 0 / 0	77 / 80
64	20,000	38,424	0	0	0 / 0 / 0	314 / 320

Here a recovery means that the fitted equation had no errors on the 2,048 held-out samples. The fitting sets contained 11,648, 49,600, and 77,376 samples respectively. All three whole-trace scans completed below their 40,000-feature cap. The correlation scans used 16,384 samples, with the 64-feature pair cap and 16-feature triple cap described above. The target counts are per source gate and target role; they need not be distinct Boolean functions. In particular, recovering the same Boolean function at two source-gate positions counts as two recovered target instances.

The observed trace begins after the input values have been masked and ends before the outputs are decoded. Thus, it does not include the gates which expose the plain inputs or outputs. The initial band is uniformly random. These choices isolate the masked computation; the complete mixing method instead has public inputs and outputs and derives its band from the input. The 20,000 moves are local mixing moves applied to the gadgetized chain. They do not include the entire sandwich, splitting, crossing, and final compression construction. These are measurements of the stated experiments, and must not be treated as results for every choice of mask schedule, band construction, or subsequent mixing.

Nevertheless, they show the problem clearly: the direct, single-state, and bounded-correlation tests can all report zero while a linear combination across the entire execution still recovers most of the tested source values. Thus, these embedded-masking circuits have not passed the whole gadget gauntlet.

We can apply the same tests to a larger band or a different encoding. Such a comparison must keep track of the physical width, the input distribution, and which parts of the execution the observer can see. Otherwise, a change in the result may come from giving the attacker a different problem.

The Trapdoor Permutation Challenge

Every measure of success we have used so far asks a question about resemblance. The heatmaps ask whether the obfuscated circuit’s behavior still looks like the original’s, compression asks whether the gate list can be squeezed back toward the original’s length, and the affine reconstruction ridge asks whether a predictor

can read the original’s intermediate state out of the obfuscated wires. These are useful questions and we have learned a great deal from them, but they share a defect. None of them gives us a statement that an attacker can falsify by handing us an answer. A heatmap that finds no correlation is evidence that *we* found no correlation, and nothing stronger. The one exception in this document was the bounty, whose win condition was to produce an equivalent circuit of 1014 gates or fewer. Thus, when the bounty was broken, there was no ambiguity about the result. This clarity is what made the bounty useful to us, even though we lost.

For this reason, we wanted another challenge with a win condition an attacker can meet. This is a direct preimage challenge: given a published target, the attacker wins by returning an input which maps to it. The attacker does not need to locate or reconstruct the original circuit inside the published one. As we noted at the very beginning, iO is powerful enough to build almost everything else, including trapdoor permutations.

The construction is a “Feistel”-shaped wrapper around the circuit we are hiding. Given a random circuit C on n wires, we build TDP_C on $2n$ wires. C ’s functionality is moved onto the high half, wires $n..2n$, and a fresh random circuit D runs on the low half. On input (x, y) the construction returns

$$\text{TDP}_C(x, y) = (\text{junk}, y \oplus C(x)),$$

with the low half carrying material that has been pushed through D and is meant to look like nothing in particular.

From here, we note several consequences that pull in different directions.

Of course, forward evaluation is available to everyone. We publish the final gate list, and a gate list is a circuit, so anyone can run it on any input with no secret at all. Thus, the “easy to compute” half of a trapdoor permutation is given by construction rather than by argument.

The complete circuit can still be reversed if its complete output is known. The challenge reveals only the answer block, however, so an attacker would also have to find the missing junk before running it backward. Recovering a preimage from this partial output, without knowing C , is the problem we ask the attacker to solve.

Fixing $y = 0$ collapses the high-half output to exactly $C(x)$, so the whole challenge becomes the single sentence “find x with $C(x) = t$ ”. More generally, we can fix y to a public constant y^* . Given a target t for the high-half output, we then seek x with $C(x) = t \oplus y^*$. In other words, this only relabels the target.

Knowing the original circuit C gives the intended trapdoor. Every r57 gate is its own inverse, so we can find the preimage by running C ’s gate list backward on $t \oplus y^*$. The proposed hiding comes from publishing the transformed circuit while keeping that original gate list secret.

SAT Solvers

We introduced the general SAT test and the circuit-to-CNF translation in *How We Measure Success*, so we will not repeat that process here. Instead, we now look at how we used this test against the TDP campaign. In these instances, X is the only free input block.

The campaign encoding

We translate the circuit into a conjunction of Boolean clauses as described above. The important detail is that the transformed circuits do not contain only r57 gates. Different kinds of gates need different numbers of clauses, so two circuits with the same number of gates do not necessarily produce SAT formulas of the same size.

There is also one small optimization which matters for the numbers below. A zero-control gate with `comp` = 0 is an unconditional NOT. We can represent this by using the negation of the Boolean variable which already represents the target, without adding a new variable or any clauses. A zero-control gate with `comp` = 1 never fires and can be dropped entirely.

For r57 itself, $o = a \oplus (b \vee \neg c)$, the direct encoding is six clauses. Here $\neg b$ means $\neg b$, and `or` is Boolean OR:

```
(-b or  a or  o)
(-b or -a or -o)
( c or  a or  o)
( c or -a or -o)
( b or -c or -a or  o)
( b or -c or  a or -o)
```

These clauses separate the gate into its three possible cases. If $b = 1$, we enforce $o = \neg a$. We do the same if $c = 0$. If $b = 0$ and $c = 1$, the gate does not flip its target, so we instead enforce $o = a$. We previously used an eight-clause truth-table encoding, but this six-clause version is smaller and propagated better in our tests. We have not isolated exactly why it propagates better. Setting $b = 1$ or $c = 0$ turns one pair into binary implications between a and o , but the other assignments still leave ternary clauses.

The published circuits also contain gates from the wider vocabulary introduced earlier. Such a gate has the form $\text{fires}(x) = \text{comp} \oplus \bigwedge_i \text{lit}_i(x)$ with $x[\text{target}] \oplus = \text{fires}(x)$, for k mixed-polarity literals. We use one of two encodings depending on k .

The **direct** form extends the six clauses above. It uses two clauses for each control literal, followed by one final pair for the case in which every control literal is true. This gives $2k + 2$ clauses and one new variable. The **auxiliary**, or Tseitin, form first introduces a variable g for the complete conjunction. It enforces g using k binary clauses $\neg g \vee \ell_i$ and one wide clause $\bigvee_i \neg \ell_i \vee g$, and

then uses four clauses for $o = a \oplus \text{comp} \oplus g$. This gives $k + 5$ clauses and two new variables.

The direct form is smaller through $k = 3$. At $k = 3$, both forms use eight clauses, but the direct form still uses one fewer variable. For larger values of k , the auxiliary form becomes smaller. We therefore use the direct form through three controls and the auxiliary form for wider gates. We can also compare this with using the direct form throughout. In particular, the direct encoding of a complemented two-control gate gives exactly the six r57 clauses shown above.

This means that we cannot compare these constructions using their gate counts alone. The number of clauses per gate changes with the gate vocabulary, as the following table shows.

Table: Per-gate CNF cost at $n = 128$, by construction

build	gates	clauses	clauses/gate
no nonlinear gadgetization	57,471	288,512	5.02
gadgetized, no Gray fold	288,334	2,176,477	7.55
gadgetized + Gray fold	989,343	5,447,206	5.51

Gadgetization without the Gray fold increases the number of clauses per gate by about $1.5\times$. This happens because 42% of its gates have at least four controls and therefore use the auxiliary encoding. The Gray fold breaks these gates back down to at most three controls, bringing the average back to about 5.5 clauses per gate. The folded circuit has 384 zero-control gates, 248,464 one-control gates, 735,689 two-control gates, and 4,806 three-control gates. This hints at the fact that using gates with *more* controls, can be effective at increasing the number of clauses, further emphasizing the limitation of relying solely on r57 gates.

For a circuit with w input wires and g r57 gates, this gives approximately $w + g$ variables and $6g$ clauses, followed by the unit clauses which pin the inputs and outputs. Once the circuit has been fragmented, we instead count the clauses from the number of controls on each gate. For instance, the full $n = 128$ construction has 989,343 gates. Of these, 384 are zero-control gates which add no variables. The formula therefore has $768 + 988,959 = 989,727$ variables. Its gate clauses are $4 \cdot 248,464 + 6 \cdot 735,689 + 8 \cdot 4,806 = 5,446,438$, and the 768 unit clauses bring the total to 5,447,206.

The challenge that broke

One preimage challenge used three n -wire registers. The first held the source input x , the second held the public value y added to the answer, and the third held auxiliary values z used by the masking. Here $n = 128$, giving 384 physical wires. The input placed x on wires 0–127, y on wires 128–255, and z on wires 256–383. Its middle output block computed

$$\text{middle} = y \oplus C(x).$$

The published circuit had 1,672,129 gates, grown from an original circuit C with 2,000 gates. The six-clause encoding produced 1,672,513 variables and 10,033,158 clauses.

We then fixed $y = 0$ and $z = 0$ using 256 unit clauses, and used another 128 unit clauses to fix the middle output to the target. Since y was zero, the SAT problem became simply

$$C(x) = \text{target}.$$

The SAT solver Kissat returned the following preimage:

```
target = 0x73f3f19f19ed8618653ee7ca759811e7
x       = 0x1318d06b8d50e2324d1b935a584b7782
```

We checked this by running the 384-wire circuit on $z \| y \| x$ with $y = z = 0$. The middle 128 output bits matched the target.

We ran many tests of similar types and found that even with more and more expansion in the latter mixing stages, the longest average SAT solve time was only around 10 hours. We use average SAT solve times as our metric here as some targets are easier to find preimages for than others.

Making the Solver Struggle

These observations lead us to two changes. First, we define the preimage challenge on one public input slice, so the auxiliary wires are not additional valid input choices. Second, we move away from the uniform r57 gate vocabulary on which most of this document was built. The first change defines the anti-inversion problem that we want to ask, while the second changes how that problem is presented to an attacker.

Slicing and the sliced sandwich

The goal of slicing is to stop the SAT solver from freely manipulating the auxiliary wires when trying to solve for preimages in the TDP challenge. One additional property we get from slicing is it stops the reversal of the published circuit from immediately giving us C^{-1} . If an attacker knows the complete output of an ordinary reversible circuit, then they can simply reverse the gate list and recover the input. We instead reveal only the output block containing $C(x)$. The other output wires contain junk which is not part of the target, so running the circuit backward would also require finding that missing junk. This removes the trivial reverse-circuit attack, although it does not by itself prove that finding a preimage is hard.

We begin with a source circuit C on n wires and build a **sliced sandwich** A on $2n$ wires. The first n wires hold the free input x , while the second n wires hold an auxiliary slice register y . S_1 and S_2 will both serve to enforce the slices. In other words, they yield the identity when the slice matches, and will permute the input bits otherwise. The sandwich has the form

$$A = [C \text{ interleaved with } S_1] ; N ; [D \text{ interleaved with } S_2]$$

We first run C on the first half while interleaving it with a slice block S_1 . We then apply the copy step N , consisting of the n CNOTs $y_i \oplus = x_i$. Finally, we run an independent random r57 circuit D on the first half while interleaving it with another slice block S_2 . We give D the same gate design and gate count as C . For the random source circuits considered here, we choose this count to be $\max(n, \text{round}(n(\log_2 n)^2))$.

Each slice block has $\max(n, \text{round}(n \log_2 n))$ gates drawn from two shapes:

$x_i \wedge = y_j$	CNOT, control in the second half	~1/3 of gates
$x_i \wedge = x_j \ \& \ y_k$	CCNOT, one control per half	the rest

Every slice gate targets the first half and reads exactly one positive literal from the second half. Therefore, each gate is individually dead when $y = 0$. We randomly interleave S_1 through C and S_2 through D while preserving the internal order of C and D .

It is easiest to trace the sandwich before we move the copy CNOTs away from the middle. Forward on the public slice, we have

$$(x, 0) \longrightarrow (C(x), 0) \longrightarrow (C(x), C(x)) \longrightarrow (\text{junk}, C(x)).$$

Thus,

$$A(x, 0) = (\text{junk}, C(x)).$$

The diagram shows this public-slice computation before we scatter the copy CNOTs. Each horizontal line stands for a bundle of n wires.

S_1 is dead while C runs. The copy step then places $C(x)$ in the second half. After that, D and S_2 may change the first half, but they cannot change the answer in the second half.

On the same slice, the reverse circuit instead gives

$$(p, 0) \longrightarrow (D^{-1}(p), 0) \longrightarrow (D^{-1}(p), D^{-1}(p)) \longrightarrow (\text{junk}, D^{-1}(p)).$$

Therefore,

$$A^{-1}(p, 0) = (\text{junk}, D^{-1}(p)).$$

The classic sliced sandwich

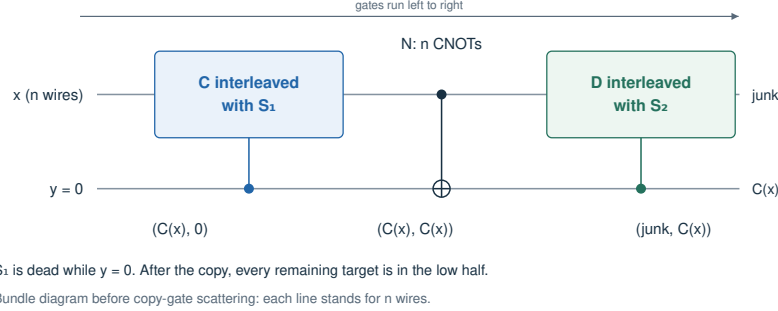


Figure 46: The classic sliced sandwich with its middle copy step

On this reverse slice, the reversed gate list exposes the random D^{-1} rather than C^{-1} . We give D the same design and size as C so that later mixing has two similar computations to work with. The copy step N initially creates an obvious column between them, so before gadgetization we assign each copy CNOT a random direction and commute it in that direction until it reaches a genuine collision. These moves preserve the function and both slice equations above, while removing the most obvious contiguous boundary.

We then gadgetize all $2n$ sandwich values. The embedded gadgetizer uses one carrier per value and a band of the same $2n$ width. We add an opening zero-slice guard controlled by this band, seed it from the input, perform the masked computation, reseed it, and add a closing guard on the junk half. This outer band slice is separate from the sandwich's n -wire register y . Thus, the width is

$$n \text{ source wires} \longrightarrow 2n \text{ sandwich wires} \longrightarrow 4n \text{ gadgetized wires.}$$

The resulting circuit is still a reversible circuit on every one of its $4n$ inputs. The circuit itself does not force any input to zero. Instead, the challenge and its verifier define the public slice by fixing both y and the outer band to zero. The complete layout is:

wires	input role	output role on the public slice
$0, \dots, n-1$	the free input x	junk
$n, \dots, 2n-1$	sandwich slice y , fixed to zero by the challenge	$C(x)$
$2n, \dots, 4n-1$	masking band, fixed to zero by the challenge	junk

On the complete public slice, the gadget G therefore satisfies

$$G(x, 0^n, 0^{2n}) = (\text{junk}, C(x), \text{junk}).$$

There are three limits to keep in mind:

1. When $y \neq 0$, the output has the form $y \oplus C'(x, y)$ because S_1 becomes live. The intended behavior is that these slices do not expose clean $C(x)$. The zero-slice equation follows gate by gate, but no per-circuit test proves that every off-slice input differs.
2. At the sandwich's functional level, D and S_2 affect only the unconstrained junk output. They add no target bits, although their encoded and mixed gate material can still change the SAT presentation. Measuring their benefit requires a matched sandwich-versus-no-sandwich comparison.
3. Fixing the auxiliary inputs prevents them from counting as additional valid preimages, but the broken SAT attack above already chose $y = z = 0$. The public pins also give the SAT solver constants which it can propagate. A separate matched pinned-versus-unpinned test is needed to measure the effect of the pins.

Slicing therefore defines the preimage problem and prevents the immediate reverse-circuit attack. It is not the defense against affine reconstruction. On the zero slice, the sandwich alone still runs C as plaintext; embedded masking is the layer which makes the internal decoding nonlinear.

Fragmentation: leaving r57 behind

Earlier, when we translated a circuit into SAT, we saw that gates with more controls require more clauses. A direct k -control gate contributes $2k + 2$ clauses, while the auxiliary encoding contributes $k + 5$. We therefore do not want to make every gate wider just for the sake of making the formula larger. At the same time, staying entirely in r57 leaves every gate in the same complemented two-control form. We need a way to increase the variety of gate shapes while also continuing to mix where their effects live.

The exact r57 split was already introduced in *Strategies*. It replaces one r57 gate with a one-control conjunction and a two-control conjunction whose firing cases are disjoint. Their XOR is the firing condition of the parent, so the replacement preserves the complete circuit exactly. The important new freedom is that these fragments are plain conjunctions. Two plain conjunctions with opposite polarities on a shared control have disjoint firing regions and can commute, even when their parent gates could not. The initial split itself does not create gates wider than r57. The three-control and wider gates appear later, when conjunction fragments are split again while crossing colliders.

We use this idea in two related fragmentation methods.

Splitting. We first apply the randomized r57 split throughout the circuit. The two fragments begin where their parent was and receive opposite travel

directions. If we left every pair beside one another, however, the original gate would still be easy to recognize. Selected splits are therefore used to make a distant join on the same target wire w . Here, a join does not mean that the two gates are moved together. The first one-control fragment stays where its parent was, while the second bracket is chosen from the gates targeting w elsewhere in the circuit. That bracket may already be a one-control conjunction. If it is instead a complemented conjunction, we apply the same randomized first-failing-literal split and use its first one-control piece. The two one-control gates remain at the ends of the resulting segment.

We then flip the control polarity of both brackets and flip every use of w as a control between them. This is equivalent to inserting a NOT on w at each end of the segment, except that both NOTs are absorbed into gates which were already present. Gates which target w inside the segment do not change.

The two absorbed NOTs cancel, so the function remains unchanged. What changes is where the explanation for the rewrite lives: the local split, the distant bracket, and the flipped uses of w now compensate for one another across the whole segment. Any complemented gate which reads w along that segment is split as part of the same process. If a join is attempted, the two-control sibling is also sent through one ordinary crossing move in its assigned direction, whether or not a distant bracket was found. On the remaining bare-split branch, the move ends without that crossing. The intended exit is exhaustion of the complemented-gate population, with a safety exit after 100 consecutive failed bracket searches. The result is a varied collection of plain conjunctions rather than one repeated complemented gate form.

The crossing walk. The crossing walk begins with those conjunction fragments and their assigned directions, so it does not split r57 gates again. A fragment first moves through every gate with which it commutes. When it reaches a true collider, we use one of three exact case-split rules. In R1 the collider writes a control of the moving fragment, so the moving fragment splits and its pieces cross. In R2 the moving fragment writes a control of the collider, so the collider splits instead. In R3 each gate reads the other's target, so one restricted residue stays behind while the remaining residues cross. Each rule divides the collision into disjoint firing cases, just as the original r57 split did, and therefore preserves the function exactly.

The new fragments usually inherit the direction of the gate sent into the collision, immediately advance through most of their next free commuting run, and continue walking. A declined or blocked crossing retreats instead of leaving the gate parked at the collision. Repeating this lets pieces of gates which originally collided move into, and sometimes past, one another. It also produces conjunctions with different polarities and with three, four, or more controls. Thus, later algebraic attacks can no longer assume that the circuit consists only of the degree-two r57 polynomial shape.

Since each extra control also raises the SAT clause cost, the walk uses a width

damper. Before a crossing which requires a split, the damper is applied to the gate being split. A gate at or below the chosen threshold is admitted outright; above it, acceptance falls exponentially with its control count. A separate hard width cap rejects any rewrite whose emitted residue would be too wide. Together, these two checks let the vocabulary grow without allowing wide gates to take over the circuit.

Contraction during the crossing walk. Every successful crossing can replace one or two gates with several fragments, so a walk which only moved forward would continue growing. We instead hold the gate count near a chosen target with a thermostat. While the circuit is below that target, another forward crossing is more likely. As it approaches and moves above the target, contraction becomes more likely. This is probabilistic rather than a hard boundary, so crossings and contractions remain interleaved throughout the walk.

One contraction option uses the record made by an earlier crossing. It first checks that every fragment produced by that crossing still exists and has not been changed by another rewrite. It then floats those pieces back around the recorded pivot and restores the original pair. If even one fragment has already taken part in another move, the record is no longer usable. Thus, this operation removes crossings which remained isolated, while crossings whose pieces fed later work cannot simply be taken back.

The other contraction option looks for two compatible gates with the same active wire, meaning the same target. Since both gates XOR into that target without reading it, they commute with one another, and their combined firing condition is the XOR of their two conjunctions. In a few exact cases this XOR has a smaller form. For example,

$$a \oplus = bc, \quad a \oplus = (-b)c$$

can be replaced by

$$a \oplus = c.$$

This works because exactly one of the first two gates fires whenever $c = 1$. The complete merge catalogue can cancel identical gates, turn a conjunction paired with its complemented form into a NOT, drop the only literal whose polarity differs, simplify a pair where one control set is the other plus one literal, and absorb a bare NOT into a complemented gate.

The same target alone is not enough. The pair must have one of these exact relationships, must lie within the search distance, and must be able to commute to a common position. The first gate is floated toward the second and then the second toward the first. We do not fragment intervening blockers to force this contraction: if another gate prevents the pair from becoming adjacent, the

attempt fails. Newly created siblings are also protected for a short period, so a split cannot be undone or merged away immediately.

Finally, the online merge refuses any pair whose result would be a complemented conjunction. In particular, the two pieces of an r57 split are not allowed to rejoin into the r57 gate from which they came. Contraction can therefore control the size without steadily rebuilding the uniform gate form which fragmentation was meant to remove. This online pairwise contraction is separate from the stronger final compression pass described next.

Compressing fragments. The frozen-database mixer was built to move in both directions. It can expand or re-spell a window to create new structure, and it can later compress a window to control the size. The compression direction also acts as an attacker test: if our own compressor can immediately undo an expansion, we should assume that an attacker can do the same. We use this same idea after leaving r57. Splitting and crossing provide the expansion side for fragments, while our fragment compressor provides the corresponding fragment-aware compression side.

The key observation is that several gates with the same active wire all XOR into the same target. If those gates can commute to one common position, then together they have the form

$$t \oplus = f_1 \oplus f_2 \oplus \cdots \oplus f_k,$$

where each f_i is one mixed-polarity conjunction. This is an exclusive OR of products, or ESOP, so the compressor can simplify the complete group instead of only looking at adjacent pairs.

The pass repeats three steps. First, it **gathers** compatible gates in one forward sweep, keeping an open group for each target. Reading a target closes that target's group, because the reader pins the value accumulated so far. Writing to any control used by the group also closes it, because the control values may not change while the gates are being moved. These two rules ensure that every member of the group can legally commute to the point where the group is closed. Second, it **reduces** the gathered ESOP. It applies the same exact pairwise identities used above until none remain. When the group's total wire support is small enough, it also expands the group into algebraic normal form (ANF), an XOR of square-free monomials, cancels duplicate monomials, and keeps that spelling only when it is smaller. Third, it **re-emits** the surviving conjunction gates together at the closing point. The whole gather, reduce, and re-emit process repeats until the gate count stops shrinking or the iteration limit is reached.

This is stronger than the online contraction used during the crossing walk. The walk can undo one intact crossing or merge one nearby compatible pair; our fragment compressor can gather and reduce a larger same-target group after the mixing is finished. It is deterministic and attacker-computable, so the size

left after this pass is the honest effective size of the fragmented circuit. Any structure which this compressor removes is structure that we assume an attacker can remove as well.

At the end of each fragmentation stage, we make one final positional float. Every gate is moved to a uniformly chosen position inside its current two-sided commutation interval. This changes neither the gates nor the circuit's function, and it never moves a gate through a true collider. It only removes positional structure which does not need to remain at the stage boundary.

The two fragmentation methods therefore do different parts of the same job. Splitting removes the uniform r57 form and couples distant parts of the circuit. The crossing walk then spreads those fragments, creates a wider range of conjunction polynomials, and uses its two contraction paths to keep that growth under control. This does not by itself prove that the SAT problem is harder; we have not isolated gate variety as a SAT variable. It does remove the restriction that every gate, every local polynomial, and every gate encoding must have the same r57 form.

The frozen table emits r57 circuits, so all database mixing is completed before these two methods. Otherwise, each successful replacement would reintroduce the structure we had just removed. There is no need to return to pure r57 after the fragmentation stages. The next section puts these parts together and compares the questions asked by the different attacks.

The Complete Mixing Method

We combine the constructions above in the following order.

1. **Build the sliced sandwich.** We place the source computation inside a reversible circuit on $2n$ wires. On the zero slice, the first n outputs are allowed to contain junk and the latter n outputs contain the desired value $C(x)$.
2. **Apply embedded masking.** This takes the sandwich to $4n$ wires. We keep balanced embedded masks open on the carriers, compute through their decodes, shuffle complete fire units, and refresh the band. Opening/closing guards and the two band seed blocks preserve the required answer on the public slice.
3. **Run database mixing.** While the circuit is still close to the r57 vocabulary, we repeatedly replace local windows with different database spellings of the same function. As in generation mixing, we want the replacements to overlap repeatedly. We grow to twice the incoming gate count and then hold near that size while continuing to replace windows.
4. **Split the remaining r57 gates.** We rewrite the uniform r57 structure into one- and two-control pieces and use the splitting construction to couple distant parts of the circuit. The resulting fragments keep their assigned

travel directions for the crossing walk.

5. **Run the crossing walk.** We shoot the resulting fragments through the circuit and fragment the gates which block them. This spreads the pieces of each earlier gate and creates a much wider range of conjunction polynomials than r57 alone provides.
6. **Compress the fragments.** Finally, we apply our whole-circuit compressor and remove whatever local redundancy it can still recover. The compressed output, rather than the larger pre-compression circuit, is the final published circuit.

This ordering matters. The frozen database is most useful before fragmentation, while the gates still resemble the r57 circuits from which the table was built. Splitting and crossing then change that structure without asking the database to translate it back into r57. The final compression pass is not another hiding layer; it makes sure that we do not count expansion which our own attacker-computable compressor can undo.

Where this leaves the solver

We judge a complete method through SAT solving, local affine reconstruction, whole-trace reconstruction, and compression. The SAT, local heatmap, and compression results below concern the full-pipeline experiments with product-share gadgetization. The whole-trace results concern the embedded-masking experiments in the gadget gauntlet. These measurements belong to those constructions and input policies.

SAT solving. In the SAT tests containing all six stages, the solver has not found a preimage within the allotted runs. This means that the complete method remains unresolved under the solvers and budgets we have tried. It does not prove that the instances are unsatisfiable or that a stronger solver could not recover a preimage.

Affine reconstruction. The affine reconstruction heatmaps we have run do not show the old interior progress diagonal. The forced input and output boundaries remain, but between them the affine predictor we tested cannot follow the original circuit’s intermediate states. This is exactly the leak which remained under the earlier linear gadgetization.

Whole-trace reconstruction (xtrace). We still fail this test. An observer who combines wire values from different times can recover source intermediate values which the local heatmaps miss. In the embedded-masking gauntlet, these recoveries remain after the tested mixing pass. Thus, removing the local affine diagonal has not removed the whole-circuit linear correlators.

Compression. The final circuits are effectively incompressible under our fragment compressor. Across four independent $n = 128$ full-pipeline outputs, the compressor retained 93.7–93.8% of the gates, removing only 6.2–6.3%. Thus,

the size of the final circuit is mostly mixing which survives our own compressor rather than easy expansion. Here, incompressibility is relative to our fragment compressor; a stronger compressor may still find structure which ours does not.

These tests ask different questions. SAT asks for a preimage, while the classic indistinguishability game for iO asks an attacker to tell which of two equal-size circuits computing the same function was obfuscated. No efficient attacker should do appreciably better than guessing. The attacker knows both candidate circuits and need not recover a preimage or compress the published circuit to distinguish them.

The remaining **xtrace** failure is therefore a failure of the hiding we need for that game. A source-dependent recovery relation can give the observer information about which computation is inside the obfuscation. The gauntlet measures those recovery relations; a complete game attack would also compare them between the two equivalent candidate circuits. We cannot treat the SAT timeouts or compression results as having settled indistinguishability. We do not yet have general iO.

Conclusion

The complete construction combines the sliced sandwich, embedded masking, database mixing, splitting, crossing, and final fragment compression. Starting from a source computation on n wires, the sandwich and gadgetization take us to $4n$ physical wires. The later stages change and spread the gate representation without requiring another increase in the wire count.

Each part addresses a different failure from the earlier methods. The nonlinear masks address the affine direction left by linear gadgetization. embedded fire keeps those masks nonlinear while we read the values. The gadget gauntlet checks both local recovery and linear relations across the entire circuit; the embedded-masking tests still have whole-trace recoveries. Database mixing repeatedly changes the local spelling while that database is still useful. Splitting and crossing leave the uniform r57 gate form and spread the resulting fragments throughout the circuit. The final compressor then removes the redundancy which remains easy for us to find.

The earlier experiments above did not yield a SAT preimage within their budgets, did not reveal the old affine diagonal, and retained most gates under our fragment compressor. These results motivate the construction, but describe the particular circuits, attacks, and budgets we tested. They are neither a proof of security nor measurements of the complete method with embedded masking substituted for the product-share construction.

Our tests have primarily been on random circuits and related families, and the **xtrace** failure remains. Thus, we have not yet achieved the indistinguishability required for general iO. Our results concern progress on particular hiding and preimage tests. We hope that these ideas can give us things like trapdoor permu-

tations, but we have not yet reached the level of fully homomorphic encryption, functional encryption, or succinct non-interactive arguments of knowledge.

September 16, 2026: Optional carrier shuffling

Embedded masking now has an optional preprocessing transform which moves a logical target between physical carrier wires during its fire block. This is different from the fire-unit order shuffles already described above. It remains the same `embedded-masking` mode: `preprocessing.shuffling_segments = 0` leaves the transform off, and values of at least eight enable it.

The emitter records exact boundaries between complete fire units and mask atoms, together with the target's live mask support. At selected boundaries, two CNOTs transfer a target value X and add a temporary mask from a band value w : $(p, q) = (X, w)$ becomes $(w, X \oplus w)$. A role-to-physical-wire map redirects subsequent gates, and the temporary masks are removed at the end of the block. The full sandwich returns the data role to its original wire after the block's final original write, preserving the public input and output positions. An experimental gauntlet arm may instead carry the layout and decode through the final map.

The implementation excludes all control dependencies of every remaining atomic unit that writes the target, including controls on dirty-helper writes. It does not split those units. Mask-coverage diagnostics are computed before physical remapping, where the original data and band role coordinates remain valid. Gauntlet comparisons can now use paired lists of mixer seeds instead of one hard-coded mixer seed.

The earlier research harness and the in-tree transform use different random streams and boundary policies, and return-home is an additional construction choice. Consequently, the historical heatmaps, SAT timings, compression ratios, and gauntlet counts above are not measurements of this new transform. See preprocessing shuffling for its equations and invariants, and the attack methodology for the distinctions a new comparison must preserve.

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